

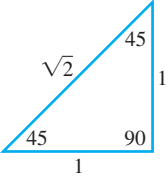
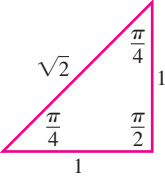
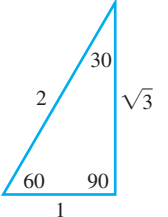
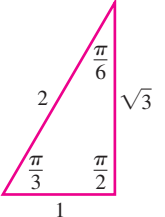
Degrees	Radians
	
	

FIGURE 1.64 The angles of two common triangles, in degrees and radians.

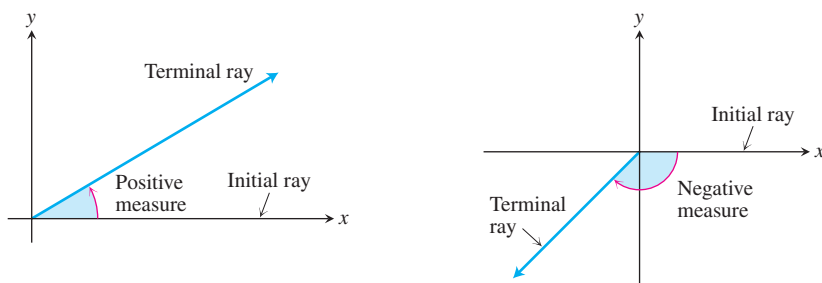


FIGURE 1.65 Angles in standard position in the xy -plane.

When angles are used to describe counterclockwise rotations, our measurements can go arbitrarily far beyond 2π radians or 360° . Similarly, angles describing clockwise rotations can have negative measures of all sizes (Figure 1.66).

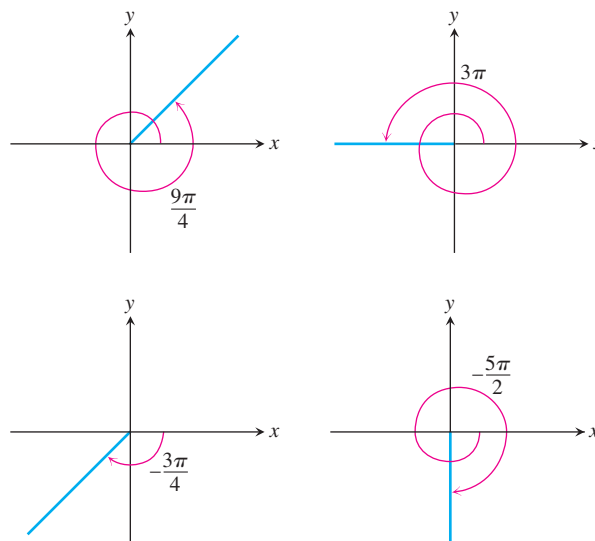


FIGURE 1.66 Nonzero radian measures can be positive or negative and can go beyond 2π .

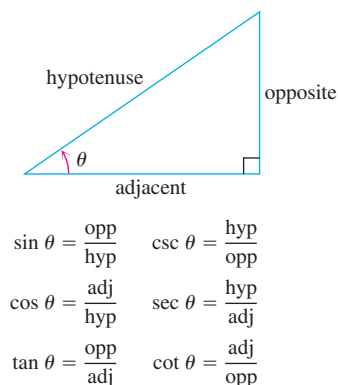


FIGURE 1.67 Trigonometric ratios of an acute angle.

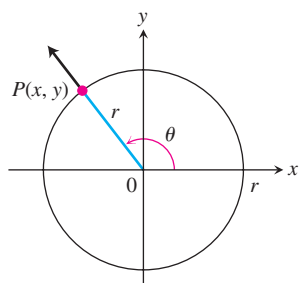


FIGURE 1.68 The trigonometric functions of a general angle θ are defined in terms of x , y , and r .

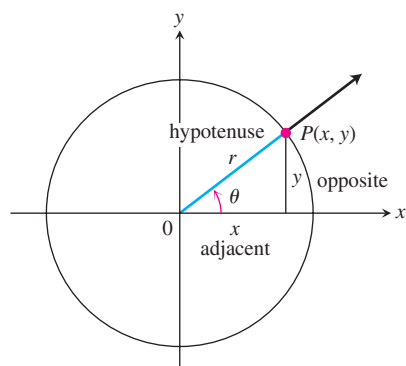


FIGURE 1.69 The new and old definitions agree for acute angles.

Angle Convention: Use Radians

From now on in this book it is assumed that all angles are measured in radians unless degrees or some other unit is stated explicitly. When we talk about the angle $\pi/3$, we mean $\pi/3$ radians (which is 60°), not $\pi/3$ degrees. When you do calculus, keep your calculator in radian mode.

The Six Basic Trigonometric Functions

You are probably familiar with defining the trigonometric functions of an acute angle in terms of the sides of a right triangle (Figure 1.67). We extend this definition to obtuse and negative angles by first placing the angle in standard position in a circle of radius r . We then define the trigonometric functions in terms of the coordinates of the point $P(x, y)$ where the angle's terminal ray intersects the circle (Figure 1.68).

$$\text{sine: } \sin \theta = \frac{y}{r} \qquad \text{cosecant: } \csc \theta = \frac{r}{y}$$

$$\text{cosine: } \cos \theta = \frac{x}{r} \qquad \text{secant: } \sec \theta = \frac{r}{x}$$

$$\text{tangent: } \tan \theta = \frac{y}{x} \qquad \text{cotangent: } \cot \theta = \frac{x}{y}$$

These extended definitions agree with the right-triangle definitions when the angle is acute (Figure 1.69).

Notice also the following definitions, whenever the quotients are defined.

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \qquad \cot \theta = \frac{1}{\tan \theta}$$

$$\sec \theta = \frac{1}{\cos \theta} \qquad \csc \theta = \frac{1}{\sin \theta}$$

As you can see, $\tan \theta$ and $\sec \theta$ are not defined if $x = 0$. This means they are not defined if θ is $\pm\pi/2, \pm3\pi/2, \dots$. Similarly, $\cot \theta$ and $\csc \theta$ are not defined for values of θ for which $y = 0$, namely $\theta = 0, \pm\pi, \pm2\pi, \dots$.

The exact values of these trigonometric ratios for some angles can be read from the triangles in Figure 1.64. For instance,

$$\begin{array}{lll} \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} & \sin \frac{\pi}{6} = \frac{1}{2} & \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \\ \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} & \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2} & \cos \frac{\pi}{3} = \frac{1}{2} \\ \tan \frac{\pi}{4} = 1 & \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} & \tan \frac{\pi}{3} = \sqrt{3} \end{array}$$

The CAST rule (Figure 1.70) is useful for remembering when the basic trigonometric functions are positive or negative. For instance, from the triangle in Figure 1.71, we see that

$$\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}, \quad \cos \frac{2\pi}{3} = -\frac{1}{2}, \quad \tan \frac{2\pi}{3} = -\sqrt{3}.$$

Using a similar method we determined the values of $\sin \theta$, $\cos \theta$, and $\tan \theta$ shown in Table 1.4.

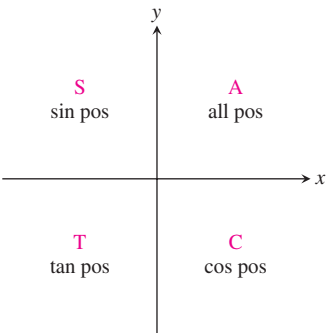


FIGURE 1.70 The CAST rule, remembered by the statement “All Students Take Calculus,” tells which trigonometric functions are positive in each quadrant.

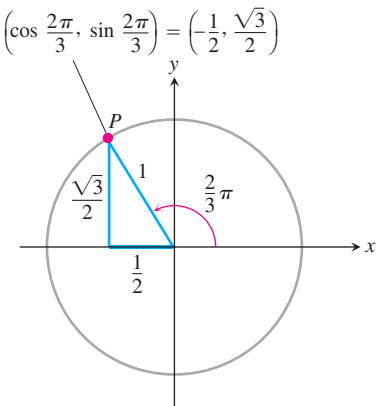


FIGURE 1.71 The triangle for calculating the sine and cosine of $2\pi/3$ radians. The side lengths come from the geometry of right triangles.

Most calculators and computers readily provide values of the trigonometric functions for angles given in either radians or degrees.

TABLE 1.4 Values of $\sin \theta$, $\cos \theta$, and $\tan \theta$ for selected values of θ															
Degrees	−180	−135	−90	−45	0	30	45	60	90	120	135	150	180	270	360
θ (radians)	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π
$\sin \theta$	0	$-\frac{\sqrt{2}}{2}$	−1	$-\frac{\sqrt{2}}{2}$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	−1	0
$\cos \theta$	−1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{3}}{2}$	−1	0	1
$\tan \theta$	0	1		−1	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$		$-\sqrt{3}$	−1	$-\frac{\sqrt{3}}{3}$	0		0

EXAMPLE 1 Finding Trigonometric Function Values

If $\tan \theta = 3/2$ and $0 < \theta < \pi/2$, find the five other trigonometric functions of θ .

Solution From $\tan \theta = 3/2$, we construct the right triangle of height 3 (opposite) and base 2 (adjacent) in Figure 1.72. The Pythagorean theorem gives the length of the hypotenuse, $\sqrt{4 + 9} = \sqrt{13}$. From the triangle we write the values of the other five trigonometric functions:

$$\cos \theta = \frac{2}{\sqrt{13}}, \quad \sin \theta = \frac{3}{\sqrt{13}}, \quad \sec \theta = \frac{\sqrt{13}}{2}, \quad \csc \theta = \frac{\sqrt{13}}{3}, \quad \cot \theta = \frac{2}{3}$$

■

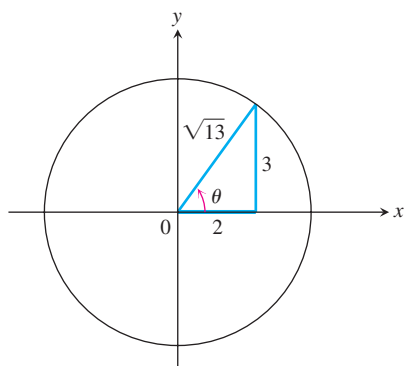


FIGURE 1.72 The triangle for calculating the trigonometric functions in Example 1.

Periodicity and Graphs of the Trigonometric Functions

When an angle of measure θ and an angle of measure $\theta + 2\pi$ are in standard position, their terminal rays coincide. The two angles therefore have the same trigonometric function values:

$$\begin{array}{lll} \cos(\theta + 2\pi) = \cos \theta & \sin(\theta + 2\pi) = \sin \theta & \tan(\theta + 2\pi) = \tan \theta \\ \sec(\theta + 2\pi) = \sec \theta & \csc(\theta + 2\pi) = \csc \theta & \cot(\theta + 2\pi) = \cot \theta \end{array}$$

Similarly, $\cos(\theta - 2\pi) = \cos \theta$, $\sin(\theta - 2\pi) = \sin \theta$, and so on. We describe this repeating behavior by saying that the six basic trigonometric functions are *periodic*.

DEFINITION Periodic Function

A function $f(x)$ is **periodic** if there is a positive number p such that $f(x + p) = f(x)$ for every value of x . The smallest such value of p is the **period** of f .

When we graph trigonometric functions in the coordinate plane, we usually denote the independent variable by x instead of θ . See Figure 1.73.

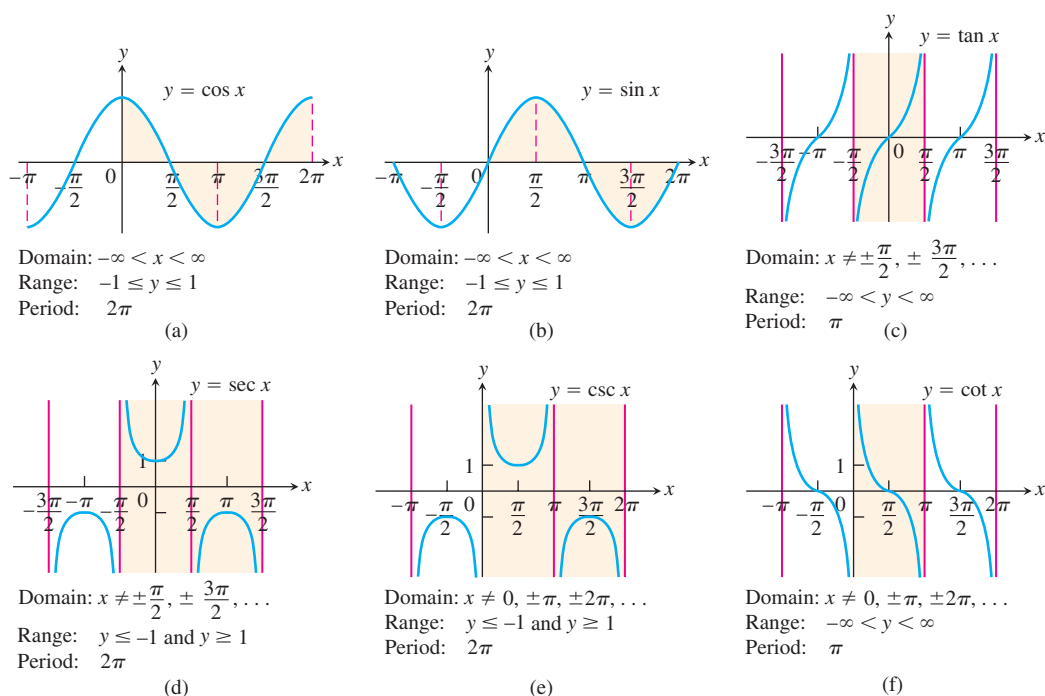


FIGURE 1.73 Graphs of the (a) cosine, (b) sine, (c) tangent, (d) secant, (e) cosecant, and (f) cotangent functions using radian measure. The shading for each trigonometric function indicates its periodicity.

Periods of Trigonometric Functions

Period π : $\tan(x + \pi) = \tan x$
 $\cot(x + \pi) = \cot x$

Period 2π : $\sin(x + 2\pi) = \sin x$
 $\cos(x + 2\pi) = \cos x$
 $\sec(x + 2\pi) = \sec x$
 $\csc(x + 2\pi) = \csc x$

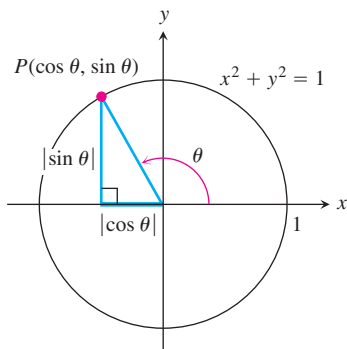


FIGURE 1.74 The reference triangle for a general angle θ .

As we can see in Figure 1.73, the tangent and cotangent functions have period $p = \pi$. The other four functions have period 2π . Periodic functions are important because many behaviors studied in science are approximately periodic. A theorem from advanced calculus says that every periodic function we want to use in mathematical modeling can be written as an algebraic combination of sines and cosines. We show how to do this in Section 11.11.

The symmetries in the graphs in Figure 1.73 reveal that the cosine and secant functions are even and the other four functions are odd:

Even	Odd
$\cos(-x) = \cos x$	$\sin(-x) = -\sin x$
$\sec(-x) = \sec x$	$\tan(-x) = -\tan x$
	$\csc(-x) = -\csc x$
	$\cot(-x) = -\cot x$

Identities

The coordinates of any point $P(x, y)$ in the plane can be expressed in terms of the point's distance from the origin and the angle that ray OP makes with the positive x -axis (Figure 1.69). Since $x/r = \cos \theta$ and $y/r = \sin \theta$, we have

$$x = r \cos \theta, \quad y = r \sin \theta.$$

When $r = 1$ we can apply the Pythagorean theorem to the reference right triangle in Figure 1.74 and obtain the equation

$$\cos^2 \theta + \sin^2 \theta = 1. \quad (1)$$

This equation, true for all values of θ , is the most frequently used identity in trigonometry. Dividing this identity in turn by $\cos^2 \theta$ and $\sin^2 \theta$ gives

$$1 + \tan^2 \theta = \sec^2 \theta.$$

$$1 + \cot^2 \theta = \csc^2 \theta.$$

The following formulas hold for all angles A and B (Exercises 53 and 54).

Addition Formulas

$$\begin{aligned} \cos(A + B) &= \cos A \cos B - \sin A \sin B \\ \sin(A + B) &= \sin A \cos B + \cos A \sin B \end{aligned} \quad (2)$$

There are similar formulas for $\cos(A - B)$ and $\sin(A - B)$ (Exercises 35 and 36). All the trigonometric identities needed in this book derive from Equations (1) and (2). For example, substituting θ for both A and B in the addition formulas gives

Double-Angle Formulas

$$\begin{aligned}\cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ \sin 2\theta &= 2 \sin \theta \cos \theta\end{aligned}\tag{3}$$

Additional formulas come from combining the equations

$$\cos^2 \theta + \sin^2 \theta = 1, \quad \cos^2 \theta - \sin^2 \theta = \cos 2\theta.$$

We add the two equations to get $2 \cos^2 \theta = 1 + \cos 2\theta$ and subtract the second from the first to get $2 \sin^2 \theta = 1 - \cos 2\theta$. This results in the following identities, which are useful in integral calculus.

Half-Angle Formulas

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}\tag{4}$$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}\tag{5}$$

The Law of Cosines

If a , b , and c are sides of a triangle ABC and if θ is the angle opposite c , then

$$c^2 = a^2 + b^2 - 2ab \cos \theta.\tag{6}$$

This equation is called the **law of cosines**.

We can see why the law holds if we introduce coordinate axes with the origin at C and the positive x -axis along one side of the triangle, as in Figure 1.75. The coordinates of A are $(b, 0)$; the coordinates of B are $(a \cos \theta, a \sin \theta)$. The square of the distance between A and B is therefore

$$\begin{aligned}c^2 &= (a \cos \theta - b)^2 + (a \sin \theta)^2 \\ &= a^2(\underbrace{\cos^2 \theta + \sin^2 \theta}_1) + b^2 - 2ab \cos \theta \\ &= a^2 + b^2 - 2ab \cos \theta.\end{aligned}$$

The law of cosines generalizes the Pythagorean theorem. If $\theta = \pi/2$, then $\cos \theta = 0$ and $c^2 = a^2 + b^2$.

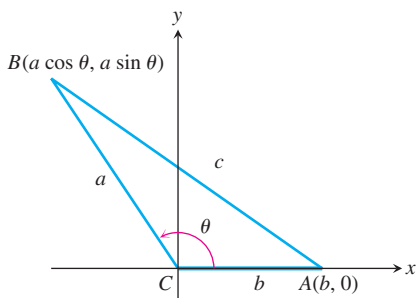
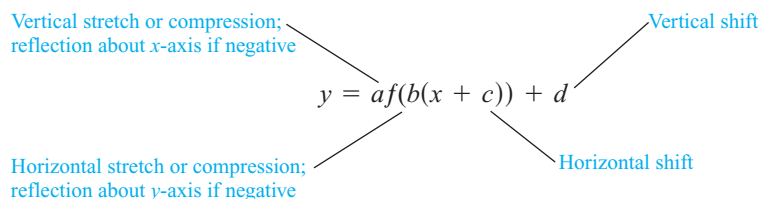


FIGURE 1.75 The square of the distance between A and B gives the law of cosines.

Transformations of Trigonometric Graphs

The rules for shifting, stretching, compressing, and reflecting the graph of a function apply to the trigonometric functions. The following diagram will remind you of the controlling parameters.



EXAMPLE 2 Modeling Temperature in Alaska

The builders of the Trans-Alaska Pipeline used insulated pads to keep the pipeline heat from melting the permanently frozen soil beneath. To design the pads, it was necessary to take into account the variation in air temperature throughout the year. The variation was represented in the calculations by a **general sine function** or **sinusoid** of the form

$$f(x) = A \sin \left[\frac{2\pi}{B} (x - C) \right] + D,$$

where $|A|$ is the *amplitude*, $|B|$ is the *period*, C is the *horizontal shift*, and D is the *vertical shift* (Figure 1.76).

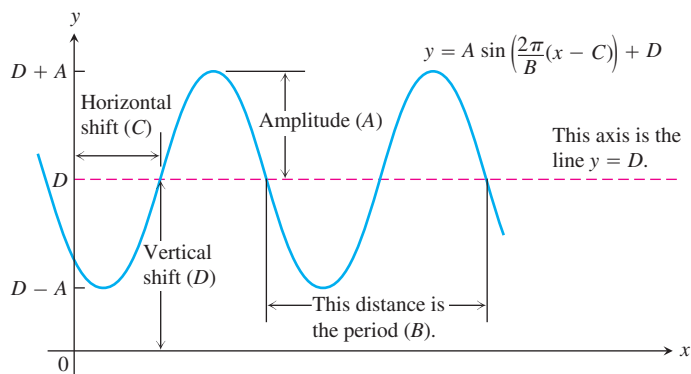


FIGURE 1.76 The general sine curve $y = A \sin [(2\pi/B)(x - C)] + D$, shown for A , B , C , and D positive (Example 2).

Figure 1.77 shows how to use such a function to represent temperature data. The data points in the figure are plots of the mean daily air temperatures for Fairbanks, Alaska, based on records of the National Weather Service from 1941 to 1970. The sine function used to fit the data is

$$f(x) = 37 \sin \left[\frac{2\pi}{365} (x - 101) \right] + 25,$$

where f is temperature in degrees Fahrenheit and x is the number of the day counting from the beginning of the year. The fit, obtained by using the sinusoidal regression feature on a calculator or computer, as we discuss in the next section, is very good at capturing the trend of the data.

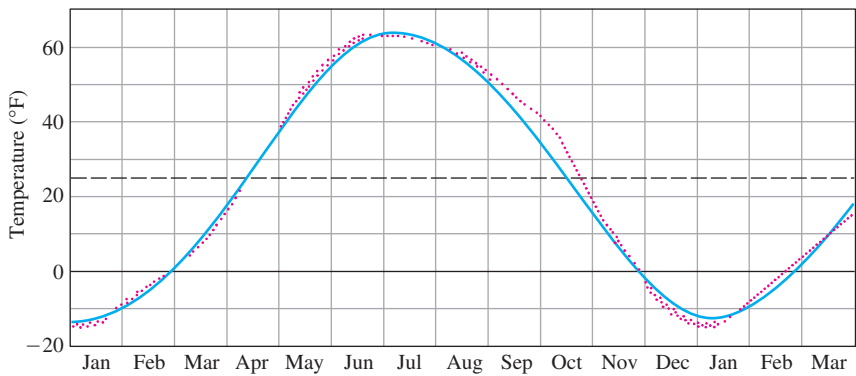


FIGURE 1.77 Normal mean air temperatures for Fairbanks, Alaska, plotted as data points (red). The approximating sine function (blue) is

$$f(x) = 37 \sin [(2\pi / 365)(x - 101)] + 25.$$



EXERCISES 1.6

Radians, Degrees, and Circular Arcs

- On a circle of radius 10 m, how long is an arc that subtends a central angle of (a) $4\pi/5$ radians? (b) 110° ?
- A central angle in a circle of radius 8 is subtended by an arc of length 10π . Find the angle's radian and degree measures.
- You want to make an 80° angle by marking an arc on the perimeter of a 12-in.-diameter disk and drawing lines from the ends of the arc to the disk's center. To the nearest tenth of an inch, how long should the arc be?
- If you roll a 1-m-diameter wheel forward 30 cm over level ground, through what angle will the wheel turn? Answer in radians (to the nearest tenth) and degrees (to the nearest degree).

Evaluating Trigonometric Functions

- Copy and complete the following table of function values. If the function is undefined at a given angle, enter "UND." Do not use a calculator or tables.

θ	$-\pi$	$-2\pi/3$	0	$\pi/2$	$3\pi/4$
$\sin \theta$					
$\cos \theta$					
$\tan \theta$					
$\cot \theta$					
$\sec \theta$					
$\csc \theta$					

- Copy and complete the following table of function values. If the function is undefined at a given angle, enter "UND." Do not use a calculator or tables.

θ	$-3\pi/2$	$-\pi/3$	$-\pi/6$	$\pi/4$	$5\pi/6$
$\sin \theta$					
$\cos \theta$					
$\tan \theta$					
$\cot \theta$					
$\sec \theta$					
$\csc \theta$					

In Exercises 7–12, one of $\sin x$, $\cos x$, and $\tan x$ is given. Find the other two if x lies in the specified interval.

- $\sin x = \frac{3}{5}$, $x \in \left[\frac{\pi}{2}, \pi\right]$
- $\tan x = 2$, $x \in \left[0, \frac{\pi}{2}\right]$
- $\cos x = \frac{1}{3}$, $x \in \left[-\frac{\pi}{2}, 0\right]$
- $\cos x = -\frac{5}{13}$, $x \in \left[\frac{\pi}{2}, \pi\right]$
- $\tan x = \frac{1}{2}$, $x \in \left[\pi, \frac{3\pi}{2}\right]$
- $\sin x = -\frac{1}{2}$, $x \in \left[\pi, \frac{3\pi}{2}\right]$

Graphing Trigonometric Functions

Graph the functions in Exercises 13–22. What is the period of each function?

13. $\sin 2x$

14. $\sin(x/2)$

15. $\cos \pi x$ 16. $\cos \frac{\pi x}{2}$
 17. $-\sin \frac{\pi x}{3}$ 18. $-\cos 2\pi x$
 19. $\cos\left(x - \frac{\pi}{2}\right)$ 20. $\sin\left(x + \frac{\pi}{2}\right)$
 21. $\sin\left(x - \frac{\pi}{4}\right) + 1$ 22. $\cos\left(x + \frac{\pi}{4}\right) - 1$

Graph the functions in Exercises 23–26 in the ts -plane (t -axis horizontal, s -axis vertical). What is the period of each function? What symmetries do the graphs have?

23. $s = \cot 2t$ 24. $s = -\tan \pi t$
 25. $s = \sec\left(\frac{\pi t}{2}\right)$ 26. $s = \csc\left(\frac{t}{2}\right)$

- T** 27. a. Graph $y = \cos x$ and $y = \sec x$ together for $-3\pi/2 \leq x \leq 3\pi/2$. Comment on the behavior of $\sec x$ in relation to the signs and values of $\cos x$.
 b. Graph $y = \sin x$ and $y = \csc x$ together for $-\pi \leq x \leq 2\pi$. Comment on the behavior of $\csc x$ in relation to the signs and values of $\sin x$.
T 28. Graph $y = \tan x$ and $y = \cot x$ together for $-7 \leq x \leq 7$. Comment on the behavior of $\cot x$ in relation to the signs and values of $\tan x$.
 29. Graph $y = \sin x$ and $y = \lfloor \sin x \rfloor$ together. What are the domain and range of $\lfloor \sin x \rfloor$?
 30. Graph $y = \sin x$ and $y = \lceil \sin x \rceil$ together. What are the domain and range of $\lceil \sin x \rceil$?

Additional Trigonometric Identities

Use the addition formulas to derive the identities in Exercises 31–36.

31. $\cos\left(x - \frac{\pi}{2}\right) = \sin x$ 32. $\cos\left(x + \frac{\pi}{2}\right) = -\sin x$
 33. $\sin\left(x + \frac{\pi}{2}\right) = \cos x$ 34. $\sin\left(x - \frac{\pi}{2}\right) = -\cos x$
 35. $\cos(A - B) = \cos A \cos B + \sin A \sin B$ (Exercise 53 provides a different derivation.)
 36. $\sin(A - B) = \sin A \cos B - \cos A \sin B$
 37. What happens if you take $B = A$ in the identity $\cos(A - B) = \cos A \cos B + \sin A \sin B$? Does the result agree with something you already know?
 38. What happens if you take $B = 2\pi$ in the addition formulas? Do the results agree with something you already know?

Using the Addition Formulas

In Exercises 39–42, express the given quantity in terms of $\sin x$ and $\cos x$.

39. $\cos(\pi + x)$ 40. $\sin(2\pi - x)$

41. $\sin\left(\frac{3\pi}{2} - x\right)$ 42. $\cos\left(\frac{3\pi}{2} + x\right)$
 43. Evaluate $\sin \frac{7\pi}{12}$ as $\sin\left(\frac{\pi}{4} + \frac{\pi}{3}\right)$.
 44. Evaluate $\cos \frac{11\pi}{12}$ as $\cos\left(\frac{\pi}{4} + \frac{2\pi}{3}\right)$.
 45. Evaluate $\cos \frac{\pi}{12}$.
 46. Evaluate $\sin \frac{5\pi}{12}$.

Using the Double-Angle Formulas

Find the function values in Exercises 47–50.

47. $\cos^2 \frac{\pi}{8}$ 48. $\cos^2 \frac{\pi}{12}$
 49. $\sin^2 \frac{\pi}{12}$ 50. $\sin^2 \frac{\pi}{8}$

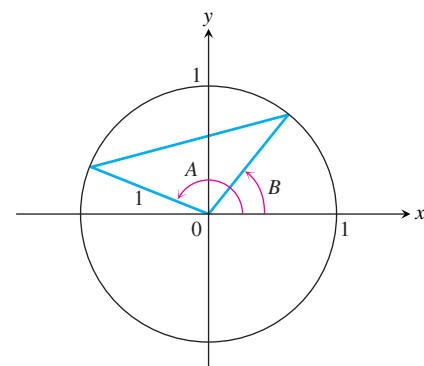
Theory and Examples

51. **The tangent sum formula** The standard formula for the tangent of the sum of two angles is

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

Derive the formula.

52. (Continuation of Exercise 51.) Derive a formula for $\tan(A - B)$.
 53. Apply the law of cosines to the triangle in the accompanying figure to derive the formula for $\cos(A - B)$.

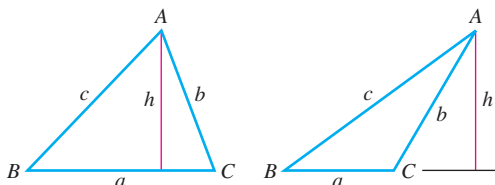


54. a. Apply the formula for $\cos(A - B)$ to the identity $\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$ to obtain the addition formula for $\sin(A + B)$.
 b. Derive the formula for $\cos(A + B)$ by substituting $-B$ for B in the formula for $\cos(A - B)$ from Exercise 53.
 55. A triangle has sides $a = 2$ and $b = 3$ and angle $C = 60^\circ$. Find the length of side c .

56. A triangle has sides $a = 2$ and $b = 3$ and angle $C = 40^\circ$. Find the length of side c .
57. **The law of sines** The *law of sines* says that if a , b , and c are the sides opposite the angles A , B , and C in a triangle, then

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}.$$

Use the accompanying figures and the identity $\sin(\pi - \theta) = \sin \theta$, if required, to derive the law.



58. A triangle has sides $a = 2$ and $b = 3$ and angle $C = 60^\circ$ (as in Exercise 55). Find the sine of angle B using the law of sines.
59. A triangle has side $c = 2$ and angles $A = \pi/4$ and $B = \pi/3$. Find the length a of the side opposite A .

- T** 60. **The approximation $\sin x \approx x$** It is often useful to know that, when x is measured in radians, $\sin x \approx x$ for numerically small values of x . In Section 3.8, we will see why the approximation holds. The approximation error is less than 1 in 5000 if $|x| < 0.1$.
- With your grapher in radian mode, graph $y = \sin x$ and $y = x$ together in a viewing window about the origin. What do you see happening as x nears the origin?
 - With your grapher in degree mode, graph $y = \sin x$ and $y = x$ together about the origin again. How is the picture different from the one obtained with radian mode?
 - A quick radian mode check** Is your calculator in radian mode? Evaluate $\sin x$ at a value of x near the origin, say $x = 0.1$. If $\sin x \approx x$, the calculator is in radian mode; if not, it isn't. Try it.

General Sine Curves

For

$$f(x) = A \sin\left(\frac{2\pi}{B}(x - C)\right) + D,$$

identify A , B , C , and D for the sine functions in Exercises 61–64 and sketch their graphs (see Figure 1.76).

61. $y = 2 \sin(x + \pi) - 1$ 62. $y = \frac{1}{2} \sin(\pi x - \pi) + \frac{1}{2}$
63. $y = -\frac{2}{\pi} \sin\left(\frac{\pi}{2}t\right) + \frac{1}{\pi}$ 64. $y = \frac{L}{2\pi} \sin \frac{2\pi t}{L}, \quad L > 0$

65. **Temperature in Fairbanks, Alaska** Find the (a) amplitude, (b) period, (c) horizontal shift, and (d) vertical shift of the general sine function

$$f(x) = 37 \sin\left(\frac{2\pi}{365}(x - 101)\right) + 25.$$

66. **Temperature in Fairbanks, Alaska** Use the equation in Exercise 65 to approximate the answers to the following questions about the temperature in Fairbanks, Alaska, shown in Figure 1.77. Assume that the year has 365 days.

- What are the highest and lowest mean daily temperatures shown?
- What is the average of the highest and lowest mean daily temperatures shown? Why is this average the vertical shift of the function?

COMPUTER EXPLORATIONS

In Exercises 67–70, you will explore graphically the general sine function

$$f(x) = A \sin\left(\frac{2\pi}{B}(x - C)\right) + D$$

as you change the values of the constants A , B , C , and D . Use a CAS or computer grapher to perform the steps in the exercises.

67. **The period B** Set the constants $A = 3$, $C = D = 0$.
- Plot $f(x)$ for the values $B = 1, 3, 2\pi, 5\pi$ over the interval $-4\pi \leq x \leq 4\pi$. Describe what happens to the graph of the general sine function as the period increases.
 - What happens to the graph for negative values of B ? Try it with $B = -3$ and $B = -2\pi$.
68. **The horizontal shift C** Set the constants $A = 3$, $B = 6$, $D = 0$.
- Plot $f(x)$ for the values $C = 0, 1$, and 2 over the interval $-4\pi \leq x \leq 4\pi$. Describe what happens to the graph of the general sine function as C increases through positive values.
 - What happens to the graph for negative values of C ?
 - What smallest positive value should be assigned to C so the graph exhibits no horizontal shift? Confirm your answer with a plot.
69. **The vertical shift D** Set the constants $A = 3$, $B = 6$, $C = 0$.
- Plot $f(x)$ for the values $D = 0, 1$, and 3 over the interval $-4\pi \leq x \leq 4\pi$. Describe what happens to the graph of the general sine function as D increases through positive values.
 - What happens to the graph for negative values of D ?
70. **The amplitude A** Set the constants $B = 6$, $C = D = 0$.
- Describe what happens to the graph of the general sine function as A increases through positive values. Confirm your answer by plotting $f(x)$ for the values $A = 1, 5$, and 9 .
 - What happens to the graph for negative values of A ?

1.7

Graphing with Calculators and Computers

A graphing calculator or a computer with graphing software enables us to graph very complicated functions with high precision. Many of these functions could not otherwise be easily graphed. However, care must be taken when using such devices for graphing purposes and we address those issues in this section. In Chapter 4 we will see how calculus helps us to be certain we are viewing accurately all the important features of a function's graph.

Graphing Windows

When using a graphing calculator or computer as a graphing tool, a portion of the graph is displayed in a rectangular **display** or **viewing window**. Often the default window gives an incomplete or misleading picture of the graph. We use the term *square window* when the units or scales on both axis are the same. This term does not mean that the display window itself is square in shape (usually it is rectangular), but means instead that the x -unit is the same as the y -unit.

When a graph is displayed in the default window, the x -unit may differ from the y -unit of scaling in order to fit the graph in the display. The viewing window in the display is set by specifying the minimum and maximum values of the independent and dependent variables. That is, an interval $a \leq x \leq b$ is specified as well as a range $c \leq y \leq d$. The machine selects a certain number of equally spaced values of x between a and b . Starting with a first value for x , if it lies within the domain of the function f being graphed, and if $f(x)$ lies inside the range $[c, d]$, then the point $(x, f(x))$ is plotted. If x lies outside the domain of f , or $f(x)$ lies outside the specified range $[c, d]$, the machine just moves on to the next x -value since it cannot plot $(x, f(x))$ in that case. The machine plots a large number of points $(x, f(x))$ in this way and approximates the curve representing the graph by drawing a short line segment between each plotted point and its next neighboring point, as we might do by hand. Usually, adjacent points are so close together that the graphical representation has the appearance of a smooth curve. Things can go wrong with this procedure and we illustrate the most common problems through the following examples.

EXAMPLE 1 Choosing a Viewing Window

Graph the function $f(x) = x^3 - 7x^2 + 28$ in each of the following display or viewing windows:

- (a) $[-10, 10]$ by $[-10, 10]$ (b) $[-4, 4]$ by $[-50, 10]$ (c) $[-4, 10]$ by $[-60, 60]$

Solution

- (a) We select $a = -10$, $b = 10$, $c = -10$, and $d = 10$ to specify the interval of x -values and the range of y -values for the window. The resulting graph is shown in Figure 1.78a. It appears that the window is cutting off the bottom part of the graph and that the interval of x -values is too large. Let's try the next window.
- (b) Now we see more features of the graph (Figure 1.78b), but the top is missing and we need to view more to the right of $x = 4$ as well. The next window should help.
- (c) Figure 1.78c shows the graph in this new viewing window. Observe that we get a more complete picture of the graph in this window and it is a reasonable graph of a third-degree polynomial. Choosing a good viewing window is a trial-and-error process which may require some troubleshooting as well.

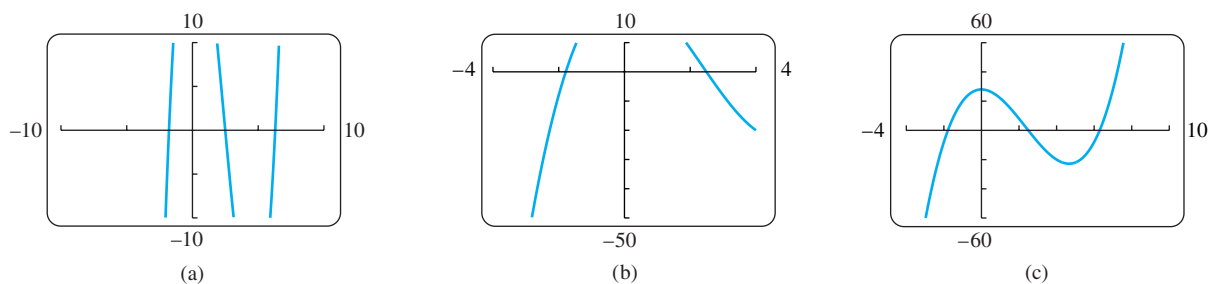


FIGURE 1.78 The graph of $f(x) = x^3 - 7x^2 + 28$ in different viewing windows (Example 1).

EXAMPLE 2 Square Windows

When a graph is displayed, the x -unit may differ from the y -unit, as in the graphs shown in Figures 1.78b and 1.78c. The result is distortion in the picture, which may be misleading. The display window can be made square by compressing or stretching the units on one axis to match the scale on the other, giving the true graph. Many systems have built-in functions to make the window “square.” If yours does not, you will have to do some calculations and set the window size manually to get a square window, or bring to your viewing some foreknowledge of the true picture.

Figure 1.79a shows the graphs of the perpendicular lines $y = x$ and $y = -x + 3\sqrt{2}$, together with the semicircle $y = \sqrt{9 - x^2}$, in a nonsquare $[-6, 6]$ by $[-6, 8]$ display window. Notice the distortion. The lines do not appear to be perpendicular, and the semicircle appears to be elliptical in shape.

Figure 1.79b shows the graphs of the same functions in a square window in which the x -units are scaled to be the same as the y -units. Notice that the $[-6, 6]$ by $[-4, 4]$ viewing window has the same x -axis in both figures, but the scaling on the x -axis has been compressed in Figure 1.79b to make the window square. Figure 1.79c gives an enlarged view with a square $[-3, 3]$ by $[0, 4]$ window.

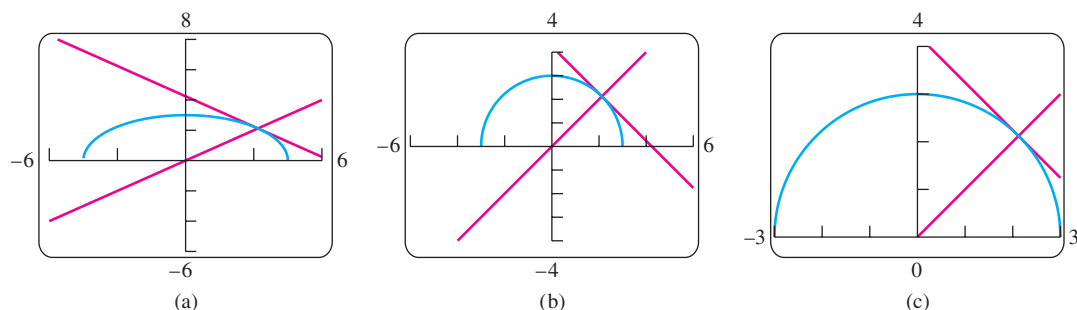


FIGURE 1.79 Graphs of the perpendicular lines $y = x$ and $y = -x + 3\sqrt{2}$, and the semicircle $y = \sqrt{9 - x^2}$, in (a) a nonsquare window, and (b) and (c) square windows (Example 2).

If the denominator of a rational function is zero at some x -value within the viewing window, a calculator or graphing computer software may produce a steep near-vertical line segment from the top to the bottom of the window. Here is an example.

EXAMPLE 3 Graph of a Rational Function

Graph the function $y = \frac{1}{2 - x}$.

Solution Figure 1.80a shows the graph in the $[-10, 10]$ by $[-10, 10]$ default square window with our computer graphing software. Notice the near-vertical line segment at $x = 2$. It is not truly a part of the graph and $x = 2$ does not belong to the domain of the function. By trial and error we can eliminate the line by changing the viewing window to the smaller $[-6, 6]$ by $[-4, 4]$ view, revealing a better graph (Figure 1.80b).

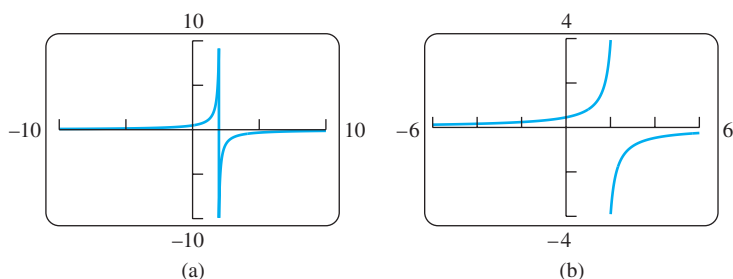


FIGURE 1.80 Graphs of the function $y = \frac{1}{2 - x}$ (Example 3).

Sometimes the graph of a trigonometric function oscillates very rapidly. When a calculator or computer software plots the points of the graph and connects them, many of the maximum and minimum points are actually missed. The resulting graph is then very misleading.

EXAMPLE 4 Graph of a Rapidly Oscillating Function

Graph the function $f(x) = \sin 100x$.

Solution Figure 1.81a shows the graph of f in the viewing window $[-12, 12]$ by $[-1, 1]$. We see that the graph looks very strange because the sine curve should oscillate periodically between -1 and 1 . This behavior is not exhibited in Figure 1.81a. We might experiment with a smaller viewing window, say $[-6, 6]$ by $[-1, 1]$, but the graph is not better (Figure 1.81b). The difficulty is that the period of the trigonometric function $y = \sin 100x$ is very small ($2\pi/100 \approx 0.063$). If we choose the much smaller viewing window $[-0.1, 0.1]$ by $[-1, 1]$ we get the graph shown in Figure 1.81c. This graph reveals the expected oscillations of a sine curve.

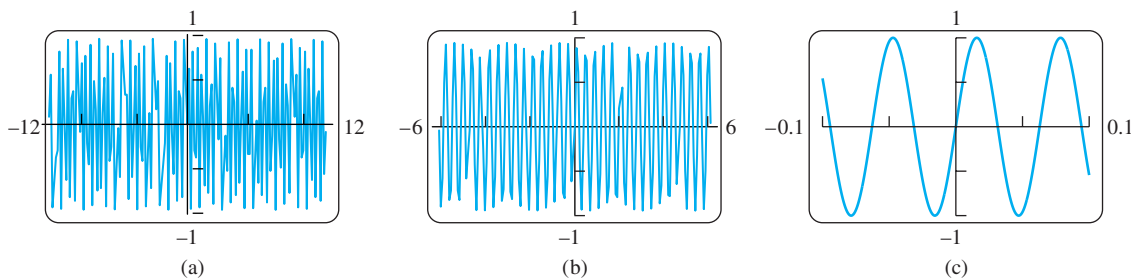


FIGURE 1.81 Graphs of the function $y = \sin 100x$ in three viewing windows. Because the period is $2\pi/100 \approx 0.063$, the smaller window in (c) best displays the true aspects of this rapidly oscillating function (Example 4).

EXAMPLE 5 Another Rapidly Oscillating Function

Graph the function $y = \cos x + \frac{1}{50} \sin 50x$.

Solution In the viewing window $[-6, 6]$ by $[-1, 1]$ the graph appears much like the cosine function with some small sharp wiggles on it (Figure 1.82a). We get a better look when we significantly reduce the window to $[-0.6, 0.6]$ by $[0.8, 1.02]$, obtaining the graph in Figure 1.82b. We now see the small but rapid oscillations of the second term, $\frac{1}{50} \sin 50x$, added to the comparatively larger values of the cosine curve.

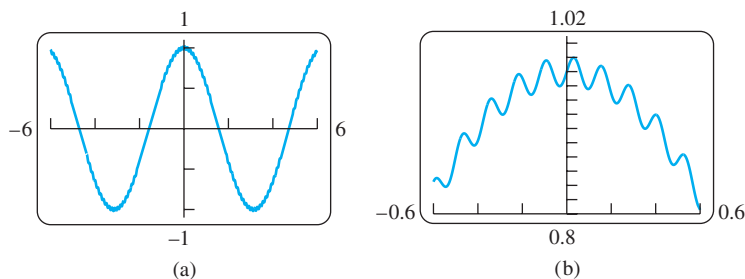


FIGURE 1.82 In (b) we see a close-up view of the function

$y = \cos x + \frac{1}{50} \sin 50x$ graphed in (a). The term $\cos x$ clearly dominates the second term, $\frac{1}{50} \sin 50x$, which produces the rapid oscillations along the cosine curve (Example 5).

EXAMPLE 6 Graphing an Odd Fractional Power

Graph the function $y = x^{1/3}$.

Solution Many graphing devices display the graph shown in Figure 1.83a. When we compare it with the graph of $y = x^{1/3} = \sqrt[3]{x}$ in Figure 1.38, we see that the left branch for $x < 0$ is missing. The reason the graphs differ is that many calculators and computer soft-

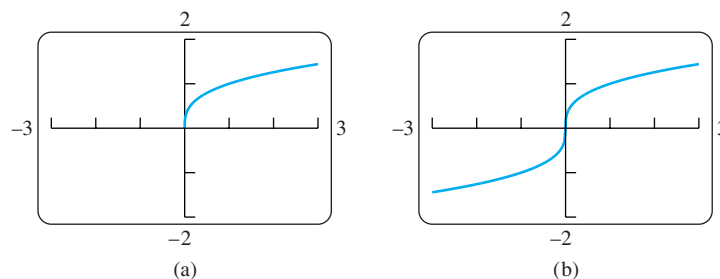


FIGURE 1.83 The graph of $y = x^{1/3}$ is missing the left branch in (a). In (b) we graph the function $f(x) = \frac{x}{|x|} \cdot |x|^{1/3}$ obtaining both branches. (See Example 6.)

ware programs calculate $x^{1/3}$ as $e^{(1/3)\ln x}$. (The exponential and logarithmic functions are studied in Chapter 7.) Since the logarithmic function is not defined for negative values of x , the computing device can only produce the right branch where $x > 0$.

To obtain the full picture showing both branches, we can graph the function

$$f(x) = \frac{x}{|x|} \cdot |x|^{1/3}.$$

This function equals $x^{1/3}$ except at $x = 0$ (where f is undefined, although $0^{1/3} = 0$). The graph of f is shown in Figure 1.83b. ■

Empirical Modeling: Capturing the Trend of Collected Data

In Example 3 of Section 1.4, we verified the reasonableness of Kepler's hypothesis that the period of a planet's orbit is proportional to its mean distance from the sun raised to the $3/2$ power. If we cannot hypothesize a relationship between a dependent variable and an independent variable, we might collect data points and try to find a curve that “fits” the data and captures the trend of the scatterplot. The process of finding a curve to fit data is called **regression analysis** and the curve is called a **regression curve**. A computer or graphing calculator finds the regression curve by finding the particular curve which minimizes the sum of the squares of the vertical distances between the data points and the curve. This method of **least squares** is discussed in the Section 14.7 exercises.

There are many useful types of regression curves, such as straight lines, power, polynomial, exponential, logarithmic, and sinusoidal curves. Many computers or graphing calculators have a regression analysis feature to fit a variety of regression curve types. The next example illustrates using a graphing calculator's linear regression feature to fit data from Table 1.5 with a linear equation.

EXAMPLE 7 Fitting a Regression Line

Starting with the data in Table 1.5, build a model for the price of a postage stamp as a function of time. After verifying that the model is “reasonable,” use it to predict the price in 2010.

Solution We are building a model for the price of a stamp since 1968. There were two increases in 1981, one of three cents followed by another of two cents. To make 1981 comparable with the other listed years, we lump them together as a single five-cent increase, giving the data in Table 1.6. Figure 1.84a gives the scatterplot for Table 1.6.

TABLE 1.5 Price of a U.S. postage stamp

Year x	Cost y
1968	0.06
1971	0.08
1974	0.10
1975	0.13
1977	0.15
1981	0.18
1981	0.20
1985	0.22
1987	0.25
1991	0.29
1995	0.32
1998	0.33
2002	0.37

TABLE 1.6 Price of a U.S postage stamp since 1968

x	0	3	6	7	9	13	17	19	23	27	30	34
y	6	8	10	13	15	20	22	25	29	32	33	37

Since the scatterplot is fairly linear, we investigate a linear model. Upon entering the data into a graphing calculator (or computer software) and selecting the linear regression option, we find the regression line to be

$$y = 0.94x + 6.10.$$

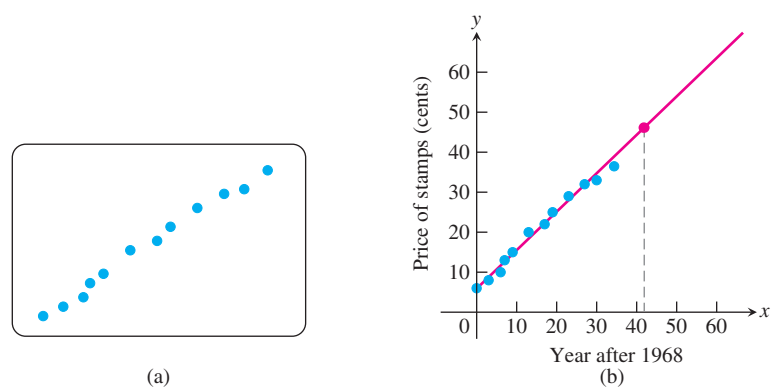


FIGURE 1.84 (a) Scatterplot of (x, y) data in Table 1.6. (b) Using the regression line to estimate the price of a stamp in 2010. (Example 7).

Figure 1.84b shows the line and scatterplot together. The fit is remarkably good, so the model seems reasonable.

Evaluating the regression line, we conclude that in 2010 ($x = 42$), the price of a stamp will be

$$y = 0.94(42) + 6.10 \approx 46 \text{ cents.}$$

The prediction is shown as the red point on the regression line in Figure 1.84b. ■

EXAMPLE 8 Finding a Curve to Predict Population Levels

We may want to predict the future size of a population, such as the number of trout or catfish living in a fish farm. Figure 1.85 shows a scatterplot of the data collected by R. Pearl for a collection of yeast cells (measured as **biomass**) growing over time (measured in hours) in a nutrient.

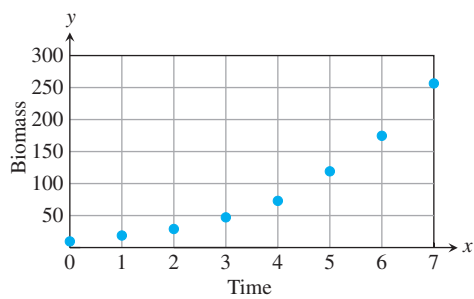


FIGURE 1.85 Biomass of a yeast culture versus elapsed time (Example 8).
(Data from R. Pearl, “The Growth of Population,” *Quart. Rev. Biol.*, Vol. 2 (1927), pp. 532–548.)

The plot of points appears to be reasonably smooth with an upward curving trend. We might attempt to capture this trend by fitting a polynomial (for example, a quadratic $y = ax^2 + bx + c$), a power curve ($y = ax^b$), or an exponential curve ($y = ae^{bx}$). Figure 1.86 shows the result of using a calculator to fit a quadratic model.

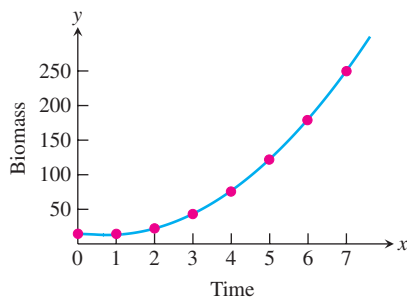


FIGURE 1.86 Fitting a quadratic to Pearl's data gives the equation $y = 6.10x^2 - 9.28x + 16.43$ and the prediction $y(17) = 1622.65$ (Example 8).

The quadratic model $y = 6.10x^2 - 9.28x + 16.43$ appears to fit the collected data reasonably well (Figure 1.86). Using this model, we predict the population after 17 hours as $y(17) = 1622.65$. Let us examine more of Pearl's data to see if our quadratic model continues to be a good one.

In Figure 1.87, we display all of Pearl's data. Now you see that the prediction of $y(17) = 1622.65$ grossly overestimates the observed population of 659.6. Why did the quadratic model fail to predict a more accurate value?

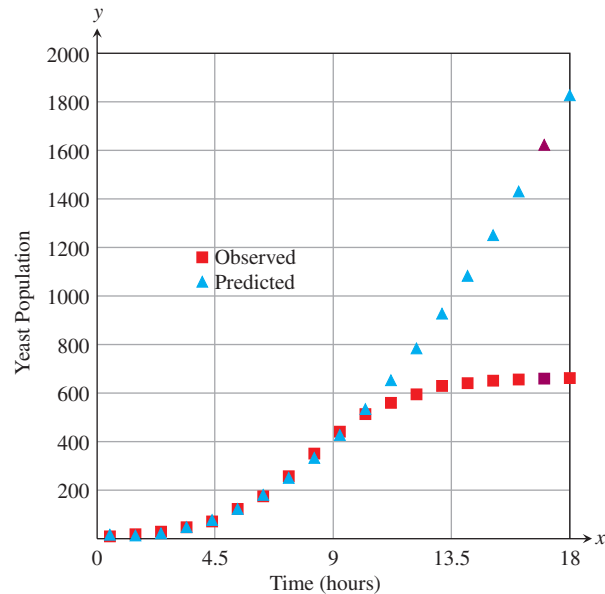


FIGURE 1.87 The rest of Pearl's data (Example 8).

The problem lies in the danger of predicting beyond the range of data used to build the empirical model. (The range of data creating our model was $0 \leq x \leq 7$.) Such *extrapolation* is especially dangerous when the model selected is not supported by some underlying rationale suggesting the form of the model. In our yeast example, why would we expect a quadratic function as underlying population growth? Why not an exponential function? In the face of this, how then do we predict future values? Often, calculus can help, and in Chapter 9 we use it to model population growth. ■

Regression Analysis

Regression analysis has four steps:

1. Plot the data (scatterplot).
2. Find a regression equation. For a line, it has the form $y = mx + b$, and for a quadratic, the form $y = ax^2 + bx + c$.
3. Superimpose the graph of the regression equation on the scatterplot to see the fit.
4. If the fit is satisfactory, use the regression equation to predict y -values for values of x not in the table.

EXERCISES 1.7

Choosing a Viewing Window

In Exercises 1–4, use a graphing calculator or computer to determine which of the given viewing windows displays the most appropriate graph of the specified function.

1. $f(x) = x^4 - 7x^2 + 6x$
 - a. $[-1, 1]$ by $[-1, 1]$
 - b. $[-2, 2]$ by $[-5, 5]$
 - c. $[-10, 10]$ by $[-10, 10]$
 - d. $[-5, 5]$ by $[-25, 15]$
2. $f(x) = x^3 - 4x^2 - 4x + 16$
 - a. $[-1, 1]$ by $[-5, 5]$
 - b. $[-3, 3]$ by $[-10, 10]$
 - c. $[-5, 5]$ by $[-10, 20]$
 - d. $[-20, 20]$ by $[-100, 100]$
3. $f(x) = 5 + 12x - x^3$
 - a. $[-1, 1]$ by $[-1, 1]$
 - b. $[-5, 5]$ by $[-10, 10]$
 - c. $[-4, 4]$ by $[-20, 20]$
 - d. $[-4, 5]$ by $[-15, 25]$
4. $f(x) = \sqrt{5 + 4x - x^2}$
 - a. $[-2, 2]$ by $[-2, 2]$
 - b. $[-2, 6]$ by $[-1, 4]$
 - c. $[-3, 7]$ by $[0, 10]$
 - d. $[-10, 10]$ by $[-10, 10]$

Determining a Viewing Window

In Exercises 5–30, determine an appropriate viewing window for the given function and use it to display its graph.

5. $f(x) = x^4 - 4x^3 + 15$
6. $f(x) = \frac{x^3}{3} - \frac{x^2}{2} - 2x + 1$
7. $f(x) = x^5 - 5x^4 + 10$
8. $f(x) = 4x^3 - x^4$
9. $f(x) = x\sqrt{9 - x^2}$
10. $f(x) = x^2(6 - x^3)$
11. $y = 2x - 3x^{2/3}$
12. $y = x^{1/3}(x^2 - 8)$
13. $y = 5x^{2/5} - 2x$
14. $y = x^{2/3}(5 - x)$
15. $y = |x^2 - 1|$
16. $y = |x^2 - x|$
17. $y = \frac{x+3}{x+2}$
18. $y = 1 - \frac{1}{x+3}$
19. $f(x) = \frac{x^2+2}{x^2+1}$
20. $f(x) = \frac{x^2-1}{x^2+1}$
21. $f(x) = \frac{x-1}{x^2-x-6}$
22. $f(x) = \frac{8}{x^2-9}$
23. $f(x) = \frac{6x^2-15x+6}{4x^2-10x}$
24. $f(x) = \frac{x^2-3}{x-2}$
25. $y = \sin 250x$
26. $y = 3 \cos 60x$
27. $y = \cos\left(\frac{x}{50}\right)$
28. $y = \frac{1}{10} \sin\left(\frac{x}{10}\right)$
29. $y = x + \frac{1}{10} \sin 30x$
30. $y = x^2 + \frac{1}{50} \cos 100x$

31. Graph the lower half of the circle defined by the equation $x^2 + 2x = 4 + 4y - y^2$.
32. Graph the upper branch of the hyperbola $y^2 - 16x^2 = 1$.
33. Graph four periods of the function $f(x) = -\tan 2x$.
34. Graph two periods of the function $f(x) = 3 \cot \frac{x}{2} + 1$.
35. Graph the function $f(x) = \sin 2x + \cos 3x$.
36. Graph the function $f(x) = \sin^3 x$.

Graphing in Dot Mode

Another way to avoid incorrect connections when using a graphing device is through the use of a “dot mode,” which plots only the points. If your graphing utility allows that mode, use it to plot the functions in Exercises 37–40.

37. $y = \frac{1}{x-3}$
38. $y = \sin \frac{1}{x}$
39. $y = x[x]$
40. $y = \frac{x^3 - 1}{x^2 - 1}$

Regression Analysis

- T** 41. Table 1.7 shows the mean annual compensation of construction workers.

TABLE 1.7 Construction workers' average annual compensation

Year	Annual compensation (dollars)
1980	22,033
1985	27,581
1988	30,466
1990	32,836
1992	34,815
1995	37,996
1999	42,236
2002	45,413

Source: U.S. Bureau of Economic Analysis.

- a. Find a linear regression equation for the data.
- b. Find the slope of the regression line. What does the slope represent?

- c. Superimpose the graph of the linear regression equation on a scatterplot of the data.
- d. Use the regression equation to predict the construction workers' average annual compensation in 2010.

- T 42.** The median price of existing single-family homes has increased consistently since 1970. The data in Table 1.8, however, show that there have been differences in various parts of the country.
- a. Find a linear regression equation for home cost in the Northeast.
 - b. What does the slope of the regression line represent?
 - c. Find a linear regression equation for home cost in the Midwest.
 - d. Where is the median price increasing more rapidly, in the Northeast or the Midwest?

TABLE 1.8 Median price of single-family homes

Year	Northeast (dollars)	Midwest (dollars)
1970	25,200	20,100
1975	39,300	30,100
1980	60,800	51,900
1985	88,900	58,900
1990	141,200	74,000
1995	197,100	88,300
2000	264,700	97,000

Source: National Association of Realtors®

- T 43. Vehicular stopping distance** Table 1.9 shows the total stopping distance of a car as a function of its speed.
- a. Find the quadratic regression equation for the data in Table 1.9.
 - b. Superimpose the graph of the quadratic regression equation on a scatterplot of the data.
 - c. Use the graph of the quadratic regression equation to predict the average total stopping distance for speeds of 72 and 85 mph. Confirm algebraically.
 - d. Now use *linear* regression to predict the average total stopping distance for speeds of 72 and 85 mph. Superimpose the regression line on a scatterplot of the data. Which gives the better fit, the line here or the graph in part (b)?

TABLE 1.9 Vehicular stopping distance

Speed (mph)	Average total stopping distance (ft)
20	42
25	56
30	73.5
35	91.5
40	116
45	142.5
50	173
55	209.5
60	248
65	292.5
70	343
75	401
80	464

Source: U.S. Bureau of Public Roads.

- T 44. Stern waves** Observations of the stern waves that follow a boat at right angles to its course have disclosed that the distance between the crests of these waves (their *wave length*) increases with the speed of the boat. Table 1.10 shows the relationship between wave length and the speed of the boat.

TABLE 1.10 Wave lengths

Wave length (m)	Speed (km/h)
0.20	1.8
0.65	3.6
1.13	5.4
2.55	7.2
4.00	9.0
5.75	10.8
7.80	12.6
10.20	14.4
12.90	16.2
16.00	18.0
18.40	19.8

- a. Find a power regression equation $y = ax^b$ for the data in Table 1.10, where x is the wave length, and y the speed of the boat.
- b. Superimpose the graph of the power regression equation on a scatterplot of the data.
- c. Use the graph of the power regression equation to predict the speed of the boat when the wave length is 11 m. Confirm algebraically.
- d. Now use *linear* regression to predict the speed when the wave length is 11 m. Superimpose the regression line on a scatterplot of the data. Which gives the better fit, the line here or the curve in part (b)?

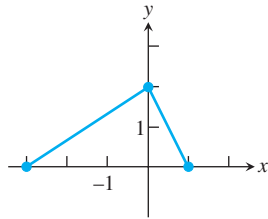
Chapter 1

Additional and Advanced Exercises

Functions and Graphs

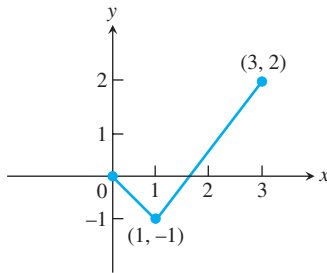
1. The graph of f is shown. Draw the graph of each function.

a. $y = f(-x)$ b. $y = -f(x)$
 c. $y = -2f(x + 1) + 1$ d. $y = 3f(x - 2) - 2$



2. A portion of the graph of a function defined on $[-3, 3]$ is shown. Complete the graph assuming that the function is

a. even. b. odd.



3. Are there two functions f and g such that $f \circ g = g \circ f$? Give reasons for your answer.
4. Are there two functions f and g with the following property? The graphs of f and g are not straight lines but the graph of $f \circ g$ is a straight line. Give reasons for your answer.
5. If $f(x)$ is odd, can anything be said of $g(x) = f(x) - 2$? What if f is even instead? Give reasons for your answer.
6. If $g(x)$ is an odd function defined for all values of x , can anything be said about $g(0)$? Give reasons for your answer.
7. Graph the equation $|x| + |y| = 1 + x$.
8. Graph the equation $y + |y| = x + |x|$.

Trigonometry

In Exercises 9–14, ABC is an arbitrary triangle with sides a , b , and c opposite angles A , B , and C , respectively.

9. Find b if $a = \sqrt{3}$, $A = \pi/3$, $B = \pi/4$.
10. Find $\sin B$ if $a = 4$, $b = 3$, $A = \pi/4$.
11. Find $\cos A$ if $a = 2$, $b = 2$, $c = 3$.
12. Find c if $a = 2$, $b = 3$, $C = \pi/4$.

13. Find $\sin B$ if $a = 2$, $b = 3$, $c = 4$.

14. Find $\sin C$ if $a = 2$, $b = 4$, $c = 5$.

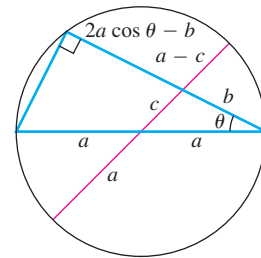
Derivations and Proofs

15. Prove the following identities.

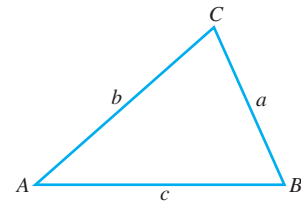
a. $\frac{1 - \cos x}{\sin x} = \frac{\sin x}{1 + \cos x}$

b. $\frac{1 - \cos x}{1 + \cos x} = \tan^2 \frac{x}{2}$

16. Explain the following “proof without words” of the law of cosines. (Source: “Proof without Words: The Law of Cosines,” Sidney H. Kung, *Mathematics Magazine*, Vol. 63, No. 5, Dec. 1990, p. 342.)



17. Show that the area of triangle ABC is given by $(1/2)ab \sin C = (1/2)bc \sin A = (1/2)ca \sin B$.



18. Show that the area of triangle ABC is given by $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a + b + c)/2$ is the semiperimeter of the triangle.
19. **Properties of inequalities** If a and b are real numbers, we say that a is less than b and write $a < b$ if (and only if) $b - a$ is positive. Use this definition to prove the following properties of inequalities.

If a , b , and c are real numbers, then:

1. $a < b \Rightarrow a + c < b + c$
 2. $a < b \Rightarrow a - c < b - c$
 3. $a < b$ and $c > 0 \Rightarrow ac < bc$
 4. $a < b$ and $c < 0 \Rightarrow bc < ac$
 (Special case: $a < b \Rightarrow -b < -a$)

5. $a > 0 \Rightarrow \frac{1}{a} > 0$
6. $0 < a < b \Rightarrow \frac{1}{b} < \frac{1}{a}$
7. $a < b < 0 \Rightarrow \frac{1}{b} < \frac{1}{a}$
20. Prove that the following inequalities hold for any real numbers a and b .
- $|a| < |b|$ if and only if $a^2 < b^2$
 - $|a - b| \geq ||a| - |b||$
- Generalizing the triangle inequality** Prove by mathematical induction that the inequalities in Exercises 21 and 22 hold for any n real numbers $a_1, a_2, \dots, a_n$. (Mathematical induction is reviewed in Appendix 1.)
21. $|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n|$
22. $|a_1 + a_2 + \dots + a_n| \geq |a_1| - |a_2| - \dots - |a_n|$
23. Show that if f is both even and odd, then $f(x) = 0$ for every x in the domain of f .
24. **a. Even-odd decompositions** Let f be a function whose domain is symmetric about the origin, that is, $-x$ belongs to the domain whenever x does. Show that f is the sum of an even function and an odd function:

$$f(x) = E(x) + O(x),$$

where E is an even function and O is an odd function. (Hint: Let $E(x) = (f(x) + f(-x))/2$. Show that $E(-x) = E(x)$, so that E is even. Then show that $O(x) = f(x) - E(x)$ is odd.)

- b. Uniqueness** Show that there is only one way to write f as the sum of an even and an odd function. (Hint: One way is given in part (a). If also $f(x) = E_1(x) + O_1(x)$ where E_1 is even and O_1 is odd, show that $E - E_1 = O_1 - O$. Then use Exercise 23 to show that $E = E_1$ and $O = O_1$.)

Grapher Explorations—Effects of Parameters

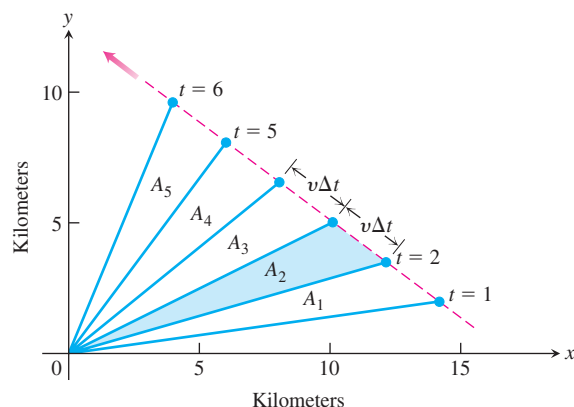
25. What happens to the graph of $y = ax^2 + bx + c$ as
- a changes while b and c remain fixed?
 - b changes (a and c fixed, $a \neq 0$)?
 - c changes (a and b fixed, $a \neq 0$)?
26. What happens to the graph of $y = a(x + b)^3 + c$ as
- a changes while b and c remain fixed?

- b changes (a and c fixed, $a \neq 0$)?
- c changes (a and b fixed, $a \neq 0$)?

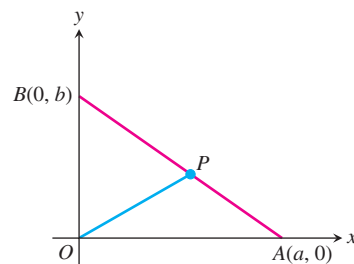
27. Find all values of the slope of the line $y = mx + 2$ for which the x -intercept exceeds $1/2$.

Geometry

28. An object's center of mass moves at a constant velocity v along a straight line past the origin. The accompanying figure shows the coordinate system and the line of motion. The dots show positions that are 1 sec apart. Why are the areas $A_1, A_2, \dots, A_5$ in the figure all equal? As in Kepler's equal area law (see Section 13.6), the line that joins the object's center of mass to the origin sweeps out equal areas in equal times.



29. **a.** Find the slope of the line from the origin to the midpoint P , of side AB in the triangle in the accompanying figure ($a, b > 0$).



- b.** When is OP perpendicular to AB ?

Chapter 1

Practice Exercises

Inequalities

In Exercises 1–4, solve the inequalities and show the solution sets on the real line.

1. $7 + 2x \geq 3$
2. $-3x < 10$
3. $\frac{1}{5}(x - 1) < \frac{1}{4}(x - 2)$
4. $\frac{x - 3}{2} \geq -\frac{4 + x}{3}$

Absolute Value

Solve the equations or inequalities in Exercises 5–8.

5. $|x + 1| = 7$
6. $|y - 3| < 4$
7. $\left|1 - \frac{x}{2}\right| > \frac{3}{2}$
8. $\left|\frac{2x + 7}{3}\right| \leq 5$

Coordinates

9. A particle in the plane moved from $A(-2, 5)$ to the y -axis in such a way that Δy equaled $3\Delta x$. What were the particle's new coordinates?
10. a. Plot the points $A(8, 1)$, $B(2, 10)$, $C(-4, 6)$, $D(2, -3)$, and $E(14/3, 6)$.
b. Find the slopes of the lines AB , BC , CD , DA , CE , and BD .
c. Do any four of the five points A , B , C , D , and E form a parallelogram?
d. Are any three of the five points collinear? How do you know?
e. Which of the lines determined by the five points pass through the origin?
11. Do the points $A(6, 4)$, $B(4, -3)$, and $C(-2, 3)$ form an isosceles triangle? A right triangle? How do you know?
12. Find the coordinates of the point on the line $y = 3x + 1$ that is equidistant from $(0, 0)$ and $(-3, 4)$.

Lines

In Exercises 13–24, write an equation for the specified line.

13. through $(1, -6)$ with slope 3
14. through $(-1, 2)$ with slope $-1/2$
15. the vertical line through $(0, -3)$

16. through $(-3, 6)$ and $(1, -2)$
17. the horizontal line through $(0, 2)$
18. through $(3, 3)$ and $(-2, 5)$
19. with slope -3 and y -intercept 3
20. through $(3, 1)$ and parallel to $2x - y = -2$
21. through $(4, -12)$ and parallel to $4x + 3y = 12$
22. through $(-2, -3)$ and perpendicular to $3x - 5y = 1$
23. through $(-1, 2)$ and perpendicular to $(1/2)x + (1/3)y = 1$
24. with x -intercept 3 and y -intercept -5

Functions and Graphs

25. Express the area and circumference of a circle as functions of the circle's radius. Then express the area as a function of the circumference.
26. Express the radius of a sphere as a function of the sphere's surface area. Then express the surface area as a function of the volume.
27. A point P in the first quadrant lies on the parabola $y = x^2$. Express the coordinates of P as functions of the angle of inclination of the line joining P to the origin.
28. A hot-air balloon rising straight up from a level field is tracked by a range finder located 500 ft from the point of liftoff. Express the balloon's height as a function of the angle the line from the range finder to the balloon makes with the ground.

In Exercises 29–32, determine whether the graph of the function is symmetric about the y -axis, the origin, or neither.

29. $y = x^{1/5}$
30. $y = x^{2/5}$
31. $y = x^2 - 2x - 1$
32. $y = e^{-x^2}$

In Exercises 33–40, determine whether the function is even, odd, or neither.

33. $y = x^2 + 1$
34. $y = x^5 - x^3 - x$
35. $y = 1 - \cos x$
36. $y = \sec x \tan x$
37. $y = \frac{x^4 + 1}{x^3 - 2x}$
38. $y = 1 - \sin x$
39. $y = x + \cos x$
40. $y = \sqrt{x^4 - 1}$

In Exercises 41–50, find the (a) domain and (b) range.

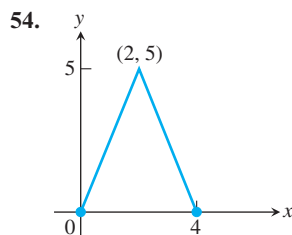
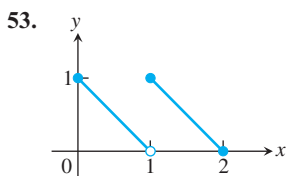
41. $y = |x| - 2$ 42. $y = -2 + \sqrt{1-x}$
 43. $y = \sqrt{16-x^2}$ 44. $y = 3^{2-x} + 1$
 45. $y = 2e^{-x} - 3$ 46. $y = \tan(2x - \pi)$
 47. $y = 2 \sin(3x + \pi) - 1$ 48. $y = x^{2/5}$
 49. $y = \ln(x-3) + 1$ 50. $y = -1 + \sqrt[3]{2-x}$

Piecewise-Defined Functions

In Exercises 51 and 52, find the (a) domain and (b) range.

51. $y = \begin{cases} \sqrt{-x}, & -4 \leq x \leq 0 \\ \sqrt{x}, & 0 < x \leq 4 \end{cases}$
 52. $y = \begin{cases} -x - 2, & -2 \leq x \leq -1 \\ x, & -1 < x \leq 1 \\ -x + 2, & 1 < x \leq 2 \end{cases}$

In Exercises 53 and 54, write a piecewise formula for the function.



Composition of Functions

In Exercises 55 and 56, find

- a. $(f \circ g)(-1)$. b. $(g \circ f)(2)$.
 c. $(f \circ f)(x)$. d. $(g \circ g)(x)$.

55. $f(x) = \frac{1}{x}$, $g(x) = \frac{1}{\sqrt{x+2}}$

56. $f(x) = 2 - x$, $g(x) = \sqrt[3]{x+1}$

In Exercises 57 and 58, (a) write a formula for $f \circ g$ and $g \circ f$ and find the (b) domain and (c) range of each.

57. $f(x) = 2 - x^2$, $g(x) = \sqrt{x+2}$

58. $f(x) = \sqrt{x}$, $g(x) = \sqrt{1-x}$

Composition with absolute values In Exercises 59–64, graph f_1 and f_2 together. Then describe how applying the absolute value function before applying f_1 affects the graph.

$f_1(x)$	$f_2(x) = f_1(x)$
59. x	$ x $
60. x^3	$ x ^3$
61. x^2	$ x ^2$
62. $\frac{1}{x}$	$\frac{1}{ x }$
63. $\sqrt{x}$	$\sqrt{ x }$
64. $\sin x$	$\sin x $

Composition with absolute values In Exercises 65–68, graph g_1 and g_2 together. Then describe how taking absolute values after applying g_1 affects the graph.

$g_1(x)$	$g_2(x) = g_1(x) $
65. x^3	$ x^3 $
66. $\sqrt{x}$	$ \sqrt{x} $
67. $4 - x^2$	$ 4 - x^2 $
68. $x^2 + x$	$ x^2 + x $

Trigonometry

In Exercises 69–72, sketch the graph of the given function. What is the period of the function?

69. $y = \cos 2x$

70. $y = \sin \frac{x}{2}$

71. $y = \sin \pi x$

72. $y = \cos \frac{\pi x}{2}$

73. Sketch the graph $y = 2 \cos\left(x - \frac{\pi}{3}\right)$.

74. Sketch the graph $y = 1 + \sin\left(x + \frac{\pi}{4}\right)$.

In Exercises 75–78, ABC is a right triangle with the right angle at C . The sides opposite angles A , B , and C are a , b , and c , respectively.

75. a. Find a and b if $c = 2$, $B = \pi/3$.

b. Find a and c if $b = 2$, $B = \pi/3$.

76. a. Express a in terms of A and c .

b. Express a in terms of A and b .

77. a. Express a in terms of B and b .

b. Express c in terms of A and a .

78. a. Express $\sin A$ in terms of a and c .

b. Express $\sin A$ in terms of b and c .

79. Height of a pole Two wires stretch from the top T of a vertical pole to points B and C on the ground, where C is 10 m closer to the base of the pole than is B . If wire BT makes an angle of 35° with the horizontal and wire CT makes an angle of 50° with the horizontal, how high is the pole?

80. Height of a weather balloon Observers at positions A and B 2 km apart simultaneously measure the angle of elevation of a weather balloon to be 40° and 70° , respectively. If the balloon is directly above a point on the line segment between A and B , find the height of the balloon.

T 81. a. Graph the function $f(x) = \sin x + \cos(x/2)$.

b. What appears to be the period of this function?

c. Confirm your finding in part (b) algebraically.

T 82. a. Graph $f(x) = \sin(1/x)$.

b. What are the domain and range of f ?

c. Is f periodic? Give reasons for your answer.

Chapter 1

Questions to Guide Your Review

1. How are the real numbers represented? What are the main categories characterizing the properties of the real number system? What are the primary subsets of the real numbers?
2. How are the rational numbers described in terms of decimal expansions? What are the irrational numbers? Give examples.
3. What are the order properties of the real numbers? How are they used in solving equations?
4. What is a number's absolute value? Give examples? How are $|-a|$, $|ab|$, $|a/b|$, and $|a + b|$ related to $|a|$ and $|b|$?
5. How are absolute values used to describe intervals or unions of intervals? Give examples.
6. How do we identify points in the plane using the Cartesian coordinate system? What is the graph of an equation in the variables x and y ?
7. How can you write an equation for a line if you know the coordinates of two points on the line? The line's slope and the coordinates of one point on the line? The line's slope and y -intercept? Give examples.
8. What are the standard equations for lines perpendicular to the coordinate axes?
9. How are the slopes of mutually perpendicular lines related? What about parallel lines? Give examples.
10. When a line is not vertical, what is the relation between its slope and its angle of inclination?
11. How do you find the distance between two points in the coordinate plane?
12. What is the standard equation of a circle with center (h, k) and radius a ? What is the unit circle and what is its equation?
13. Describe the steps you would take to graph the circle $x^2 + y^2 + 4x - 6y + 12 = 0$.
14. What inequality describes the points in the coordinate plane that lie inside the circle of radius a centered at the point (h, k) ? That lie inside or on the circle? That lie outside the circle? That lie outside or on the circle?
15. If a , b , and c are constants and $a \neq 0$, what can you say about the graph of the equation $y = ax^2 + bx + c$? In particular, how would you go about sketching the curve $y = 2x^2 + 4x$?
16. What is a function? What is its domain? Its range? What is an arrow diagram for a function? Give examples.
17. What is the graph of a real-valued function of a real variable? What is the vertical line test?
18. What is a piecewise-defined function? Give examples.
19. What are the important types of functions frequently encountered in calculus? Give an example of each type.
20. In terms of its graph, what is meant by an increasing function? A decreasing function? Give an example of each.
21. What is an even function? An odd function? What symmetry properties do the graphs of such functions have? What advantage can we take of this? Given an example of a function that is neither even nor odd.
22. What does it mean to say that y is proportional to x ? To $x^{3/2}$? What is the geometric interpretation of proportionality? How can this interpretation be used to test a proposed proportionality?
23. If f and g are real-valued functions, how are the domains of $f + g$, $f - g$, fg , and f/g related to the domains of f and g ? Give examples.
24. When is it possible to compose one function with another? Give examples of composites and their values at various points. Does the order in which functions are composed ever matter?
25. How do you change the equation $y = f(x)$ to shift its graph vertically up or down by a factor $k > 0$? Horizontally to the left or right? Give examples.
26. How do you change the equation $y = f(x)$ to compress or stretch the graph by $c > 1$? Reflect the graph across a coordinate axis? Give examples.
27. What is the standard equation of an ellipse with center (h, k) ? What is its major axis? Its minor axis? Give examples.
28. What is radian measure? How do you convert from radians to degrees? Degrees to radians?
29. Graph the six basic trigonometric functions. What symmetries do the graphs have?
30. What is a periodic function? Give examples. What are the periods of the six basic trigonometric functions?

31. Starting with the identity $\sin^2 \theta + \cos^2 \theta = 1$ and the formulas for $\cos(A + B)$ and $\sin(A + B)$, show how a variety of other trigonometric identities may be derived.
32. How does the formula for the general sine function $f(x) = A \sin((2\pi/B)(x - C)) + D$ relate to the shifting, stretching, compressing, and reflection of its graph? Give examples. Graph the general sine curve and identify the constants A , B , C , and D .
33. Name three issues that arise when functions are graphed using a calculator or computer with graphing software. Give examples.

Chapter 1

Technology Application Projects

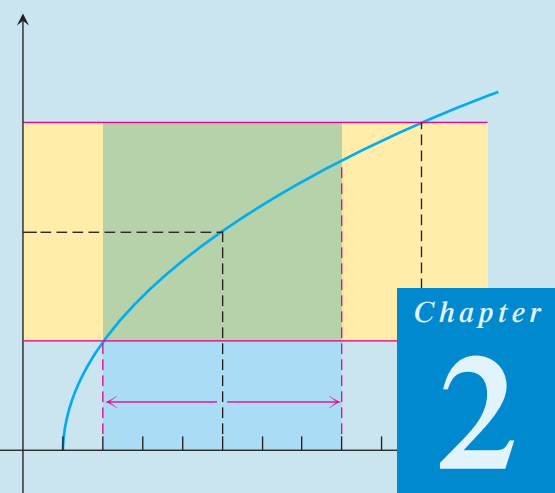
An Overview of Mathematica

An overview of *Mathematica* sufficient to complete the *Mathematica* modules appearing on the Web site.

Mathematica/Maple Module

Modeling Change: Springs, Driving Safety, Radioactivity, Trees, Fish, and Mammals.

Construct and interpret mathematical models, analyze and improve them, and make predictions using them.



LIMITS AND CONTINUITY

OVERVIEW The concept of a limit is a central idea that distinguishes calculus from algebra and trigonometry. It is fundamental to finding the tangent to a curve or the velocity of an object.

In this chapter we develop the limit, first intuitively and then formally. We use limits to describe the way a function f varies. Some functions vary continuously; small changes in x produce only small changes in $f(x)$. Other functions can have values that jump or vary erratically. The notion of limit gives a precise way to distinguish between these behaviors. The geometric application of using limits to define the tangent to a curve leads at once to the important concept of the derivative of a function. The derivative, which we investigate thoroughly in Chapter 3, quantifies the way a function's values change.

2.1

Rates of Change and Limits

In this section, we introduce average and instantaneous rates of change. These lead to the main idea of the section, the idea of limit.

Average and Instantaneous Speed

A moving body's **average speed** during an interval of time is found by dividing the distance covered by the time elapsed. The unit of measure is length per unit time: kilometers per hour, feet per second, or whatever is appropriate to the problem at hand.

EXAMPLE 1 Finding an Average Speed

A rock breaks loose from the top of a tall cliff. What is its average speed

- (a) during the first 2 sec of fall?
- (b) during the 1-sec interval between second 1 and second 2?

Solution In solving this problem we use the fact, discovered by Galileo in the late sixteenth century, that a solid object dropped from rest (not moving) to fall freely near the surface of the earth will fall a distance proportional to the square of the time it has been falling. (This assumes negligible air resistance to slow the object down and that gravity is

HISTORICAL BIOGRAPHY*
Galileo Galilei
(1564–1642)

the only force acting on the falling body. We call this type of motion **free fall**.) If y denotes the distance fallen in feet after t seconds, then Galileo’s law is

$$y = 16t^2,$$

where 16 is the constant of proportionality.

The average speed of the rock during a given time interval is the change in distance, Δy , divided by the length of the time interval, Δt .

- (a) For the first 2 sec:

$$\frac{\Delta y}{\Delta t} = \frac{16(2)^2 - 16(0)^2}{2 - 0} = 32 \frac{\text{ft}}{\text{sec}}$$
- (b) From sec 1 to sec 2:

$$\frac{\Delta y}{\Delta t} = \frac{16(2)^2 - 16(1)^2}{2 - 1} = 48 \frac{\text{ft}}{\text{sec}}$$

The next example examines what happens when we look at the average speed of a falling object over shorter and shorter time intervals.

EXAMPLE 2 Finding an Instantaneous Speed

Find the speed of the falling rock at $t = 1$ and $t = 2$ sec.

Solution We can calculate the average speed of the rock over a time interval $[t_0, t_0 + h]$, having length $\Delta t = h$, as

$$\frac{\Delta y}{\Delta t} = \frac{16(t_0 + h)^2 - 16t_0^2}{h}. \tag{1}$$

We cannot use this formula to calculate the “instantaneous” speed at t_0 by substituting $h = 0$, because we cannot divide by zero. But we *can* use it to calculate average speeds over increasingly short time intervals starting at $t_0 = 1$ and $t_0 = 2$. When we do so, we see a pattern (Table 2.1).

TABLE 2.1 Average speeds over short time intervals		
Average speed: $\frac{\Delta y}{\Delta t} = \frac{16(t_0 + h)^2 - 16t_0^2}{h}$		
Length of time interval h	Average speed over interval of length h starting at $t_0 = 1$	Average speed over interval of length h starting at $t_0 = 2$
1	48	80
0.1	33.6	65.6
0.01	32.16	64.16
0.001	32.016	64.016
0.0001	32.0016	64.0016

The average speed on intervals starting at $t_0 = 1$ seems to approach a limiting value of 32 as the length of the interval decreases. This suggests that the rock is falling at a speed of 32 ft/sec at $t_0 = 1$ sec. Let’s confirm this algebraically.

To learn more about the historical figures and the development of the major elements and topics of calculus, visit www.aw-bc.com/thomas.

If we set $t_0 = 1$ and then expand the numerator in Equation (1) and simplify, we find that

$$\begin{aligned}\frac{\Delta y}{\Delta t} &= \frac{16(1+h)^2 - 16(1)^2}{h} = \frac{16(1+2h+h^2) - 16}{h} \\ &= \frac{32h + 16h^2}{h} = 32 + 16h.\end{aligned}$$

For values of h different from 0, the expressions on the right and left are equivalent and the average speed is $32 + 16h$ ft/sec. We can now see why the average speed has the limiting value $32 + 16(0) = 32$ ft/sec as h approaches 0.

Similarly, setting $t_0 = 2$ in Equation (1), the procedure yields

$$\frac{\Delta y}{\Delta t} = 64 + 16h$$

for values of h different from 0. As h gets closer and closer to 0, the average speed at $t_0 = 2$ sec has the limiting value 64 ft/sec. ■

Average Rates of Change and Secant Lines

Given an arbitrary function $y = f(x)$, we calculate the average rate of change of y with respect to x over the interval $[x_1, x_2]$ by dividing the change in the value of y , $\Delta y = f(x_2) - f(x_1)$, by the length $\Delta x = x_2 - x_1 = h$ of the interval over which the change occurs.

DEFINITION Average Rate of Change over an Interval

The **average rate of change** of $y = f(x)$ with respect to x over the interval $[x_1, x_2]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1 + h) - f(x_1)}{h}, \quad h \neq 0.$$

Geometrically, the rate of change of f over $[x_1, x_2]$ is the slope of the line through the points $P(x_1, f(x_1))$ and $Q(x_2, f(x_2))$ (Figure 2.1). In geometry, a line joining two points of a curve is a **secant** to the curve. Thus, the average rate of change of f from x_1 to x_2 is identical with the slope of secant PQ .

Experimental biologists often want to know the rates at which populations grow under controlled laboratory conditions.

EXAMPLE 3 The Average Growth Rate of a Laboratory Population

Figure 2.2 shows how a population of fruit flies (*Drosophila*) grew in a 50-day experiment. The number of flies was counted at regular intervals, the counted values plotted with respect to time, and the points joined by a smooth curve (colored blue in Figure 2.2). Find the average growth rate from day 23 to day 45.

Solution There were 150 flies on day 23 and 340 flies on day 45. Thus the number of flies increased by $340 - 150 = 190$ in $45 - 23 = 22$ days. The average rate of change of the population from day 23 to day 45 was

$$\text{Average rate of change: } \frac{\Delta p}{\Delta t} = \frac{340 - 150}{45 - 23} = \frac{190}{22} \approx 8.6 \text{ flies/day.}$$

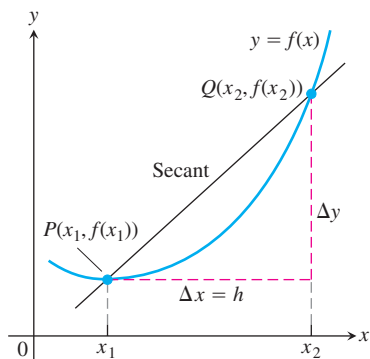


FIGURE 2.1 A secant to the graph $y = f(x)$. Its slope is $\Delta y/\Delta x$, the average rate of change of f over the interval $[x_1, x_2]$.

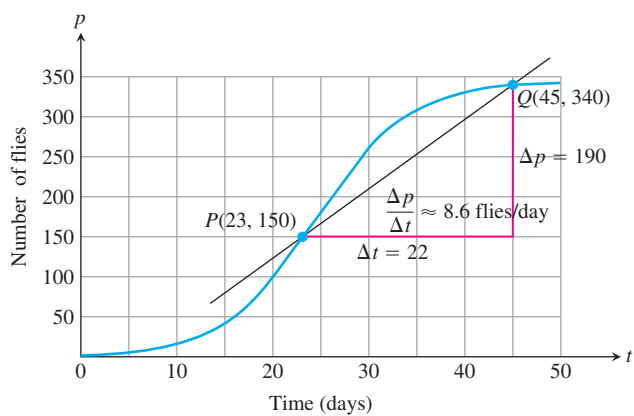


FIGURE 2.2 Growth of a fruit fly population in a controlled experiment. The average rate of change over 22 days is the slope $\Delta p/\Delta t$ of the secant line.

This average is the slope of the secant through the points P and Q on the graph in Figure 2.2.

The average rate of change from day 23 to day 45 calculated in Example 3 does not tell us how fast the population was changing on day 23 itself. For that we need to examine time intervals closer to the day in question.

EXAMPLE 4 The Growth Rate on Day 23

How fast was the number of flies in the population of Example 3 growing on day 23?

Solution To answer this question, we examine the average rates of change over increasingly short time intervals starting at day 23. In geometric terms, we find these rates by calculating the slopes of secants from P to Q , for a sequence of points Q approaching P along the curve (Figure 2.3).

Q	Slope of $PQ = \Delta p/\Delta t$ (flies/day)
(45, 340)	$\frac{340 - 150}{45 - 23} \approx 8.6$
(40, 330)	$\frac{330 - 150}{40 - 23} \approx 10.6$
(35, 310)	$\frac{310 - 150}{35 - 23} \approx 13.3$
(30, 265)	$\frac{265 - 150}{30 - 23} \approx 16.4$

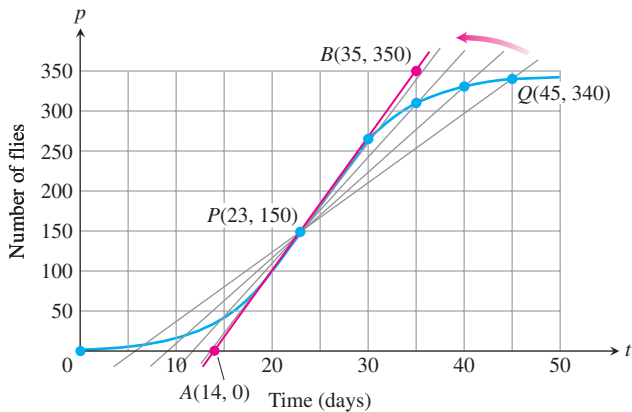


FIGURE 2.3 The positions and slopes of four secants through the point P on the fruit fly graph (Example 4).

The values in the table show that the secant slopes rise from 8.6 to 16.4 as the t -coordinate of Q decreases from 45 to 30, and we would expect the slopes to rise slightly higher as t continued on toward 23. Geometrically, the secants rotate about P and seem to approach the red line in the figure, a line that goes through P in the same direction that the curve goes through P . We will see that this line is called the *tangent* to the curve at P . Since the line appears to pass through the points $(14, 0)$ and $(35, 350)$, it has slope

$$\frac{350 - 0}{35 - 14} = 16.7 \text{ flies/day (approximately).}$$

On day 23 the population was increasing at a rate of about 16.7 flies/day. ■

The rates at which the rock in Example 2 was falling at the instants $t = 1$ and $t = 2$ and the rate at which the population in Example 4 was changing on day $t = 23$ are called *instantaneous rates of change*. As the examples suggest, we find instantaneous rates as limiting values of average rates. In Example 4, we also pictured the tangent line to the population curve on day 23 as a limiting position of secant lines. Instantaneous rates and tangent lines, intimately connected, appear in many other contexts. To talk about the two constructively, and to understand the connection further, we need to investigate the process by which we determine limiting values, or *limits*, as we will soon call them.

Limits of Function Values

Our examples have suggested the limit idea. Let's begin with an informal definition of limit, postponing the precise definition until we've gained more insight.

Let $f(x)$ be defined on an open interval about x_0 , *except possibly at x_0 itself*. If $f(x)$ gets arbitrarily close to L (as close to L as we like) for all x sufficiently close to x_0 , we say that f approaches the **limit** L as x approaches x_0 , and we write

$$\lim_{x \rightarrow x_0} f(x) = L,$$

which is read “the limit of $f(x)$ as x approaches x_0 is L ”. Essentially, the definition says that the values of $f(x)$ are close to the number L whenever x is close to x_0 (on either side of x_0). This definition is “informal” because phrases like *arbitrarily close* and *sufficiently close* are imprecise; their meaning depends on the context. To a machinist manufacturing a piston, *close* may mean *within a few thousandths of an inch*. To an astronomer studying distant galaxies, *close* may mean *within a few thousand light-years*. The definition is clear enough, however, to enable us to recognize and evaluate limits of specific functions. We will need the precise definition of Section 2.3, however, when we set out to prove theorems about limits.

EXAMPLE 5 Behavior of a Function Near a Point

How does the function

$$f(x) = \frac{x^2 - 1}{x - 1}$$

behave near $x = 1$?

Solution The given formula defines f for all real numbers x except $x = 1$ (we cannot divide by zero). For any $x \neq 1$, we can simplify the formula by factoring the numerator and canceling common factors:

$$f(x) = \frac{(x - 1)(x + 1)}{x - 1} = x + 1 \quad \text{for} \quad x \neq 1.$$

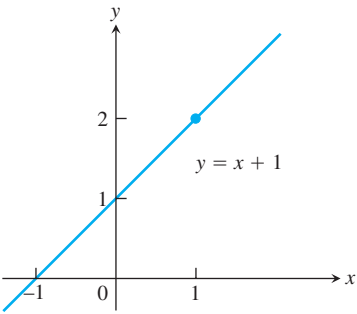
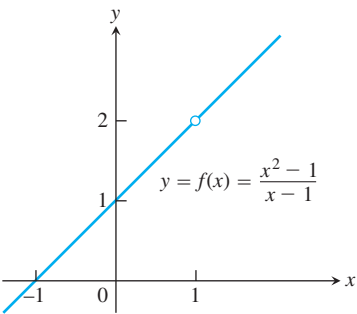


FIGURE 2.4 The graph of f is identical with the line $y = x + 1$ except at $x = 1$, where f is not defined (Example 5).

The graph of f is thus the line $y = x + 1$ with the point $(1, 2)$ removed. This removed point is shown as a “hole” in Figure 2.4. Even though $f(1)$ is not defined, it is clear that we can make the value of $f(x)$ as close as we want to 2 by choosing x close enough to 1 (Table 2.2).

TABLE 2.2 The closer x gets to 1, the closer $f(x) = (x^2 - 1)/(x - 1)$ seems to get to 2

Values of x below and above 1	$f(x) = \frac{x^2 - 1}{x - 1} = x + 1, \quad x \neq 1$
0.9	1.9
1.1	2.1
0.99	1.99
1.01	2.01
0.999	1.999
1.001	2.001
0.999999	1.999999
1.000001	2.000001

We say that $f(x)$ approaches the *limit* 2 as x approaches 1, and write

$$\lim_{x \rightarrow 1} f(x) = 2, \quad \text{or} \quad \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

EXAMPLE 6 The Limit Value Does Not Depend on How the Function Is Defined at x_0

The function f in Figure 2.5 has limit 2 as $x \rightarrow 1$ even though f is not defined at $x = 1$. The function g has limit 2 as $x \rightarrow 1$ even though $2 \neq g(1)$. The function h is the only one

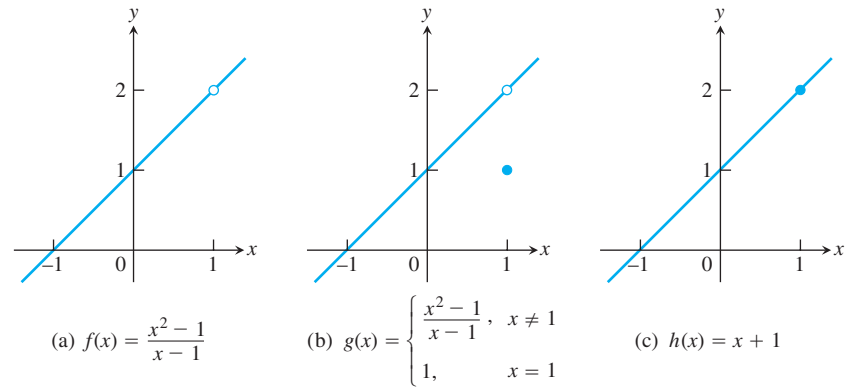


FIGURE 2.5 The limits of $f(x)$, $g(x)$, and $h(x)$ all equal 2 as x approaches 1. However, only $h(x)$ has the same function value as its limit at $x = 1$ (Example 6).

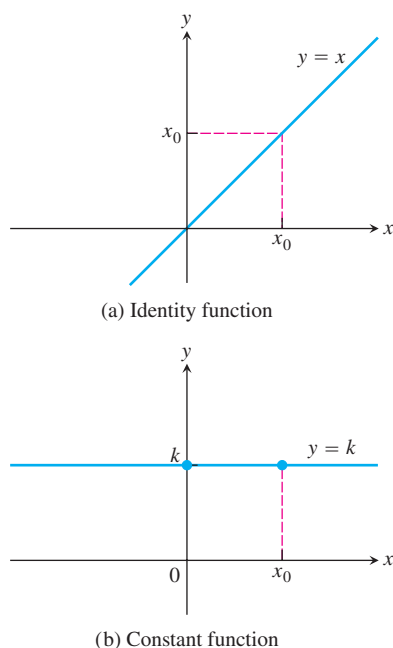


FIGURE 2.6 The functions in Example 8.

whose limit as $x \rightarrow 1$ equals its value at $x = 1$. For h , we have $\lim_{x \rightarrow 1} h(x) = h(1)$. This equality of limit and function value is special, and we return to it in Section 2.6. ■

Sometimes $\lim_{x \rightarrow x_0} f(x)$ can be evaluated by calculating $f(x_0)$. This holds, for example, whenever $f(x)$ is an algebraic combination of polynomials and trigonometric functions for which $f(x_0)$ is defined. (We will say more about this in Sections 2.2 and 2.6.)

EXAMPLE 7 Finding Limits by Calculating $f(x_0)$

- (a) $\lim_{x \rightarrow 2} (4) = 4$
- (b) $\lim_{x \rightarrow -13} (4) = 4$
- (c) $\lim_{x \rightarrow 3} x = 3$
- (d) $\lim_{x \rightarrow 2} (5x - 3) = 10 - 3 = 7$
- (e) $\lim_{x \rightarrow -2} \frac{3x + 4}{x + 5} = \frac{-6 + 4}{-2 + 5} = -\frac{2}{3}$

EXAMPLE 8 The Identity and Constant Functions Have Limits at Every Point

- (a) If f is the **identity function** $f(x) = x$, then for any value of x_0 (Figure 2.6a),

$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} x = x_0.$$

- (b) If f is the **constant function** $f(x) = k$ (function with the constant value k), then for any value of x_0 (Figure 2.6b),

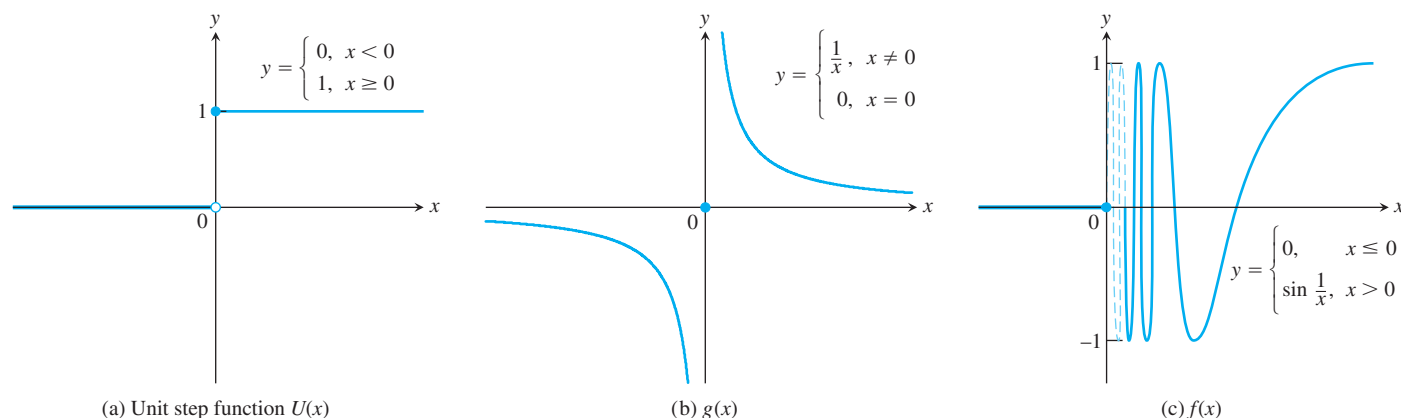
$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} k = k.$$

For instance,

$$\lim_{x \rightarrow 3} x = 3 \quad \text{and} \quad \lim_{x \rightarrow -7} (4) = \lim_{x \rightarrow 2} (4) = 4.$$

We prove these results in Example 3 in Section 2.3. ■

Some ways that limits can fail to exist are illustrated in Figure 2.7 and described in the next example.

FIGURE 2.7 None of these functions has a limit as x approaches 0 (Example 9).

EXAMPLE 9 A Function May Fail to Have a Limit at a Point in Its Domain

Discuss the behavior of the following functions as $x \rightarrow 0$.

- (a) $U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$
- (b) $g(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$
- (c) $f(x) = \begin{cases} 0, & x \leq 0 \\ \sin \frac{1}{x}, & x > 0 \end{cases}$

Solution

- (a) It *jumps*: The **unit step function** $U(x)$ has no limit as $x \rightarrow 0$ because its values jump at $x = 0$. For negative values of x arbitrarily close to zero, $U(x) = 0$. For positive values of x arbitrarily close to zero, $U(x) = 1$. There is no *single* value L approached by $U(x)$ as $x \rightarrow 0$ (Figure 2.7a).
- (b) It *grows too large to have a limit*: $g(x)$ has no limit as $x \rightarrow 0$ because the values of g grow arbitrarily large in absolute value as $x \rightarrow 0$ and do not stay close to *any* real number (Figure 2.7b).
- (c) It *oscillates too much to have a limit*: $f(x)$ has no limit as $x \rightarrow 0$ because the function's values oscillate between $+1$ and -1 in every open interval containing 0. The values do not stay close to any one number as $x \rightarrow 0$ (Figure 2.7c). ■

Using Calculators and Computers to Estimate Limits

Tables 2.1 and 2.2 illustrate using a calculator or computer to guess a limit numerically as x gets closer and closer to x_0 . That procedure would also be successful for the limits of functions like those in Example 7 (these are *continuous* functions and we study them in Section 2.6). However, calculators and computers can give *false values and misleading impressions* for functions that are undefined at a point or fail to have a limit there. The differential calculus will help us know when a calculator or computer is providing strange or ambiguous information about a function's behavior near some point (see Sections 4.4 and 4.6). For now, we simply need to be attentive to the fact that pitfalls may occur when using computing devices to guess the value of a limit. Here's one example.

EXAMPLE 10 Guessing a Limit

Guess the value of $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}$.

Solution Table 2.3 lists values of the function for several values near $x = 0$. As x approaches 0 through the values ± 1 , ± 0.5 , ± 0.10 , and ± 0.01 , the function seems to approach the number 0.05.

As we take even smaller values of x , ± 0.0005 , ± 0.0001 , ± 0.00001 , and ± 0.000001 , the function appears to approach the value 0.

So what is the answer? Is it 0.05 or 0, or some other value? The calculator/computer values are ambiguous, but the theorems on limits presented in the next section will confirm the correct limit value to be $0.05 (= 1/20)$. Problems such as these demonstrate the

TABLE 2.3 Computer values of $f(x) = \frac{\sqrt{x^2 + 100} - 10}{x^2}$ Near $x = 0$

x	$f(x)$	
± 1	0.049876	} approaches 0.05?
± 0.5	0.049969	
± 0.1	0.049999	
± 0.01	0.050000	
± 0.0005	0.080000	} approaches 0?
± 0.0001	0.000000	
± 0.00001	0.000000	
± 0.000001	0.000000	

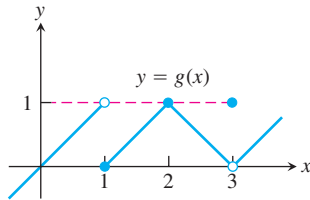
power of mathematical reasoning, once it is developed, over the conclusions we might draw from making a few observations. Both approaches have advantages and disadvantages in revealing nature’s realities. ■

EXERCISES 2.1

Limits from Graphs

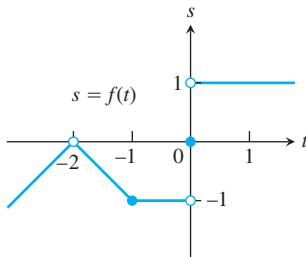
1. For the function $g(x)$ graphed here, find the following limits or explain why they do not exist.

a. $\lim_{x \rightarrow 1} g(x)$ b. $\lim_{x \rightarrow 2} g(x)$ c. $\lim_{x \rightarrow 3} g(x)$



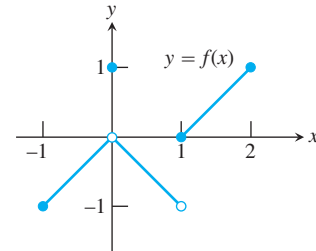
2. For the function $f(t)$ graphed here, find the following limits or explain why they do not exist.

a. $\lim_{t \rightarrow -2} f(t)$ b. $\lim_{t \rightarrow -1} f(t)$ c. $\lim_{t \rightarrow 0} f(t)$



3. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

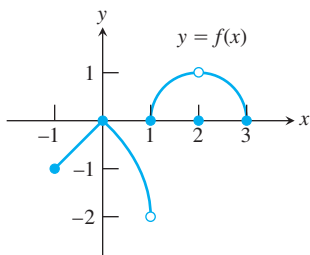
- a. $\lim_{x \rightarrow 0} f(x)$ exists.
 b. $\lim_{x \rightarrow 0} f(x) = 0$.
 c. $\lim_{x \rightarrow 0} f(x) = 1$.
 d. $\lim_{x \rightarrow 1} f(x) = 1$.
 e. $\lim_{x \rightarrow 1} f(x) = 0$.
 f. $\lim_{x \rightarrow x_0} f(x)$ exists at every point x_0 in $(-1, 1)$.



4. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?

- a. $\lim_{x \rightarrow 2} f(x)$ does not exist.
 b. $\lim_{x \rightarrow 2} f(x) = 2$.

- c. $\lim_{x \rightarrow 1} f(x)$ does not exist.
 d. $\lim_{x \rightarrow x_0} f(x)$ exists at every point x_0 in $(-1, 1)$.
 e. $\lim_{x \rightarrow x_0} f(x)$ exists at every point x_0 in $(1, 3)$.



Existence of Limits

In Exercises 5 and 6, explain why the limits do not exist.

5. $\lim_{x \rightarrow 0} \frac{x}{|x|}$ 6. $\lim_{x \rightarrow 1} \frac{1}{x-1}$
7. Suppose that a function $f(x)$ is defined for all real values of x except $x = x_0$. Can anything be said about the existence of $\lim_{x \rightarrow x_0} f(x)$? Give reasons for your answer.
8. Suppose that a function $f(x)$ is defined for all x in $[-1, 1]$. Can anything be said about the existence of $\lim_{x \rightarrow 0} f(x)$? Give reasons for your answer.
9. If $\lim_{x \rightarrow 1} f(x) = 5$, must f be defined at $x = 1$? If it is, must $f(1) = 5$? Can we conclude *anything* about the values of f at $x = 1$? Explain.
10. If $f(1) = 5$, must $\lim_{x \rightarrow 1} f(x)$ exist? If it does, then must $\lim_{x \rightarrow 1} f(x) = 5$? Can we conclude *anything* about $\lim_{x \rightarrow 1} f(x)$? Explain.

Estimating Limits

T You will find a graphing calculator useful for Exercises 11–20.

11. Let $f(x) = (x^2 - 9)/(x + 3)$.
- Make a table of the values of f at the points $x = -3.1, -3.01, -3.001$, and so on as far as your calculator can go. Then estimate $\lim_{x \rightarrow -3} f(x)$. What estimate do you arrive at if you evaluate f at $x = -2.9, -2.99, -2.999, \dots$ instead?
 - Support your conclusions in part (a) by graphing f near $x_0 = -3$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow -3$.
 - Find $\lim_{x \rightarrow -3} f(x)$ algebraically, as in Example 5.
12. Let $g(x) = (x^2 - 2)/(x - \sqrt{2})$.
- Make a table of the values of g at the points $x = 1.4, 1.41, 1.414$, and so on through successive decimal approximations of $\sqrt{2}$. Estimate $\lim_{x \rightarrow \sqrt{2}} g(x)$.
 - Support your conclusion in part (a) by graphing g near $x_0 = \sqrt{2}$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow \sqrt{2}$.
 - Find $\lim_{x \rightarrow \sqrt{2}} g(x)$ algebraically.
13. Let $G(x) = (x + 6)/(x^2 + 4x - 12)$.
- Make a table of the values of G at $x = -5.9, -5.99, -5.999$, and so on. Then estimate $\lim_{x \rightarrow -6} G(x)$. What estimate do you arrive at if you evaluate G at $x = -6.1, -6.01, -6.001, \dots$ instead?
 - Support your conclusions in part (a) by graphing G and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow -6$.
 - Find $\lim_{x \rightarrow -6} G(x)$ algebraically.
14. Let $h(x) = (x^2 - 2x - 3)/(x^2 - 4x + 3)$.
- Make a table of the values of h at $x = 2.9, 2.99, 2.999$, and so on. Then estimate $\lim_{x \rightarrow 3} h(x)$. What estimate do you arrive at if you evaluate h at $x = 3.1, 3.01, 3.001, \dots$ instead?
 - Support your conclusions in part (a) by graphing h near $x_0 = 3$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow 3$.
 - Find $\lim_{x \rightarrow 3} h(x)$ algebraically.
15. Let $f(x) = (x^2 - 1)/(|x| - 1)$.
- Make tables of the values of f at values of x that approach $x_0 = -1$ from above and below. Then estimate $\lim_{x \rightarrow -1} f(x)$.
 - Support your conclusion in part (a) by graphing f near $x_0 = -1$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow -1$.
 - Find $\lim_{x \rightarrow -1} f(x)$ algebraically.
16. Let $F(x) = (x^2 + 3x + 2)/(2 - |x|)$.
- Make tables of values of F at values of x that approach $x_0 = -2$ from above and below. Then estimate $\lim_{x \rightarrow -2} F(x)$.
 - Support your conclusion in part (a) by graphing F near $x_0 = -2$ and using Zoom and Trace to estimate y -values on the graph as $x \rightarrow -2$.
 - Find $\lim_{x \rightarrow -2} F(x)$ algebraically.
17. Let $g(\theta) = (\sin \theta)/\theta$.
- Make a table of the values of g at values of θ that approach $\theta_0 = 0$ from above and below. Then estimate $\lim_{\theta \rightarrow 0} g(\theta)$.
 - Support your conclusion in part (a) by graphing g near $\theta_0 = 0$.
18. Let $G(t) = (1 - \cos t)/t^2$.
- Make tables of values of G at values of t that approach $t_0 = 0$ from above and below. Then estimate $\lim_{t \rightarrow 0} G(t)$.
 - Support your conclusion in part (a) by graphing G near $t_0 = 0$.
19. Let $f(x) = x^{1/(1-x)}$.
- Make tables of values of f at values of x that approach $x_0 = 1$ from above and below. Does f appear to have a limit as $x \rightarrow 1$? If so, what is it? If not, why not?
 - Support your conclusions in part (a) by graphing f near $x_0 = 1$.

20. Let $f(x) = (3^x - 1)/x$.

- Make tables of values of f at values of x that approach $x_0 = 0$ from above and below. Does f appear to have a limit as $x \rightarrow 0$? If so, what is it? If not, why not?
- Support your conclusions in part (a) by graphing f near $x_0 = 0$.

Limits by Substitution

In Exercises 21–28, find the limits by substitution. *Support your answers with a computer or calculator if available.*

21. $\lim_{x \rightarrow 2} 2x$

22. $\lim_{x \rightarrow 0} 2x$

23. $\lim_{x \rightarrow 1/3} (3x - 1)$

24. $\lim_{x \rightarrow 1} \frac{-1}{(3x - 1)}$

25. $\lim_{x \rightarrow -1} 3x(2x - 1)$

26. $\lim_{x \rightarrow -1} \frac{3x^2}{2x - 1}$

27. $\lim_{x \rightarrow \pi/2} x \sin x$

28. $\lim_{x \rightarrow \pi} \frac{\cos x}{1 - \pi}$

Average Rates of Change

In Exercises 29–34, find the average rate of change of the function over the given interval or intervals.

29. $f(x) = x^3 + 1$;

- $[2, 3]$
- $[-1, 1]$

30. $g(x) = x^2$;

- $[-1, 1]$
- $[-2, 0]$

31. $h(t) = \cot t$;

- $[\pi/4, 3\pi/4]$
- $[\pi/6, \pi/2]$

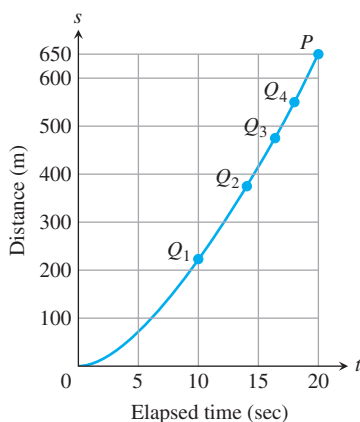
32. $g(t) = 2 + \cos t$;

- $[0, \pi]$
- $[-\pi, \pi]$

33. $R(\theta) = \sqrt{4\theta + 1}$; $[0, 2]$

34. $P(\theta) = \theta^3 - 4\theta^2 + 5\theta$; $[1, 2]$

35. **A Ford Mustang Cobra's speed** The accompanying figure shows the time-to-distance graph for a 1994 Ford Mustang Cobra accelerating from a standstill.



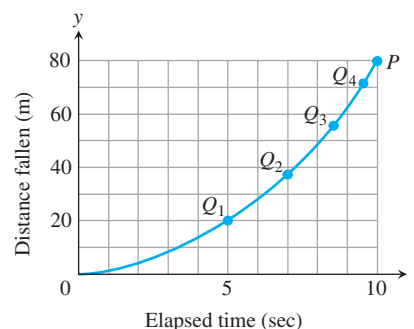
- Estimate the slopes of secants PQ_1 , PQ_2 , PQ_3 , and PQ_4 , arranging them in order in a table like the one in Figure 2.3. What are the appropriate units for these slopes?

- Then estimate the Cobra's speed at time $t = 20$ sec.

36. The accompanying figure shows the plot of distance fallen versus time for an object that fell from the lunar landing module a distance 80 m to the surface of the moon.

- Estimate the slopes of the secants PQ_1 , PQ_2 , PQ_3 , and PQ_4 , arranging them in a table like the one in Figure 2.3.

- About how fast was the object going when it hit the surface?



T 37. The profits of a small company for each of the first five years of its operation are given in the following table:

Year	Profit in \$1000s
1990	6
1991	27
1992	62
1993	111
1994	174

- Plot points representing the profit as a function of year, and join them by as smooth a curve as you can.
- What is the average rate of increase of the profits between 1992 and 1994?
- Use your graph to estimate the rate at which the profits were changing in 1992.

T 38. Make a table of values for the function $F(x) = (x + 2)/(x - 2)$ at the points $x = 1.2$, $x = 11/10$, $x = 101/100$, $x = 1001/1000$, $x = 10001/10000$, and $x = 1$.

- Find the average rate of change of $F(x)$ over the intervals $[1, x]$ for each $x \neq 1$ in your table.
- Extending the table if necessary, try to determine the rate of change of $F(x)$ at $x = 1$.

T 39. Let $g(x) = \sqrt{x}$ for $x \geq 0$.

- Find the average rate of change of $g(x)$ with respect to x over the intervals $[1, 2]$, $[1, 1.5]$ and $[1, 1 + h]$.
- Make a table of values of the average rate of change of g with respect to x over the interval $[1, 1 + h]$ for some values of h

approaching zero, say $h = 0.1, 0.01, 0.001, 0.0001, 0.00001$, and 0.000001 .

- c. What does your table indicate is the rate of change of $g(x)$ with respect to x at $x = 1$?
- d. Calculate the limit as h approaches zero of the average rate of change of $g(x)$ with respect to x over the interval $[1, 1 + h]$.

T 40. Let $f(t) = 1/t$ for $t \neq 0$.

- a. Find the average rate of change of f with respect to t over the intervals (i) from $t = 2$ to $t = 3$, and (ii) from $t = 2$ to $t = T$.
- b. Make a table of values of the average rate of change of f with respect to t over the interval $[2, T]$, for some values of T approaching 2, say $T = 2.1, 2.01, 2.001, 2.0001, 2.00001$, and 2.000001 .
- c. What does your table indicate is the rate of change of f with respect to t at $t = 2$?

- d. Calculate the limit as T approaches 2 of the average rate of change of f with respect to t over the interval from 2 to T . You will have to do some algebra before you can substitute $T = 2$.

COMPUTER EXPLORATIONS

Graphical Estimates of Limits

In Exercises 41–46, use a CAS to perform the following steps:

- a. Plot the function near the point x_0 being approached.
- b. From your plot guess the value of the limit.

$$41. \lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$$

$$43. \lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - 1}{x}$$

$$45. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x}$$

$$42. \lim_{x \rightarrow -1} \frac{x^3 - x^2 - 5x - 3}{(x + 1)^2}$$

$$44. \lim_{x \rightarrow 3} \frac{x^2 - 9}{\sqrt{x^2 + 7} - 4}$$

$$46. \lim_{x \rightarrow 0} \frac{2x^2}{3 - 3 \cos x}$$

2.2

Calculating Limits Using the Limit Laws

HISTORICAL ESSAY*

Limits

In Section 2.1 we used graphs and calculators to guess the values of limits. This section presents theorems for calculating limits. The first three let us build on the results of Example 8 in the preceding section to find limits of polynomials, rational functions, and powers. The fourth and fifth prepare for calculations later in the text.

The Limit Laws

The next theorem tells how to calculate limits of functions that are arithmetic combinations of functions whose limits we already know.

THEOREM 1 **Limit Laws**

If L , M , c and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M, \quad \text{then}$$

1. Sum Rule: $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$

The limit of the sum of two functions is the sum of their limits.

2. Difference Rule: $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$

The limit of the difference of two functions is the difference of their limits.

3. Product Rule: $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$

The limit of a product of two functions is the product of their limits.

To learn more about the historical figures and the development of the major elements and topics of calculus, visit www.aw-bc.com/thomas.

4. Constant Multiple Rule: $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$

The limit of a constant times a function is the constant times the limit of the function.

5. Quotient Rule: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$

The limit of a quotient of two functions is the quotient of their limits, provided the limit of the denominator is not zero.

6. Power Rule: If r and s are integers with no common factor and $s \neq 0$, then

$$\lim_{x \rightarrow c} (f(x))^{r/s} = L^{r/s}$$

provided that $L^{r/s}$ is a real number. (If s is even, we assume that $L > 0$.)

The limit of a rational power of a function is that power of the limit of the function, provided the latter is a real number.

It is easy to convince ourselves that the properties in Theorem 1 are true (although these intuitive arguments do not constitute proofs). If x is sufficiently close to c , then $f(x)$ is close to L and $g(x)$ is close to M , from our informal definition of a limit. It is then reasonable that $f(x) + g(x)$ is close to $L + M$; $f(x) - g(x)$ is close to $L - M$; $f(x)g(x)$ is close to LM ; $kf(x)$ is close to kL ; and that $f(x)/g(x)$ is close to L/M if M is not zero. We prove the Sum Rule in Section 2.3, based on a precise definition of limit. Rules 2–5 are proved in Appendix 2. Rule 6 is proved in more advanced texts.

Here are some examples of how Theorem 1 can be used to find limits of polynomial and rational functions.

EXAMPLE 1 Using the Limit Laws

Use the observations $\lim_{x \rightarrow c} k = k$ and $\lim_{x \rightarrow c} x = c$ (Example 8 in Section 2.1) and the properties of limits to find the following limits.

$$(a) \lim_{x \rightarrow c} (x^3 + 4x^2 - 3) \quad (b) \lim_{x \rightarrow c} \frac{x^4 + x^2 - 1}{x^2 + 5} \quad (c) \lim_{x \rightarrow -2} \sqrt{4x^2 - 3}$$

Solution

$$(a) \lim_{x \rightarrow c} (x^3 + 4x^2 - 3) = \lim_{x \rightarrow c} x^3 + \lim_{x \rightarrow c} 4x^2 - \lim_{x \rightarrow c} 3 \quad \text{Sum and Difference Rules}$$

$$= c^3 + 4c^2 - 3 \quad \text{Product and Multiple Rules}$$

$$(b) \lim_{x \rightarrow c} \frac{x^4 + x^2 - 1}{x^2 + 5} = \frac{\lim_{x \rightarrow c} (x^4 + x^2 - 1)}{\lim_{x \rightarrow c} (x^2 + 5)} \quad \text{Quotient Rule}$$

$$= \frac{\lim_{x \rightarrow c} x^4 + \lim_{x \rightarrow c} x^2 - \lim_{x \rightarrow c} 1}{\lim_{x \rightarrow c} x^2 + \lim_{x \rightarrow c} 5} \quad \text{Sum and Difference Rules}$$

$$= \frac{c^4 + c^2 - 1}{c^2 + 5} \quad \text{Power or Product Rule}$$

$$\begin{aligned}
 \text{(c)} \quad \lim_{x \rightarrow -2} \sqrt{4x^2 - 3} &= \sqrt{\lim_{x \rightarrow -2} (4x^2 - 3)} && \text{Power Rule with } r/s = 1/2 \\
 &= \sqrt{\lim_{x \rightarrow -2} 4x^2 - \lim_{x \rightarrow -2} 3} && \text{Difference Rule} \\
 &= \sqrt{4(-2)^2 - 3} && \text{Product and Multiple Rules} \\
 &= \sqrt{16 - 3} \\
 &= \sqrt{13}
 \end{aligned}$$

Two consequences of Theorem 1 further simplify the task of calculating limits of polynomials and rational functions. To evaluate the limit of a polynomial function as x approaches c , merely substitute c for x in the formula for the function. To evaluate the limit of a rational function as x approaches a point c at which the denominator is not zero, substitute c for x in the formula for the function. (See Examples 1a and 1b.)

THEOREM 2 Limits of Polynomials Can Be Found by Substitution

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$, then

$$\lim_{x \rightarrow c} P(x) = P(c) = a_n c^n + a_{n-1} c^{n-1} + \cdots + a_0.$$

THEOREM 3 Limits of Rational Functions Can Be Found by Substitution If the Limit of the Denominator Is Not Zero

If $P(x)$ and $Q(x)$ are polynomials and $Q(c) \neq 0$, then

$$\lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \frac{P(c)}{Q(c)}.$$

EXAMPLE 2 Limit of a Rational Function

$$\lim_{x \rightarrow -1} \frac{x^3 + 4x^2 - 3}{x^2 + 5} = \frac{(-1)^3 + 4(-1)^2 - 3}{(-1)^2 + 5} = \frac{0}{6} = 0$$

This result is similar to the second limit in Example 1 with $c = -1$, now done in one step.

Identifying Common Factors

It can be shown that if $Q(x)$ is a polynomial and $Q(c) = 0$, then $(x - c)$ is a factor of $Q(x)$. Thus, if the numerator and denominator of a rational function of x are both zero at $x = c$, they have $(x - c)$ as a common factor.

Eliminating Zero Denominators Algebraically

Theorem 3 applies only if the denominator of the rational function is not zero at the limit point c . If the denominator is zero, canceling common factors in the numerator and denominator may reduce the fraction to one whose denominator is no longer zero at c . If this happens, we can find the limit by substitution in the simplified fraction.

EXAMPLE 3 Canceling a Common Factor

Evaluate

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}.$$

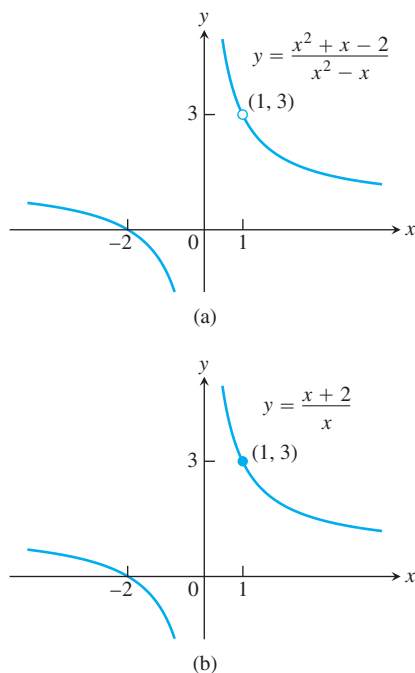


FIGURE 2.8 The graph of $f(x) = (x^2 + x - 2)/(x^2 - x)$ in part (a) is the same as the graph of $g(x) = (x + 2)/x$ in part (b) except at $x = 1$, where f is undefined. The functions have the same limit as $x \rightarrow 1$ (Example 3).

Solution We cannot substitute $x = 1$ because it makes the denominator zero. We test the numerator to see if it, too, is zero at $x = 1$. It is, so it has a factor of $(x - 1)$ in common with the denominator. Canceling the $(x - 1)$'s gives a simpler fraction with the same values as the original for $x \neq 1$:

$$\frac{x^2 + x - 2}{x^2 - x} = \frac{(x - 1)(x + 2)}{x(x - 1)} = \frac{x + 2}{x}, \quad \text{if } x \neq 1.$$

Using the simpler fraction, we find the limit of these values as $x \rightarrow 1$ by substitution:

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x} = \lim_{x \rightarrow 1} \frac{x + 2}{x} = \frac{1 + 2}{1} = 3.$$

See Figure 2.8.

EXAMPLE 4 Creating and Canceling a Common Factor

Evaluate

$$\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}.$$

Solution This is the limit we considered in Example 10 of the preceding section. We cannot substitute $x = 0$, and the numerator and denominator have no obvious common factors. We can create a common factor by multiplying both numerator and denominator by the expression $\sqrt{x^2 + 100} + 10$ (obtained by changing the sign after the square root). The preliminary algebra rationalizes the numerator:

$$\begin{aligned} \frac{\sqrt{x^2 + 100} - 10}{x^2} &= \frac{\sqrt{x^2 + 100} - 10}{x^2} \cdot \frac{\sqrt{x^2 + 100} + 10}{\sqrt{x^2 + 100} + 10} \\ &= \frac{x^2 + 100 - 100}{x^2(\sqrt{x^2 + 100} + 10)} \\ &= \frac{x^2}{x^2(\sqrt{x^2 + 100} + 10)} && \text{Common factor } x^2 \\ &= \frac{1}{\sqrt{x^2 + 100} + 10}. && \text{Cancel } x^2 \text{ for } x \neq 0 \end{aligned}$$

Therefore,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2} &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x^2 + 100} + 10} \\ &= \frac{1}{\sqrt{0^2 + 100} + 10} && \text{Denominator not 0 at } x = 0; \text{ substitute} \\ &= \frac{1}{20} = 0.05. \end{aligned}$$

This calculation provides the correct answer to the ambiguous computer results in Example 10 of the preceding section.

The Sandwich Theorem

The following theorem will enable us to calculate a variety of limits in subsequent chapters. It is called the Sandwich Theorem because it refers to a function f whose values are

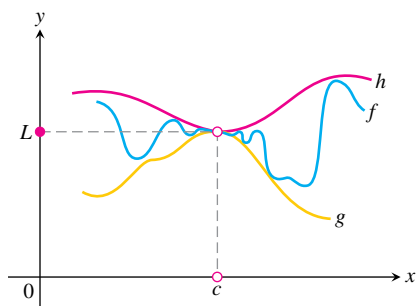


FIGURE 2.9 The graph of f is sandwiched between the graphs of g and h .

sandwiched between the values of two other functions g and h that have the same limit L at a point c . Being trapped between the values of two functions that approach L , the values of f must also approach L (Figure 2.9). You will find a proof in Appendix 2.

THEOREM 4 The Sandwich Theorem

Suppose that $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing c , except possibly at $x = c$ itself. Suppose also that

$$\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L.$$

Then $\lim_{x \rightarrow c} f(x) = L$.

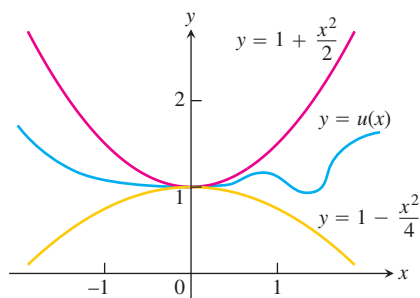


FIGURE 2.10 Any function $u(x)$ whose graph lies in the region between $y = 1 + (x^2/2)$ and $y = 1 - (x^2/4)$ has limit 1 as $x \rightarrow 0$ (Example 5).

The Sandwich Theorem is sometimes called the Squeeze Theorem or the Pinching Theorem.

EXAMPLE 5 Applying the Sandwich Theorem

Given that

$$1 - \frac{x^2}{4} \leq u(x) \leq 1 + \frac{x^2}{2} \quad \text{for all } x \neq 0,$$

find $\lim_{x \rightarrow 0} u(x)$, no matter how complicated u is.

Solution Since

$$\lim_{x \rightarrow 0} (1 - (x^2/4)) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} (1 + (x^2/2)) = 1,$$

the Sandwich Theorem implies that $\lim_{x \rightarrow 0} u(x) = 1$ (Figure 2.10). ■

EXAMPLE 6 More Applications of the Sandwich Theorem

(a) (Figure 2.11a). It follows from the definition of $\sin \theta$ that $-|\theta| \leq \sin \theta \leq |\theta|$ for all θ , and since $\lim_{\theta \rightarrow 0} (-|\theta|) = \lim_{\theta \rightarrow 0} |\theta| = 0$, we have

$$\lim_{\theta \rightarrow 0} \sin \theta = 0.$$

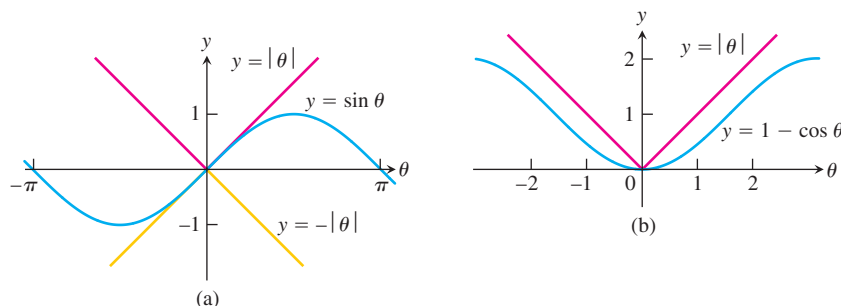


FIGURE 2.11 The Sandwich Theorem confirms that (a) $\lim_{\theta \rightarrow 0} \sin \theta = 0$ and (b) $\lim_{\theta \rightarrow 0} (1 - \cos \theta) = 0$ (Example 6).

- (b) (Figure 2.11b). From the definition of $\cos \theta$, $0 \leq 1 - \cos \theta \leq |\theta|$ for all θ , and we have $\lim_{\theta \rightarrow 0} (1 - \cos \theta) = 0$ or

$$\lim_{\theta \rightarrow 0} \cos \theta = 1.$$

- (c) For any function $f(x)$, if $\lim_{x \rightarrow c} |f(x)| = 0$, then $\lim_{x \rightarrow c} f(x) = 0$. The argument: $-|f(x)| \leq f(x) \leq |f(x)|$ and $-|f(x)|$ and $|f(x)|$ have limit 0 as $x \rightarrow c$. ■

Another important property of limits is given by the next theorem. A proof is given in the next section.

THEOREM 5 If $f(x) \leq g(x)$ for all x in some open interval containing c , except possibly at $x = c$ itself, and the limits of f and g both exist as x approaches c , then

$$\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x).$$

The assertion resulting from replacing the less than or equal to $\leq$ inequality by the strict $<$ inequality in Theorem 5 is false. Figure 2.11a shows that for $\theta \neq 0$, $-|\theta| < \sin \theta < |\theta|$, but in the limit as $\theta \rightarrow 0$, equality holds.

EXERCISES 2.2

Limit Calculations

Find the limits in Exercises 1–18.

1. $\lim_{x \rightarrow -7} (2x + 5)$
2. $\lim_{x \rightarrow 12} (10 - 3x)$
3. $\lim_{x \rightarrow 2} (-x^2 + 5x - 2)$
4. $\lim_{x \rightarrow -2} (x^3 - 2x^2 + 4x + 8)$
5. $\lim_{t \rightarrow 6} 8(t - 5)(t - 7)$
6. $\lim_{s \rightarrow 2/3} 3s(2s - 1)$
7. $\lim_{x \rightarrow 2} \frac{x + 3}{x + 6}$
8. $\lim_{x \rightarrow 5} \frac{4}{x - 7}$
9. $\lim_{y \rightarrow -5} \frac{y^2}{5 - y}$
10. $\lim_{y \rightarrow 2} \frac{y + 2}{y^2 + 5y + 6}$
11. $\lim_{x \rightarrow -1} 3(2x - 1)^2$
12. $\lim_{x \rightarrow -4} (x + 3)^{1984}$
13. $\lim_{y \rightarrow -3} (5 - y)^{4/3}$
14. $\lim_{z \rightarrow 0} (2z - 8)^{1/3}$
15. $\lim_{h \rightarrow 0} \frac{3}{\sqrt{3h + 1} + 1}$
16. $\lim_{h \rightarrow 0} \frac{5}{\sqrt{5h + 4} + 2}$
17. $\lim_{h \rightarrow 0} \frac{\sqrt{3h + 1} - 1}{h}$
18. $\lim_{h \rightarrow 0} \frac{\sqrt{5h + 4} - 2}{h}$

Find the limits in Exercises 19–36.

19. $\lim_{x \rightarrow 5} \frac{x - 5}{x^2 - 25}$
20. $\lim_{x \rightarrow -3} \frac{x + 3}{x^2 + 4x + 3}$
21. $\lim_{x \rightarrow -5} \frac{x^2 + 3x - 10}{x + 5}$
22. $\lim_{x \rightarrow 2} \frac{x^2 - 7x + 10}{x - 2}$
23. $\lim_{t \rightarrow 1} \frac{t^2 + t - 2}{t^2 - 1}$
24. $\lim_{t \rightarrow -1} \frac{t^2 + 3t + 2}{t^2 - t - 2}$
25. $\lim_{x \rightarrow -2} \frac{-2x - 4}{x^3 + 2x^2}$
26. $\lim_{y \rightarrow 0} \frac{5y^3 + 8y^2}{3y^4 - 16y^2}$
27. $\lim_{u \rightarrow 1} \frac{u^4 - 1}{u^3 - 1}$
28. $\lim_{v \rightarrow 2} \frac{v^3 - 8}{v^4 - 16}$
29. $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$
30. $\lim_{x \rightarrow 4} \frac{4x - x^2}{2 - \sqrt{x}}$
31. $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x + 3} - 2}$
32. $\lim_{x \rightarrow -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1}$
33. $\lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 12} - 4}{x - 2}$
34. $\lim_{x \rightarrow -2} \frac{x + 2}{\sqrt{x^2 + 5} - 3}$

$$35. \lim_{x \rightarrow -3} \frac{2 - \sqrt{x^2 - 5}}{x + 3}$$

$$36. \lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}$$

Using Limit Rules

37. Suppose $\lim_{x \rightarrow 0} f(x) = 1$ and $\lim_{x \rightarrow 0} g(x) = -5$. Name the rules in Theorem 1 that are used to accomplish steps (a), (b), and (c) of the following calculation.

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{2f(x) - g(x)}{(f(x) + 7)^{2/3}} &= \frac{\lim_{x \rightarrow 0} (2f(x) - g(x))}{\lim_{x \rightarrow 0} (f(x) + 7)^{2/3}} & (a) \\ &= \frac{\lim_{x \rightarrow 0} 2f(x) - \lim_{x \rightarrow 0} g(x)}{\left(\lim_{x \rightarrow 0} (f(x) + 7)\right)^{2/3}} & (b) \\ &= \frac{2 \lim_{x \rightarrow 0} f(x) - \lim_{x \rightarrow 0} g(x)}{\left(\lim_{x \rightarrow 0} f(x) + \lim_{x \rightarrow 0} 7\right)^{2/3}} & (c) \\ &= \frac{(2)(1) - (-5)}{(1 + 7)^{2/3}} = \frac{7}{4} \end{aligned}$$

38. Let $\lim_{x \rightarrow 1} h(x) = 5$, $\lim_{x \rightarrow 1} p(x) = 1$, and $\lim_{x \rightarrow 1} r(x) = 2$. Name the rules in Theorem 1 that are used to accomplish steps (a), (b), and (c) of the following calculation.

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{\sqrt{5h(x)}}{p(x)(4 - r(x))} &= \frac{\lim_{x \rightarrow 1} \sqrt{5h(x)}}{\lim_{x \rightarrow 1} (p(x)(4 - r(x)))} & (a) \\ &= \frac{\sqrt{\lim_{x \rightarrow 1} 5h(x)}}{\left(\lim_{x \rightarrow 1} p(x)\right)\left(\lim_{x \rightarrow 1} (4 - r(x))\right)} & (b) \\ &= \frac{\sqrt{5 \lim_{x \rightarrow 1} h(x)}}{\left(\lim_{x \rightarrow 1} p(x)\right)\left(\lim_{x \rightarrow 1} 4 - \lim_{x \rightarrow 1} r(x)\right)} & (c) \\ &= \frac{\sqrt{(5)(5)}}{(1)(4 - 2)} = \frac{5}{2} \end{aligned}$$

39. Suppose $\lim_{x \rightarrow c} f(x) = 5$ and $\lim_{x \rightarrow c} g(x) = -2$. Find

$$\begin{array}{ll} \text{a. } \lim_{x \rightarrow c} f(x)g(x) & \text{b. } \lim_{x \rightarrow c} 2f(x)g(x) \\ \text{c. } \lim_{x \rightarrow c} (f(x) + 3g(x)) & \text{d. } \lim_{x \rightarrow c} \frac{f(x)}{f(x) - g(x)} \end{array}$$

40. Suppose $\lim_{x \rightarrow 4} f(x) = 0$ and $\lim_{x \rightarrow 4} g(x) = -3$. Find

$$\begin{array}{ll} \text{a. } \lim_{x \rightarrow 4} (g(x) + 3) & \text{b. } \lim_{x \rightarrow 4} xf(x) \\ \text{c. } \lim_{x \rightarrow 4} (g(x))^2 & \text{d. } \lim_{x \rightarrow 4} \frac{g(x)}{f(x) - 1} \end{array}$$

41. Suppose $\lim_{x \rightarrow b} f(x) = 7$ and $\lim_{x \rightarrow b} g(x) = -3$. Find

$$\begin{array}{ll} \text{a. } \lim_{x \rightarrow b} (f(x) + g(x)) & \text{b. } \lim_{x \rightarrow b} f(x) \cdot g(x) \\ \text{c. } \lim_{x \rightarrow b} 4g(x) & \text{d. } \lim_{x \rightarrow b} f(x)/g(x) \end{array}$$

42. Suppose that $\lim_{x \rightarrow -2} p(x) = 4$, $\lim_{x \rightarrow -2} r(x) = 0$, and $\lim_{x \rightarrow -2} s(x) = -3$. Find

$$\begin{array}{l} \text{a. } \lim_{x \rightarrow -2} (p(x) + r(x) + s(x)) \\ \text{b. } \lim_{x \rightarrow -2} p(x) \cdot r(x) \cdot s(x) \\ \text{c. } \lim_{x \rightarrow -2} (-4p(x) + 5r(x))/s(x) \end{array}$$

Limits of Average Rates of Change

Because of their connection with secant lines, tangents, and instantaneous rates, limits of the form

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

occur frequently in calculus. In Exercises 43–48, evaluate this limit for the given value of x and function f .

- $$\begin{array}{ll} 43. f(x) = x^2, & x = 1 \\ 44. f(x) = x^2, & x = -2 \\ 45. f(x) = 3x - 4, & x = 2 \\ 46. f(x) = 1/x, & x = -2 \\ 47. f(x) = \sqrt{x}, & x = 7 \\ 48. f(x) = \sqrt{3x + 1}, & x = 0 \end{array}$$

Using the Sandwich Theorem

49. If $\sqrt{5 - 2x^2} \leq f(x) \leq \sqrt{5 - x^2}$ for $-1 \leq x \leq 1$, find $\lim_{x \rightarrow 0} f(x)$.

50. If $2 - x^2 \leq g(x) \leq 2 \cos x$ for all x , find $\lim_{x \rightarrow 0} g(x)$.

51. a. It can be shown that the inequalities

$$1 - \frac{x^2}{6} < \frac{x \sin x}{2 - 2 \cos x} < 1$$

hold for all values of x close to zero. What, if anything, does this tell you about

$$\lim_{x \rightarrow 0} \frac{x \sin x}{2 - 2 \cos x}?$$

Give reasons for your answer.

- T** b. Graph

$y = 1 - (x^2/6)$, $y = (x \sin x)/(2 - 2 \cos x)$, and $y = 1$ together for $-2 \leq x \leq 2$. Comment on the behavior of the graphs as $x \rightarrow 0$.

52. a. Suppose that the inequalities

$$\frac{1}{2} - \frac{x^2}{24} < \frac{1 - \cos x}{x^2} < \frac{1}{2}$$

hold for values of x close to zero. (They do, as you will see in Section 11.9.) What, if anything, does this tell you about

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}?$$

Give reasons for your answer.

- b. Graph the equations $y = (1/2) - (x^2/24)$, $y = (1 - \cos x)/x^2$, and $y = 1/2$ together for $-2 \leq x \leq 2$. Comment on the behavior of the graphs as $x \rightarrow 0$.

Theory and Examples

53. If $x^4 \leq f(x) \leq x^2$ for x in $[-1, 1]$ and $x^2 \leq f(x) \leq x^4$ for $x < -1$ and $x > 1$, at what points c do you automatically know $\lim_{x \rightarrow c} f(x)$? What can you say about the value of the limit at these points?

54. Suppose that $g(x) \leq f(x) \leq h(x)$ for all $x \neq 2$ and suppose that

$$\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} h(x) = -5.$$

Can we conclude anything about the values of f , g , and h at $x = 2$? Could $f(2) = 0$? Could $\lim_{x \rightarrow 2} f(x) = 0$? Give reasons for your answers.

55. If $\lim_{x \rightarrow 4} \frac{f(x) - 5}{x - 2} = 1$, find $\lim_{x \rightarrow 4} f(x)$.

56. If $\lim_{x \rightarrow -2} \frac{f(x)}{x^2} = 1$, find

a. $\lim_{x \rightarrow -2} f(x)$ b. $\lim_{x \rightarrow -2} \frac{f(x)}{x}$

57. a. If $\lim_{x \rightarrow 2} \frac{f(x) - 5}{x - 2} = 3$, find $\lim_{x \rightarrow 2} f(x)$.

b. If $\lim_{x \rightarrow 2} \frac{f(x) - 5}{x - 2} = 4$, find $\lim_{x \rightarrow 2} f(x)$.

58. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 1$, find

a. $\lim_{x \rightarrow 0} f(x)$ b. $\lim_{x \rightarrow 0} \frac{f(x)}{x}$

T 59. a. Graph $g(x) = x \sin(1/x)$ to estimate $\lim_{x \rightarrow 0} g(x)$, zooming in on the origin as necessary.

b. Confirm your estimate in part (a) with a proof.

T 60. a. Graph $h(x) = x^2 \cos(1/x^3)$ to estimate $\lim_{x \rightarrow 0} h(x)$, zooming in on the origin as necessary.

b. Confirm your estimate in part (a) with a proof.

2.3

The Precise Definition of a Limit

Now that we have gained some insight into the limit concept, working intuitively with the informal definition, we turn our attention to its precise definition. We replace vague phrases like “gets arbitrarily close to” in the informal definition with specific conditions that can be applied to any particular example. With a precise definition we will be able to prove conclusively the limit properties given in the preceding section, and we can establish other particular limits important to the study of calculus.

To show that the limit of $f(x)$ as $x \rightarrow x_0$ equals the number L , we need to show that the gap between $f(x)$ and L can be made “as small as we choose” if x is kept “close enough” to x_0 . Let us see what this would require if we specified the size of the gap between $f(x)$ and L .

EXAMPLE 1 A Linear Function

Consider the function $y = 2x - 1$ near $x_0 = 4$. Intuitively it is clear that y is close to 7 when x is close to 4, so $\lim_{x \rightarrow 4} (2x - 1) = 7$. However, how close to $x_0 = 4$ does x have to be so that $y = 2x - 1$ differs from 7 by, say, less than 2 units?

Solution We are asked: For what values of x is $|y - 7| < 2$? To find the answer we first express $|y - 7|$ in terms of x :

$$|y - 7| = |(2x - 1) - 7| = |2x - 8|.$$

The question then becomes: what values of x satisfy the inequality $|2x - 8| < 2$? To find out, we solve the inequality:

$$\begin{aligned} |2x - 8| &< 2 \\ -2 &< 2x - 8 < 2 \\ 6 &< 2x < 10 \\ 3 &< x < 5 \\ -1 &< x - 4 < 1. \end{aligned}$$

Keeping x within 1 unit of $x_0 = 4$ will keep y within 2 units of $y_0 = 7$ (Figure 2.12). ■

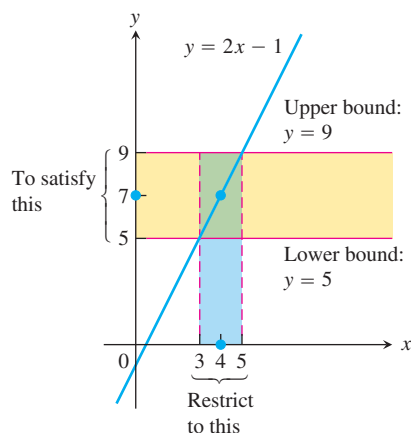


FIGURE 2.12 Keeping x within 1 unit of $x_0 = 4$ will keep y within 2 units of $y_0 = 7$ (Example 1).

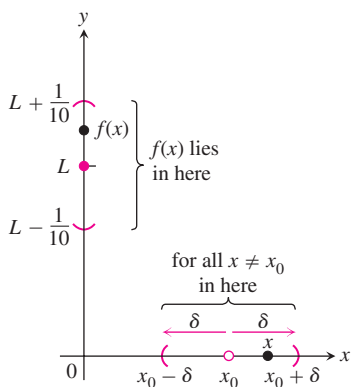


FIGURE 2.13 How should we define $\delta > 0$ so that keeping x within the interval $(x_0 - \delta, x_0 + \delta)$ will keep $f(x)$ within the interval $\left(L - \frac{1}{10}, L + \frac{1}{10}\right)$?

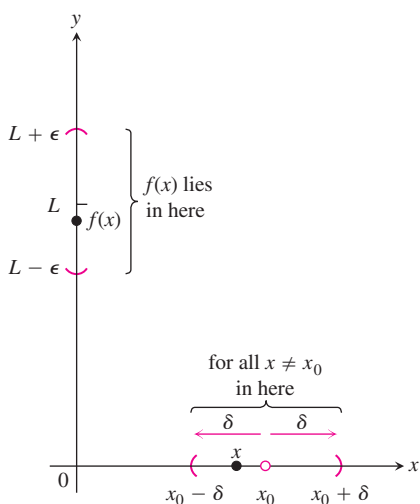


FIGURE 2.14 The relation of δ and ϵ in the definition of limit.

In the previous example we determined how close x must be to a particular value x_0 to ensure that the outputs $f(x)$ of some function lie within a prescribed interval about a limit value L . To show that the limit of $f(x)$ as $x \rightarrow x_0$ actually equals L , we must be able to show that the gap between $f(x)$ and L can be made less than *any prescribed error*, no matter how small, by holding x close enough to x_0 .

Definition of Limit

Suppose we are watching the values of a function $f(x)$ as x approaches x_0 (without taking on the value of x_0 itself). Certainly we want to be able to say that $f(x)$ stays within one-tenth of a unit of L as soon as x stays within some distance δ of x_0 (Figure 2.13). But that in itself is not enough, because as x continues on its course toward x_0 , what is to prevent $f(x)$ from jittering about within the interval from $L - (1/10)$ to $L + (1/10)$ without tending toward L ?

We can be told that the error can be no more than $1/100$ or $1/1000$ or $1/100,000$. Each time, we find a new δ -interval about x_0 so that keeping x within that interval satisfies the new error tolerance. And each time the possibility exists that $f(x)$ jitters away from L at some stage.

The figures on the next page illustrate the problem. You can think of this as a quarrel between a skeptic and a scholar. The skeptic presents ϵ -challenges to prove that the limit does not exist or, more precisely, that there is room for doubt, and the scholar answers every challenge with a δ -interval around x_0 .

How do we stop this seemingly endless series of challenges and responses? By proving that for every error tolerance ϵ that the challenger can produce, we can find, calculate, or conjure a matching distance δ that keeps x “close enough” to x_0 to keep $f(x)$ within that tolerance of L (Figure 2.14). This leads us to the precise definition of a limit.

DEFINITION Limit of a Function

Let $f(x)$ be defined on an open interval about x_0 , except possibly at x_0 itself. We say that the **limit of $f(x)$ as x approaches x_0 is the number L** , and write

$$\lim_{x \rightarrow x_0} f(x) = L,$$

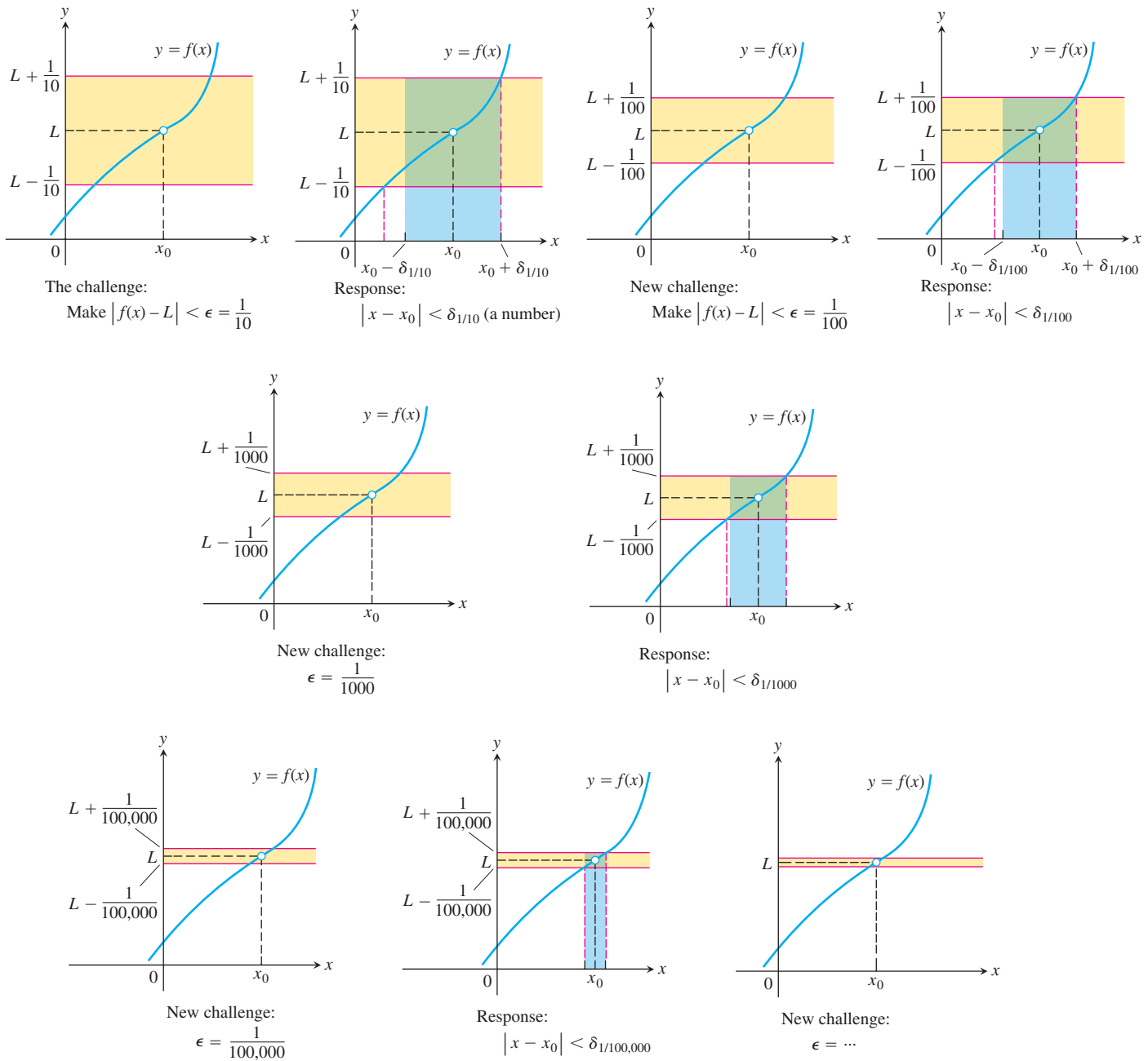
if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all x ,

$$0 < |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

One way to think about the definition is to suppose we are machining a generator shaft to a close tolerance. We may try for diameter L , but since nothing is perfect, we must be satisfied with a diameter $f(x)$ somewhere between $L - \epsilon$ and $L + \epsilon$. The δ is the measure of how accurate our control setting for x must be to guarantee this degree of accuracy in the diameter of the shaft. Notice that as the tolerance for error becomes stricter, we may have to adjust δ . That is, the value of δ , how tight our control setting must be, depends on the value of ϵ , the error tolerance.

Examples: Testing the Definition

The formal definition of limit does not tell how to find the limit of a function, but it enables us to verify that a suspected limit is correct. The following examples show how the



definition can be used to verify limit statements for specific functions. (The first two examples correspond to parts of Examples 7 and 8 in Section 2.1.) However, the real purpose of the definition is not to do calculations like this, but rather to prove general theorems so that the calculation of specific limits can be simplified.

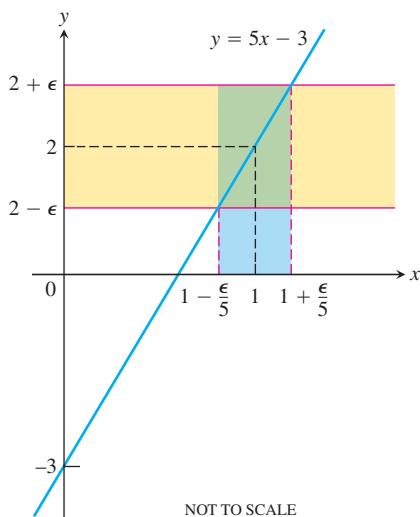


FIGURE 2.15 If $f(x) = 5x - 3$, then $0 < |x - 1| < \epsilon/5$ guarantees that $|f(x) - 2| < \epsilon$ (Example 2).

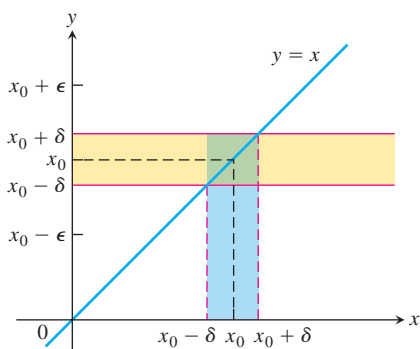


FIGURE 2.16 For the function $f(x) = x$, we find that $0 < |x - x_0| < \delta$ will guarantee $|f(x) - x_0| < \epsilon$ whenever $\delta \leq \epsilon$ (Example 3a).

EXAMPLE 2 Testing the Definition

Show that

$$\lim_{x \rightarrow 1} (5x - 3) = 2.$$

Solution Set $x_0 = 1$, $f(x) = 5x - 3$, and $L = 2$ in the definition of limit. For any given $\epsilon > 0$, we have to find a suitable $\delta > 0$ so that if $x \neq 1$ and x is within distance δ of $x_0 = 1$, that is, whenever

$$0 < |x - 1| < \delta,$$

it is true that $f(x)$ is within distance ϵ of $L = 2$, so

$$|f(x) - 2| < \epsilon.$$

We find δ by working backward from the ϵ -inequality:

$$\begin{aligned} |(5x - 3) - 2| &= |5x - 5| < \epsilon \\ 5|x - 1| &< \epsilon \\ |x - 1| &< \epsilon/5. \end{aligned}$$

Thus, we can take $\delta = \epsilon/5$ (Figure 2.15). If $0 < |x - 1| < \delta = \epsilon/5$, then

$$|(5x - 3) - 2| = |5x - 5| = 5|x - 1| < 5(\epsilon/5) = \epsilon,$$

which proves that $\lim_{x \rightarrow 1} (5x - 3) = 2$.

The value of $\delta = \epsilon/5$ is not the only value that will make $0 < |x - 1| < \delta$ imply $|5x - 5| < \epsilon$. Any smaller positive δ will do as well. The definition does not ask for a “best” positive δ , just one that will work. ■

EXAMPLE 3 Limits of the Identity and Constant Functions

Prove:

$$(a) \lim_{x \rightarrow x_0} x = x_0 \quad (b) \lim_{x \rightarrow x_0} k = k \quad (k \text{ constant}).$$

Solution

(a) Let $\epsilon > 0$ be given. We must find $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \quad \text{implies} \quad |x - x_0| < \epsilon.$$

The implication will hold if δ equals ϵ or any smaller positive number (Figure 2.16). This proves that $\lim_{x \rightarrow x_0} x = x_0$.

(b) Let $\epsilon > 0$ be given. We must find $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \quad \text{implies} \quad |k - k| < \epsilon.$$

Since $k - k = 0$, we can use any positive number for δ and the implication will hold (Figure 2.17). This proves that $\lim_{x \rightarrow x_0} k = k$. ■

Finding Deltas Algebraically for Given Epsilons

In Examples 2 and 3, the interval of values about x_0 for which $|f(x) - L|$ was less than ϵ was symmetric about x_0 and we could take δ to be half the length of that interval. When

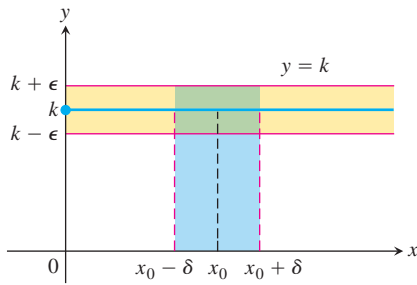


FIGURE 2.17 For the function $f(x) = k$, we find that $|f(x) - k| < \epsilon$ for any positive δ (Example 3b).

such symmetry is absent, as it usually is, we can take δ to be the distance from x_0 to the interval's *nearer* endpoint.

EXAMPLE 4 Finding Delta Algebraically

For the limit $\lim_{x \rightarrow 5} \sqrt{x-1} = 2$, find a $\delta > 0$ that works for $\epsilon = 1$. That is, find a $\delta > 0$ such that for all x

$$0 < |x - 5| < \delta \quad \Rightarrow \quad |\sqrt{x-1} - 2| < 1.$$

Solution We organize the search into two steps, as discussed below.

1. Solve the inequality $|\sqrt{x-1} - 2| < 1$ to find an interval containing $x_0 = 5$ on which the inequality holds for all $x \neq x_0$.

$$\begin{aligned} |\sqrt{x-1} - 2| &< 1 \\ -1 &< \sqrt{x-1} - 2 < 1 \\ 1 &< \sqrt{x-1} < 3 \\ 1 &< x-1 < 9 \\ 2 &< x < 10 \end{aligned}$$

The inequality holds for all x in the open interval $(2, 10)$, so it holds for all $x \neq 5$ in this interval as well (see Figure 2.19).

2. Find a value of $\delta > 0$ to place the centered interval $5 - \delta < x < 5 + \delta$ (centered at $x_0 = 5$) inside the interval $(2, 10)$. The distance from 5 to the nearer endpoint of $(2, 10)$ is 3 (Figure 2.18). If we take $\delta = 3$ or any smaller positive number, then the inequality $0 < |x - 5| < \delta$ will automatically place x between 2 and 10 to make $|\sqrt{x-1} - 2| < 1$ (Figure 2.19)

$$0 < |x - 5| < 3 \quad \Rightarrow \quad |\sqrt{x-1} - 2| < 1.$$



FIGURE 2.18 An open interval of radius 3 about $x_0 = 5$ will lie inside the open interval $(2, 10)$.

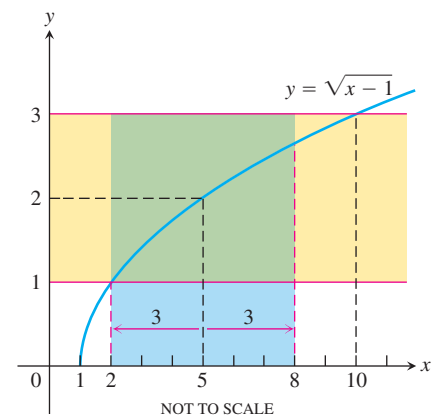


FIGURE 2.19 The function and intervals in Example 4.

How to Find Algebraically a δ for a Given f , L , x_0 , and $\epsilon > 0$

The process of finding a $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon$$

can be accomplished in two steps.

1. Solve the inequality $|f(x) - L| < \epsilon$ to find an open interval (a, b) containing x_0 on which the inequality holds for all $x \neq x_0$.
2. Find a value of $\delta > 0$ that places the open interval $(x_0 - \delta, x_0 + \delta)$ centered at x_0 inside the interval (a, b) . The inequality $|f(x) - L| < \epsilon$ will hold for all $x \neq x_0$ in this δ -interval.

EXAMPLE 5 Finding Delta Algebraically

Prove that $\lim_{x \rightarrow 2} f(x) = 4$ if

$$f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2. \end{cases}$$

Solution Our task is to show that given $\epsilon > 0$ there exists a $\delta > 0$ such that for all x

$$0 < |x - 2| < \delta \quad \Rightarrow \quad |f(x) - 4| < \epsilon.$$

1. Solve the inequality $|f(x) - 4| < \epsilon$ to find an open interval containing $x_0 = 2$ on which the inequality holds for all $x \neq x_0$.

For $x \neq x_0 = 2$, we have $f(x) = x^2$, and the inequality to solve is $|x^2 - 4| < \epsilon$:

$$\begin{aligned} |x^2 - 4| &< \epsilon \\ -\epsilon &< x^2 - 4 < \epsilon \\ 4 - \epsilon &< x^2 < 4 + \epsilon \\ \sqrt{4 - \epsilon} &< |x| < \sqrt{4 + \epsilon} && \text{Assumes } \epsilon < 4; \text{ see below.} \\ \sqrt{4 - \epsilon} &< x < \sqrt{4 + \epsilon} && \text{An open interval about } x_0 = 2 \\ &&& \text{that solves the inequality} \end{aligned}$$

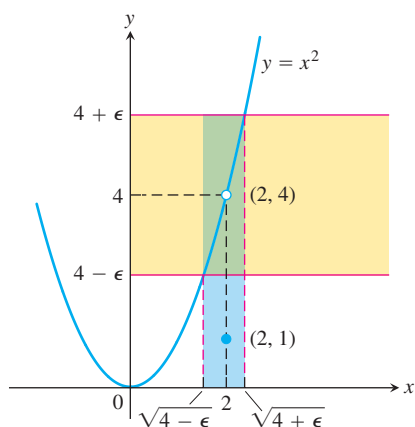


FIGURE 2.20 An interval containing $x = 2$ so that the function in Example 5 satisfies $|f(x) - 4| < \epsilon$.

The inequality $|f(x) - 4| < \epsilon$ holds for all $x \neq 2$ in the open interval $(\sqrt{4 - \epsilon}, \sqrt{4 + \epsilon})$ (Figure 2.20).

2. Find a value of $\delta > 0$ that places the centered interval $(2 - \delta, 2 + \delta)$ inside the interval $(\sqrt{4 - \epsilon}, \sqrt{4 + \epsilon})$.

Take δ to be the distance from $x_0 = 2$ to the nearer endpoint of $(\sqrt{4 - \epsilon}, \sqrt{4 + \epsilon})$. In other words, take $\delta = \min \{2 - \sqrt{4 - \epsilon}, \sqrt{4 + \epsilon} - 2\}$, the *minimum* (the smaller) of the two numbers $2 - \sqrt{4 - \epsilon}$ and $\sqrt{4 + \epsilon} - 2$. If δ has this or any smaller positive value, the inequality $0 < |x - 2| < \delta$ will automatically place x between $\sqrt{4 - \epsilon}$ and $\sqrt{4 + \epsilon}$ to make $|f(x) - 4| < \epsilon$. For all x ,

$$0 < |x - 2| < \delta \quad \Rightarrow \quad |f(x) - 4| < \epsilon.$$

This completes the proof.

Why was it all right to assume $\epsilon < 4$? Because, in finding a δ such that for all x , $0 < |x - 2| < \delta$ implied $|f(x) - 4| < \epsilon < 4$, we found a δ that would work for any larger ϵ as well.

Finally, notice the freedom we gained in letting $\delta = \min \{2 - \sqrt{4 - \epsilon}, \sqrt{4 + \epsilon} - 2\}$. We did not have to spend time deciding which, if either, number was the smaller of the two. We just let δ represent the smaller and went on to finish the argument. ■

Using the Definition to Prove Theorems

We do not usually rely on the formal definition of limit to verify specific limits such as those in the preceding examples. Rather we appeal to general theorems about limits, in particular the theorems of Section 2.2. The definition is used to prove these theorems (Appendix 2). As an example, we prove part 1 of Theorem 1, the Sum Rule.

EXAMPLE 6 Proving the Rule for the Limit of a Sum

Given that $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, prove that

$$\lim_{x \rightarrow c} (f(x) + g(x)) = L + M.$$

Solution Let $\epsilon > 0$ be given. We want to find a positive number δ such that for all x

$$0 < |x - c| < \delta \quad \Rightarrow \quad |f(x) + g(x) - (L + M)| < \epsilon.$$

Regrouping terms, we get

$$\begin{aligned} |f(x) + g(x) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M|. \end{aligned}$$

Triangle Inequality:
 $|a + b| \leq |a| + |b|$

Since $\lim_{x \rightarrow c} f(x) = L$, there exists a number $\delta_1 > 0$ such that for all x

$$0 < |x - c| < \delta_1 \quad \Rightarrow \quad |f(x) - L| < \epsilon/2.$$

Similarly, since $\lim_{x \rightarrow c} g(x) = M$, there exists a number $\delta_2 > 0$ such that for all x

$$0 < |x - c| < \delta_2 \quad \Rightarrow \quad |g(x) - M| < \epsilon/2.$$

Let $\delta = \min \{\delta_1, \delta_2\}$, the smaller of δ_1 and δ_2 . If $0 < |x - c| < \delta$ then $|x - c| < \delta_1$, so $|f(x) - L| < \epsilon/2$, and $|x - c| < \delta_2$, so $|g(x) - M| < \epsilon/2$. Therefore

$$|f(x) + g(x) - (L + M)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This shows that $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$. ■

Let's also prove Theorem 5 of Section 2.2.

EXAMPLE 7 Given that $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, and that $f(x) \leq g(x)$ for all x in an open interval containing c (except possibly c itself), prove that $L \leq M$.

Solution We use the method of proof by contradiction. Suppose, on the contrary, that $L > M$. Then by the limit of a difference property in Theorem 1,

$$\lim_{x \rightarrow c} (g(x) - f(x)) = M - L.$$

Therefore, for any $\epsilon > 0$, there exists $\delta > 0$ such that

$$|(g(x) - f(x)) - (M - L)| < \epsilon \quad \text{whenever} \quad 0 < |x - c| < \delta.$$

Since $L - M > 0$ by hypothesis, we take $\epsilon = L - M$ in particular and we have a number $\delta > 0$ such that

$$|(g(x) - f(x)) - (M - L)| < L - M \quad \text{whenever} \quad 0 < |x - c| < \delta.$$

Since $a \leq |a|$ for any number a , we have

$$(g(x) - f(x)) - (M - L) < L - M \quad \text{whenever} \quad 0 < |x - c| < \delta$$

which simplifies to

$$g(x) < f(x) \quad \text{whenever} \quad 0 < |x - c| < \delta.$$

But this contradicts $f(x) \leq g(x)$. Thus the inequality $L > M$ must be false. Therefore $L \leq M$. ■

EXERCISES 2.3

Centering Intervals About a Point

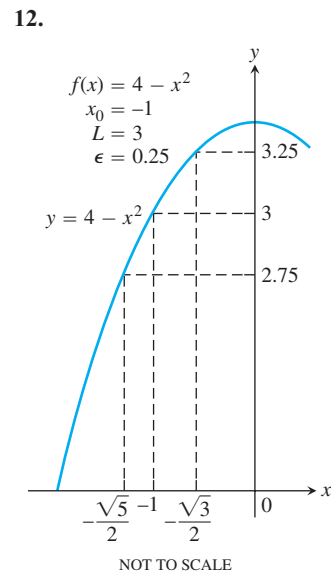
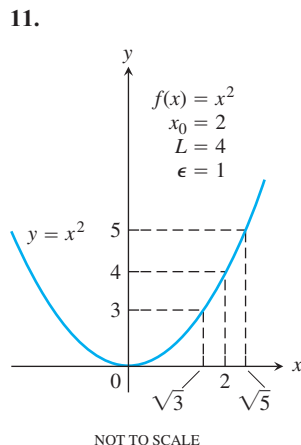
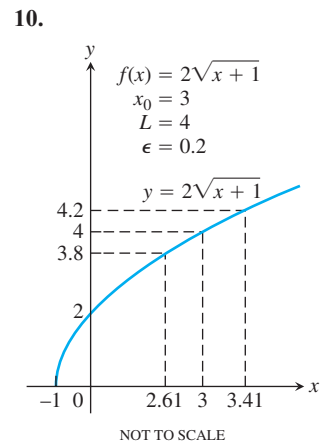
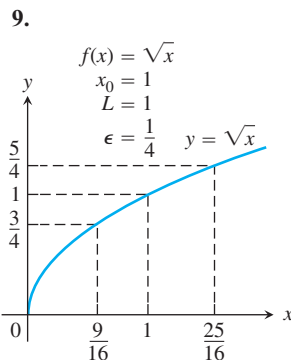
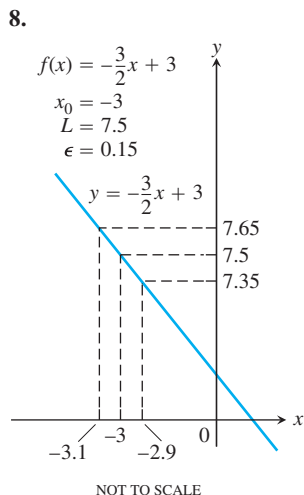
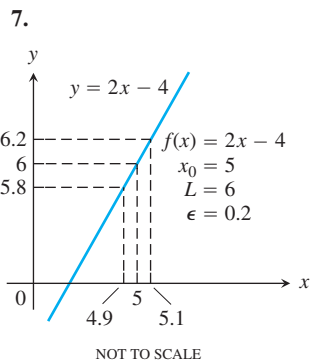
In Exercises 1–6, sketch the interval (a, b) on the x -axis with the point x_0 inside. Then find a value of $\delta > 0$ such that for all x , $0 < |x - x_0| < \delta \Rightarrow a < x < b$.

1. $a = 1$, $b = 7$, $x_0 = 5$
2. $a = 1$, $b = 7$, $x_0 = 2$
3. $a = -7/2$, $b = -1/2$, $x_0 = -3$
4. $a = -7/2$, $b = -1/2$, $x_0 = -3/2$
5. $a = 4/9$, $b = 4/7$, $x_0 = 1/2$
6. $a = 2.7591$, $b = 3.2391$, $x_0 = 3$

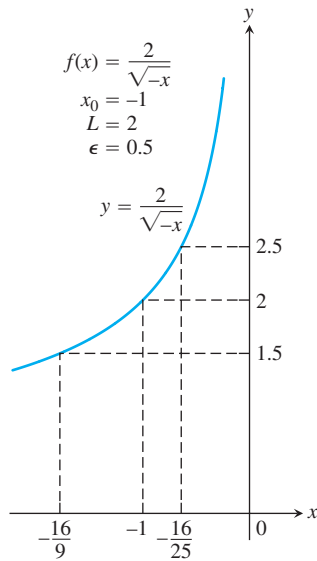
Finding Deltas Graphically

In Exercises 7–14, use the graphs to find a $\delta > 0$ such that for all x

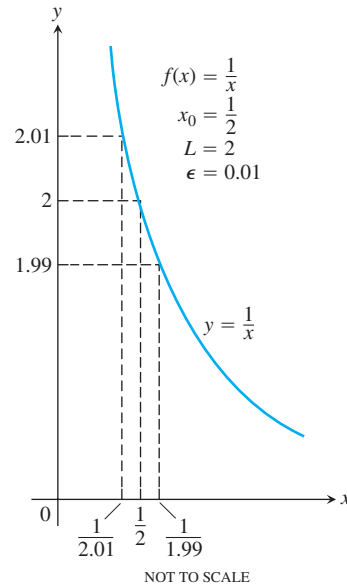
$$0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \epsilon.$$



13.



14.



Finding Deltas Algebraically

Each of Exercises 15–30 gives a function $f(x)$ and numbers L , x_0 and $\epsilon > 0$. In each case, find an open interval about x_0 on which the inequality $|f(x) - L| < \epsilon$ holds. Then give a value for $\delta > 0$ such that for all x satisfying $0 < |x - x_0| < \delta$ the inequality $|f(x) - L| < \epsilon$ holds.

15. $f(x) = x + 1$, $L = 5$, $x_0 = 4$, $\epsilon = 0.01$
16. $f(x) = 2x - 2$, $L = -6$, $x_0 = -2$, $\epsilon = 0.02$
17. $f(x) = \sqrt{x + 1}$, $L = 1$, $x_0 = 0$, $\epsilon = 0.1$
18. $f(x) = \sqrt{x}$, $L = 1/2$, $x_0 = 1/4$, $\epsilon = 0.1$
19. $f(x) = \sqrt{19 - x}$, $L = 3$, $x_0 = 10$, $\epsilon = 1$
20. $f(x) = \sqrt{x - 7}$, $L = 4$, $x_0 = 23$, $\epsilon = 1$
21. $f(x) = 1/x$, $L = 1/4$, $x_0 = 4$, $\epsilon = 0.05$
22. $f(x) = x^2$, $L = 3$, $x_0 = \sqrt{3}$, $\epsilon = 0.1$
23. $f(x) = x^2$, $L = 4$, $x_0 = -2$, $\epsilon = 0.5$
24. $f(x) = 1/x$, $L = -1$, $x_0 = -1$, $\epsilon = 0.1$
25. $f(x) = x^2 - 5$, $L = 11$, $x_0 = 4$, $\epsilon = 1$
26. $f(x) = 120/x$, $L = 5$, $x_0 = 24$, $\epsilon = 1$
27. $f(x) = mx$, $m > 0$, $L = 2m$, $x_0 = 2$, $\epsilon = 0.03$
28. $f(x) = mx$, $m > 0$, $L = 3m$, $x_0 = 3$, $\epsilon = c > 0$
29. $f(x) = mx + b$, $m > 0$, $L = (m/2) + b$, $x_0 = 1/2$, $\epsilon = c > 0$
30. $f(x) = mx + b$, $m > 0$, $L = m + b$, $x_0 = 1$, $\epsilon = 0.05$

More on Formal Limits

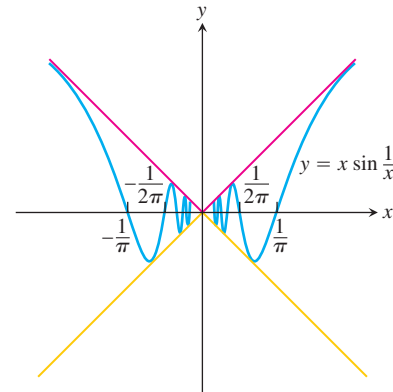
Each of Exercises 31–36 gives a function $f(x)$, a point x_0 , and a positive number ϵ . Find $L = \lim_{x \rightarrow x_0} f(x)$. Then find a number $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \implies |f(x) - L| < \epsilon.$$

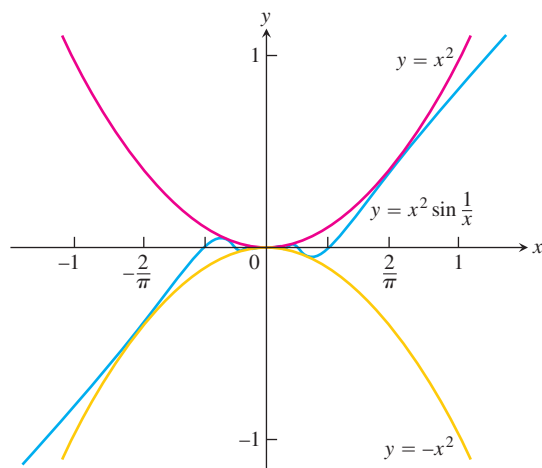
31. $f(x) = 3 - 2x$, $x_0 = 3$, $\epsilon = 0.02$
32. $f(x) = -3x - 2$, $x_0 = -1$, $\epsilon = 0.03$
33. $f(x) = \frac{x^2 - 4}{x - 2}$, $x_0 = 2$, $\epsilon = 0.05$
34. $f(x) = \frac{x^2 + 6x + 5}{x + 5}$, $x_0 = -5$, $\epsilon = 0.05$
35. $f(x) = \sqrt{1 - 5x}$, $x_0 = -3$, $\epsilon = 0.5$
36. $f(x) = 4/x$, $x_0 = 2$, $\epsilon = 0.4$

Prove the limit statements in Exercises 37–50.

37. $\lim_{x \rightarrow 4} (9 - x) = 5$
38. $\lim_{x \rightarrow 3} (3x - 7) = 2$
39. $\lim_{x \rightarrow 9} \sqrt{x - 5} = 2$
40. $\lim_{x \rightarrow 0} \sqrt{4 - x} = 2$
41. $\lim_{x \rightarrow 1} f(x) = 1$ if $f(x) = \begin{cases} x^2, & x \neq 1 \\ 2, & x = 1 \end{cases}$
42. $\lim_{x \rightarrow -2} f(x) = 4$ if $f(x) = \begin{cases} x^2, & x \neq -2 \\ 1, & x = -2 \end{cases}$
43. $\lim_{x \rightarrow 1} \frac{1}{x} = 1$
44. $\lim_{x \rightarrow \sqrt{3}} \frac{1}{x^2} = \frac{1}{3}$
45. $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3} = -6$
46. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$
47. $\lim_{x \rightarrow 1} f(x) = 2$ if $f(x) = \begin{cases} 4 - 2x, & x < 1 \\ 6x - 4, & x \geq 1 \end{cases}$
48. $\lim_{x \rightarrow 0} f(x) = 0$ if $f(x) = \begin{cases} 2x, & x < 0 \\ x/2, & x \geq 0 \end{cases}$
49. $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$



50. $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$



Theory and Examples

51. Define what it means to say that $\lim_{x \rightarrow 0} g(x) = k$.
52. Prove that $\lim_{x \rightarrow c} f(x) = L$ if and only if $\lim_{h \rightarrow 0} f(h + c) = L$.
53. **A wrong statement about limits** Show by example that the following statement is wrong.

The number L is the limit of $f(x)$ as x approaches x_0 if $f(x)$ gets closer to L as x approaches x_0 .

Explain why the function in your example does not have the given value of L as a limit as $x \rightarrow x_0$.

54. **Another wrong statement about limits** Show by example that the following statement is wrong.

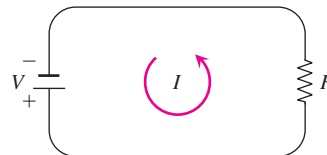
The number L is the limit of $f(x)$ as x approaches x_0 if, given any $\epsilon > 0$, there exists a value of x for which $|f(x) - L| < \epsilon$.

Explain why the function in your example does not have the given value of L as a limit as $x \rightarrow x_0$.

- T** 55. **Grinding engine cylinders** Before contracting to grind engine cylinders to a cross-sectional area of 9 in^2 , you need to know how much deviation from the ideal cylinder diameter of $x_0 = 3.385 \text{ in.}$ you can allow and still have the area come within 0.01 in^2 of the required 9 in^2 . To find out, you let $A = \pi(x/2)^2$ and look for the interval in which you must hold x to make $|A - 9| \leq 0.01$. What interval do you find?

56. **Manufacturing electrical resistors** Ohm's law for electrical circuits like the one shown in the accompanying figure states that $V = RI$. In this equation, V is a constant voltage, I is the current in amperes, and R is the resistance in ohms. Your firm has been asked to supply the resistors for a circuit in which V will be 120

volts and I is to be $5 \pm 0.1 \text{ amp.}$ In what interval does R have to lie for I to be within 0.1 amp of the value $I_0 = 5$?



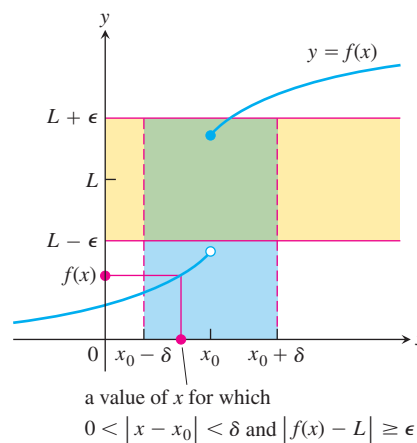
When Is a Number L Not the Limit of $f(x)$ as $x \rightarrow x_0$?

We can prove that $\lim_{x \rightarrow x_0} f(x) \neq L$ by providing an $\epsilon > 0$ such that no possible $\delta > 0$ satisfies the condition

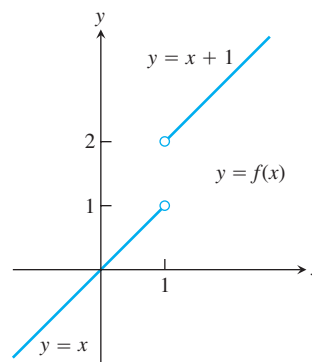
$$\text{For all } x, \quad 0 < |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

We accomplish this for our candidate ϵ by showing that for each $\delta > 0$ there exists a value of x such that

$$0 < |x - x_0| < \delta \quad \text{and} \quad |f(x) - L| \geq \epsilon.$$



57. Let $f(x) = \begin{cases} x, & x < 1 \\ x + 1, & x > 1 \end{cases}$.



- a. Let $\epsilon = 1/2$. Show that no possible $\delta > 0$ satisfies the following condition:

$$\text{For all } x, \quad 0 < |x - 1| < \delta \quad \Rightarrow \quad |f(x) - 2| < 1/2.$$

That is, for each $\delta > 0$ show that there is a value of x such that

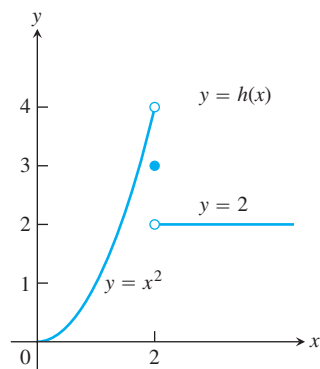
$$0 < |x - 1| < \delta \quad \text{and} \quad |f(x) - 2| \geq 1/2.$$

This will show that $\lim_{x \rightarrow 1} f(x) \neq 2$.

- b. Show that $\lim_{x \rightarrow 1} f(x) \neq 1$.

- c. Show that $\lim_{x \rightarrow 1} f(x) \neq 1.5$.

58. Let
$$h(x) = \begin{cases} x^2, & x < 2 \\ 3, & x = 2 \\ 2, & x > 2. \end{cases}$$

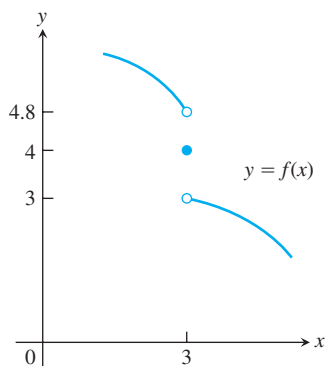


Show that

- a. $\lim_{x \rightarrow 2} h(x) \neq 4$
 b. $\lim_{x \rightarrow 2} h(x) \neq 3$
 c. $\lim_{x \rightarrow 2} h(x) \neq 2$

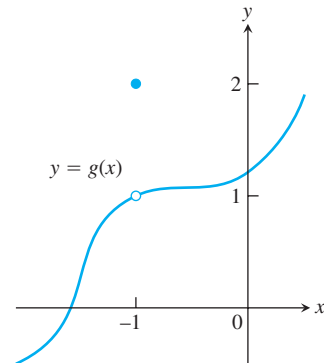
59. For the function graphed here, explain why

- a. $\lim_{x \rightarrow 3} f(x) \neq 4$
 b. $\lim_{x \rightarrow 3} f(x) \neq 4.8$
 c. $\lim_{x \rightarrow 3} f(x) \neq 3$



60. a. For the function graphed here, show that $\lim_{x \rightarrow -1} g(x) \neq 2$.

- b. Does $\lim_{x \rightarrow -1} g(x)$ appear to exist? If so, what is the value of the limit? If not, why not?



COMPUTER EXPLORATIONS

In Exercises 61–66, you will further explore finding deltas graphically. Use a CAS to perform the following steps:

- a. Plot the function $y = f(x)$ near the point x_0 being approached.
 b. Guess the value of the limit L and then evaluate the limit symbolically to see if you guessed correctly.
 c. Using the value $\epsilon = 0.2$, graph the banding lines $y_1 = L - \epsilon$ and $y_2 = L + \epsilon$ together with the function f near x_0 .
 d. From your graph in part (c), estimate a $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

Test your estimate by plotting f , y_1 , and y_2 over the interval $0 < |x - x_0| < \delta$. For your viewing window use $x_0 - 2\delta \leq x \leq x_0 + 2\delta$ and $L - 2\epsilon \leq y \leq L + 2\epsilon$. If any function values lie outside the interval $[L - \epsilon, L + \epsilon]$, your choice of δ was too large. Try again with a smaller estimate.

- e. Repeat parts (c) and (d) successively for $\epsilon = 0.1, 0.05$, and 0.001 .

61. $f(x) = \frac{x^4 - 81}{x - 3}, \quad x_0 = 3$

62. $f(x) = \frac{5x^3 + 9x^2}{2x^5 + 3x^2}, \quad x_0 = 0$

63. $f(x) = \frac{\sin 2x}{3x}, \quad x_0 = 0$

64. $f(x) = \frac{x(1 - \cos x)}{x - \sin x}, \quad x_0 = 0$

65. $f(x) = \frac{\sqrt[3]{x} - 1}{x - 1}, \quad x_0 = 1$

66. $f(x) = \frac{3x^2 - (7x + 1)\sqrt{x} + 5}{x - 1}, \quad x_0 = 1$

2.4 One-Sided Limits and Limits at Infinity

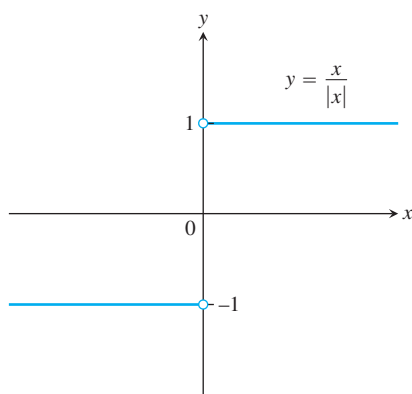


FIGURE 2.21 Different right-hand and left-hand limits at the origin.

In this section we extend the limit concept to *one-sided limits*, which are limits as x approaches the number x_0 from the left-hand side (where $x < x_0$) or the right-hand side ($x > x_0$) only. We also analyze the graphs of certain rational functions as well as other functions with limit behavior as $x \rightarrow \pm\infty$.

One-Sided Limits

To have a limit L as x approaches c , a function f must be defined on *both sides* of c and its values $f(x)$ must approach L as x approaches c from either side. Because of this, ordinary limits are called **two-sided**.

If f fails to have a two-sided limit at c , it may still have a one-sided limit, that is, a limit if the approach is only from one side. If the approach is from the right, the limit is a **right-hand limit**. From the left, it is a **left-hand limit**.

The function $f(x) = x/|x|$ (Figure 2.21) has limit 1 as x approaches 0 from the right, and limit -1 as x approaches 0 from the left. Since these one-sided limit values are not the same, there is no single number that $f(x)$ approaches as x approaches 0. So $f(x)$ does not have a (two-sided) limit at 0.

Intuitively, if $f(x)$ is defined on an interval (c, b) , where $c < b$, and approaches arbitrarily close to L as x approaches c from within that interval, then f has **right-hand limit** L at c . We write

$$\lim_{x \rightarrow c^+} f(x) = L.$$

The symbol “ $x \rightarrow c^+$ ” means that we consider only values of x greater than c .

Similarly, if $f(x)$ is defined on an interval (a, c) , where $a < c$ and approaches arbitrarily close to M as x approaches c from within that interval, then f has **left-hand limit** M at c . We write

$$\lim_{x \rightarrow c^-} f(x) = M.$$

The symbol “ $x \rightarrow c^-$ ” means that we consider only x values less than c .

These informal definitions are illustrated in Figure 2.22. For the function $f(x) = x/|x|$ in Figure 2.21 we have

$$\lim_{x \rightarrow 0^+} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 0^-} f(x) = -1.$$

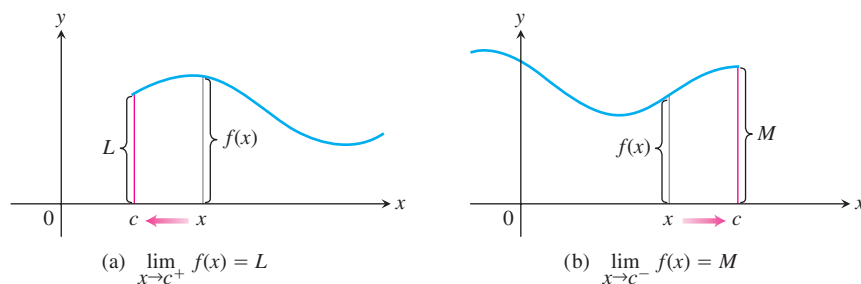


FIGURE 2.22 (a) Right-hand limit as x approaches c . (b) Left-hand limit as x approaches c .

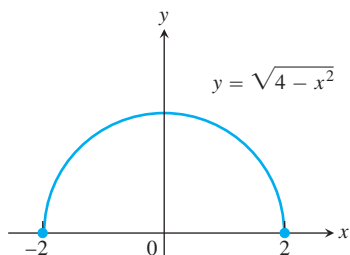


FIGURE 2.23 $\lim_{x \rightarrow -2^-} \sqrt{4 - x^2} = 0$ and $\lim_{x \rightarrow -2^+} \sqrt{4 - x^2} = 0$ (Example 1).

EXAMPLE 1 One-Sided Limits for a Semicircle

The domain of $f(x) = \sqrt{4 - x^2}$ is $[-2, 2]$; its graph is the semicircle in Figure 2.23. We have

$$\lim_{x \rightarrow -2^+} \sqrt{4 - x^2} = 0 \quad \text{and} \quad \lim_{x \rightarrow 2^-} \sqrt{4 - x^2} = 0.$$

The function does not have a left-hand limit at $x = -2$ or a right-hand limit at $x = 2$. It does not have ordinary two-sided limits at either -2 or 2 . ■

One-sided limits have all the properties listed in Theorem 1 in Section 2.2. The right-hand limit of the sum of two functions is the sum of their right-hand limits, and so on. The theorems for limits of polynomials and rational functions hold with one-sided limits, as does the Sandwich Theorem and Theorem 5. One-sided limits are related to limits in the following way.

THEOREM 6

A function $f(x)$ has a limit as x approaches c if and only if it has left-hand and right-hand limits there and these one-sided limits are equal:

$$\lim_{x \rightarrow c} f(x) = L \quad \Leftrightarrow \quad \lim_{x \rightarrow c^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c^+} f(x) = L.$$

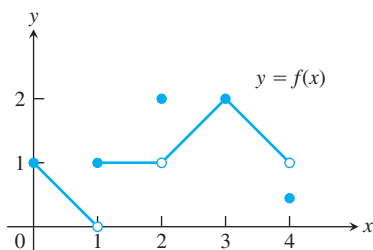


FIGURE 2.24 Graph of the function in Example 2.

EXAMPLE 2 Limits of the Function Graphed in Figure 2.24

- At $x = 0$: $\lim_{x \rightarrow 0^+} f(x) = 1$,
 $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0} f(x)$ do not exist. The function is not defined to the left of $x = 0$.
- At $x = 1$: $\lim_{x \rightarrow 1^-} f(x) = 0$ even though $f(1) = 1$,
 $\lim_{x \rightarrow 1^+} f(x) = 1$,
 $\lim_{x \rightarrow 1} f(x)$ does not exist. The right- and left-hand limits are not equal.
- At $x = 2$: $\lim_{x \rightarrow 2^-} f(x) = 1$,
 $\lim_{x \rightarrow 2^+} f(x) = 1$,
 $\lim_{x \rightarrow 2} f(x) = 1$ even though $f(2) = 2$.
- At $x = 3$: $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3} f(x) = f(3) = 2$.
- At $x = 4$: $\lim_{x \rightarrow 4^-} f(x) = 1$ even though $f(4) \neq 1$,
 $\lim_{x \rightarrow 4^+} f(x)$ and $\lim_{x \rightarrow 4} f(x)$ do not exist. The function is not defined to the right of $x = 4$.

At every other point c in $[0, 4]$, $f(x)$ has limit $f(c)$. ■

Precise Definitions of One-Sided Limits

The formal definition of the limit in Section 2.3 is readily modified for one-sided limits.

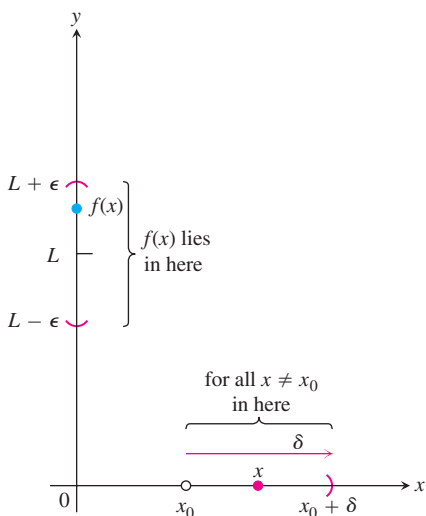


FIGURE 2.25 Intervals associated with the definition of right-hand limit.

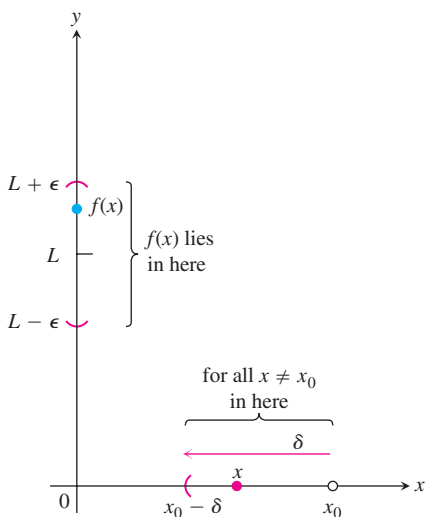


FIGURE 2.26 Intervals associated with the definition of left-hand limit.

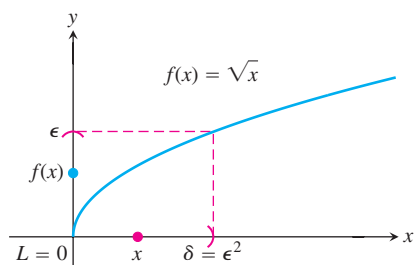


FIGURE 2.27 $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$ in Example 3.

DEFINITIONS Right-Hand, Left-Hand Limits

We say that $f(x)$ has **right-hand limit** L at x_0 , and write

$$\lim_{x \rightarrow x_0^+} f(x) = L \quad (\text{See Figure 2.25})$$

if for every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x

$$x_0 < x < x_0 + \delta \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

We say that f has **left-hand limit** L at x_0 , and write

$$\lim_{x \rightarrow x_0^-} f(x) = L \quad (\text{See Figure 2.26})$$

if for every number $\epsilon > 0$ there exists a corresponding number $\delta > 0$ such that for all x

$$x_0 - \delta < x < x_0 \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

EXAMPLE 3 Applying the Definition to Find Delta

Prove that

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0.$$

Solution Let $\epsilon > 0$ be given. Here $x_0 = 0$ and $L = 0$, so we want to find a $\delta > 0$ such that for all x

$$0 < x < \delta \quad \Rightarrow \quad |\sqrt{x} - 0| < \epsilon,$$

or

$$0 < x < \delta \quad \Rightarrow \quad \sqrt{x} < \epsilon.$$

Squaring both sides of this last inequality gives

$$x < \epsilon^2 \quad \text{if} \quad 0 < x < \delta.$$

If we choose $\delta = \epsilon^2$ we have

$$0 < x < \delta = \epsilon^2 \quad \Rightarrow \quad \sqrt{x} < \epsilon,$$

or

$$0 < x < \epsilon^2 \quad \Rightarrow \quad |\sqrt{x} - 0| < \epsilon.$$

According to the definition, this shows that $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$ (Figure 2.27). ■

The functions examined so far have had some kind of limit at each point of interest. In general, that need not be the case.

EXAMPLE 4 A Function Oscillating Too Much

Show that $y = \sin(1/x)$ has no limit as x approaches zero from either side (Figure 2.28).

Solution As x approaches zero, its reciprocal, $1/x$, grows without bound and the values of $\sin(1/x)$ cycle repeatedly from -1 to 1 . There is no single number L that the function's

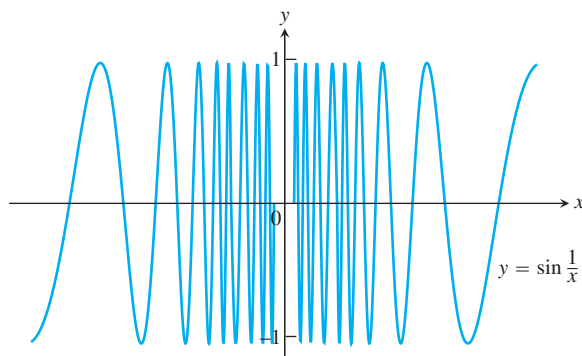


FIGURE 2.28 The function $y = \sin(1/x)$ has neither a right-hand nor a left-hand limit as x approaches zero (Example 4).

values stay increasingly close to as x approaches zero. This is true even if we restrict x to positive values or to negative values. The function has neither a right-hand limit nor a left-hand limit at $x = 0$. ■

Limits Involving $(\sin \theta)/\theta$

A central fact about $(\sin \theta)/\theta$ is that in radian measure its limit as $\theta \rightarrow 0$ is 1. We can see this in Figure 2.29 and confirm it algebraically using the Sandwich Theorem.

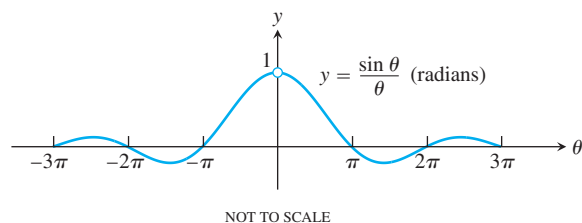


FIGURE 2.29 The graph of $f(\theta) = (\sin \theta)/\theta$.

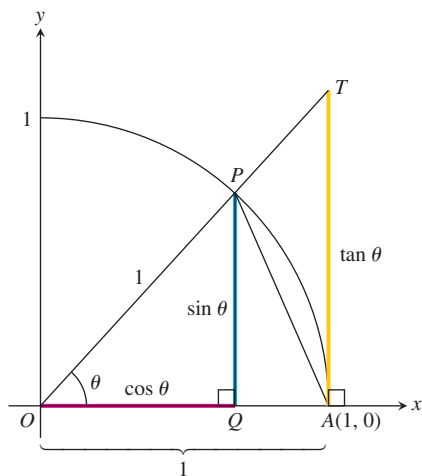


FIGURE 2.30 The figure for the proof of Theorem 7. $TA/OA = \tan \theta$, but $OA = 1$, so $TA = \tan \theta$.

THEOREM 7

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad (\theta \text{ in radians}) \quad (1)$$

Proof The plan is to show that the right-hand and left-hand limits are both 1. Then we will know that the two-sided limit is 1 as well.

To show that the right-hand limit is 1, we begin with positive values of θ less than $\pi/2$ (Figure 2.30). Notice that

$$\text{Area } \triangle OAP < \text{area sector } OAP < \text{area } \triangle OAT.$$

Equation (2) is where radian measure comes in: The area of sector OAP is $\theta/2$ only if θ is measured in radians.

We can express these areas in terms of θ as follows:

$$\begin{aligned}\text{Area } \triangle OAP &= \frac{1}{2} \text{ base} \times \text{height} = \frac{1}{2} (1)(\sin \theta) = \frac{1}{2} \sin \theta \\ \text{Area sector } OAP &= \frac{1}{2} r^2 \theta = \frac{1}{2} (1)^2 \theta = \frac{\theta}{2} \\ \text{Area } \triangle OAT &= \frac{1}{2} \text{ base} \times \text{height} = \frac{1}{2} (1)(\tan \theta) = \frac{1}{2} \tan \theta.\end{aligned}\tag{2}$$

Thus,

$$\frac{1}{2} \sin \theta < \frac{1}{2} \theta < \frac{1}{2} \tan \theta.$$

This last inequality goes the same way if we divide all three terms by the number $(1/2) \sin \theta$, which is positive since $0 < \theta < \pi/2$:

$$1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta}.$$

Taking reciprocals reverses the inequalities:

$$1 > \frac{\sin \theta}{\theta} > \cos \theta.$$

Since $\lim_{\theta \rightarrow 0^+} \cos \theta = 1$ (Example 6b, Section 2.2), the Sandwich Theorem gives

$$\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1.$$

Recall that $\sin \theta$ and θ are both *odd functions* (Section 1.4). Therefore, $f(\theta) = (\sin \theta)/\theta$ is an *even function*, with a graph symmetric about the y -axis (see Figure 2.29). This symmetry implies that the left-hand limit at 0 exists and has the same value as the right-hand limit:

$$\lim_{\theta \rightarrow 0^-} \frac{\sin \theta}{\theta} = 1 = \lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta},$$

so $\lim_{\theta \rightarrow 0} (\sin \theta)/\theta = 1$ by Theorem 6. ■

EXAMPLE 5 Using $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Show that (a) $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0$ and (b) $\lim_{x \rightarrow 0} \frac{\sin 2x}{5x} = \frac{2}{5}$.

Solution

(a) Using the half-angle formula $\cos h = 1 - 2 \sin^2(h/2)$, we calculate

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} &= \lim_{h \rightarrow 0} -\frac{2 \sin^2(h/2)}{h} \\ &= -\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \sin \theta && \text{Let } \theta = h/2. \\ &= -(1)(0) = 0.\end{aligned}$$

- (b) Equation (1) does not apply to the original fraction. We need a $2x$ in the denominator, not a $5x$. We produce it by multiplying numerator and denominator by $2/5$:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin 2x}{5x} &= \lim_{x \rightarrow 0} \frac{(2/5) \cdot \sin 2x}{(2/5) \cdot 5x} \\ &= \frac{2}{5} \lim_{x \rightarrow 0} \frac{\sin 2x}{2x} \\ &= \frac{2}{5} (1) = \frac{2}{5}\end{aligned}$$

Now, Eq. (1) applies with $\theta = 2x$.

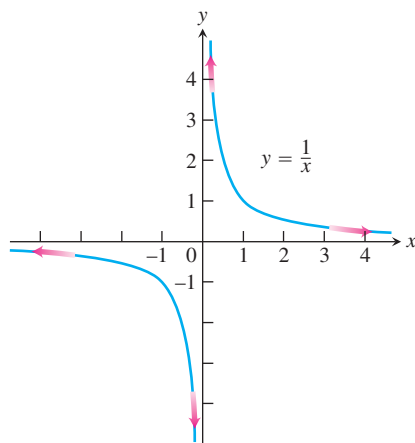


FIGURE 2.31 The graph of $y = 1/x$.

Finite Limits as $x \rightarrow \pm \infty$

The symbol for infinity (∞) does not represent a real number. We use ∞ to describe the behavior of a function when the values in its domain or range outgrow all finite bounds. For example, the function $f(x) = 1/x$ is defined for all $x \neq 0$ (Figure 2.31). When x is positive and becomes increasingly large, $1/x$ becomes increasingly small. When x is negative and its magnitude becomes increasingly large, $1/x$ again becomes small. We summarize these observations by saying that $f(x) = 1/x$ has limit 0 as $x \rightarrow \pm \infty$ or that 0 is a *limit of $f(x) = 1/x$ at infinity and negative infinity*. Here is a precise definition.

DEFINITIONS Limit as x approaches ∞ or $-\infty$

1. We say that $f(x)$ has the **limit L as x approaches infinity** and write

$$\lim_{x \rightarrow \infty} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number M such that for all x

$$x > M \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

2. We say that $f(x)$ has the **limit L as x approaches minus infinity** and write

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number N such that for all x

$$x < N \quad \Rightarrow \quad |f(x) - L| < \epsilon.$$

Intuitively, $\lim_{x \rightarrow \infty} f(x) = L$ if, as x moves increasingly far from the origin in the positive direction, $f(x)$ gets arbitrarily close to L . Similarly, $\lim_{x \rightarrow -\infty} f(x) = L$ if, as x moves increasingly far from the origin in the negative direction, $f(x)$ gets arbitrarily close to L .

The strategy for calculating limits of functions as $x \rightarrow \pm \infty$ is similar to the one for finite limits in Section 2.2. There we first found the limits of the constant and identity functions $y = k$ and $y = x$. We then extended these results to other functions by applying a theorem about limits of algebraic combinations. Here we do the same thing, except that the starting functions are $y = k$ and $y = 1/x$ instead of $y = k$ and $y = x$.

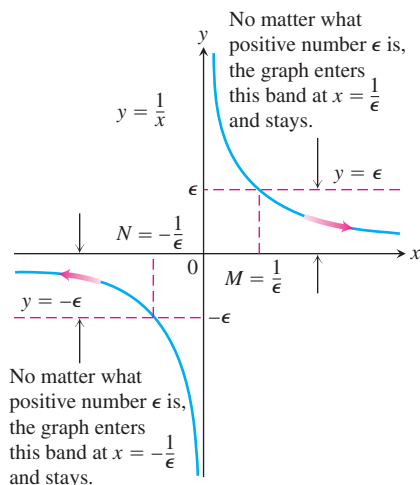


FIGURE 2.32 The geometry behind the argument in Example 6.

The basic facts to be verified by applying the formal definition are

$$\lim_{x \rightarrow \pm\infty} k = k \quad \text{and} \quad \lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0. \quad (3)$$

We prove the latter and leave the former to Exercises 71 and 72.

EXAMPLE 6 Limits at Infinity for $f(x) = \frac{1}{x}$

Show that

(a) $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$

(b) $\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$

Solution

(a) Let $\epsilon > 0$ be given. We must find a number M such that for all x

$$x > M \quad \Rightarrow \quad \left| \frac{1}{x} - 0 \right| = \left| \frac{1}{x} \right| < \epsilon.$$

The implication will hold if $M = 1/\epsilon$ or any larger positive number (Figure 2.32). This proves $\lim_{x \rightarrow \infty} (1/x) = 0$.

(b) Let $\epsilon > 0$ be given. We must find a number N such that for all x

$$x < N \quad \Rightarrow \quad \left| \frac{1}{x} - 0 \right| = \left| \frac{1}{x} \right| < \epsilon.$$

The implication will hold if $N = -1/\epsilon$ or any number less than $-1/\epsilon$ (Figure 2.32). This proves $\lim_{x \rightarrow -\infty} (1/x) = 0$. ■

Limits at infinity have properties similar to those of finite limits.

THEOREM 8 Limit Laws as $x \rightarrow \pm\infty$

If L , M , and k , are real numbers and

$$\lim_{x \rightarrow \pm\infty} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow \pm\infty} g(x) = M, \quad \text{then}$$

1. *Sum Rule:*

$$\lim_{x \rightarrow \pm\infty} (f(x) + g(x)) = L + M$$

2. *Difference Rule:*

$$\lim_{x \rightarrow \pm\infty} (f(x) - g(x)) = L - M$$

3. *Product Rule:*

$$\lim_{x \rightarrow \pm\infty} (f(x) \cdot g(x)) = L \cdot M$$

4. *Constant Multiple Rule:*

$$\lim_{x \rightarrow \pm\infty} (k \cdot f(x)) = k \cdot L$$

5. *Quotient Rule:*

$$\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$$

6. *Power Rule:* If r and s are integers with no common factors, $s \neq 0$, then

$$\lim_{x \rightarrow \pm\infty} (f(x))^{r/s} = L^{r/s}$$

provided that $L^{r/s}$ is a real number. (If s is even, we assume that $L > 0$.)

These properties are just like the properties in Theorem 1, Section 2.2, and we use them the same way.

EXAMPLE 7 Using Theorem 8

$$\begin{aligned} \text{(a)} \quad \lim_{x \rightarrow \infty} \left(5 + \frac{1}{x} \right) &= \lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} \frac{1}{x} && \text{Sum Rule} \\ &= 5 + 0 = 5 && \text{Known limits} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lim_{x \rightarrow -\infty} \frac{\pi\sqrt{3}}{x^2} &= \lim_{x \rightarrow -\infty} \pi\sqrt{3} \cdot \frac{1}{x} \cdot \frac{1}{x} \\ &= \lim_{x \rightarrow -\infty} \pi\sqrt{3} \cdot \lim_{x \rightarrow -\infty} \frac{1}{x} \cdot \lim_{x \rightarrow -\infty} \frac{1}{x} && \text{Product rule} \\ &= \pi\sqrt{3} \cdot 0 \cdot 0 = 0 && \text{Known limits} \end{aligned}$$

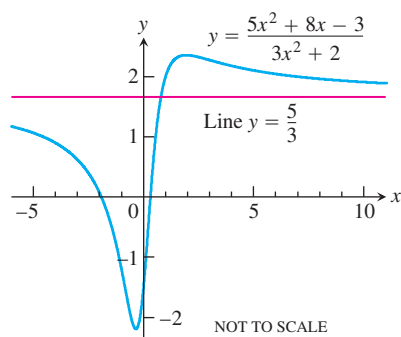


FIGURE 2.33 The graph of the function in Example 8. The graph approaches the line $y = 5/3$ as $|x|$ increases.

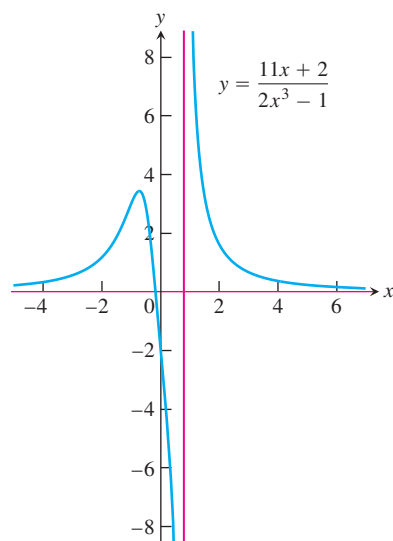


FIGURE 2.34 The graph of the function in Example 9. The graph approaches the x -axis as $|x|$ increases.

Limits at Infinity of Rational Functions

To determine the limit of a rational function as $x \rightarrow \pm\infty$, we can divide the numerator and denominator by the highest power of x in the denominator. What happens then depends on the degrees of the polynomials involved.

EXAMPLE 8 Numerator and Denominator of Same Degree

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2} &= \lim_{x \rightarrow \infty} \frac{5 + (8/x) - (3/x^2)}{3 + (2/x^2)} && \text{Divide numerator and denominator by } x^2. \\ &= \frac{5 + 0 - 0}{3 + 0} = \frac{5}{3} && \text{See Fig. 2.33.} \end{aligned}$$

EXAMPLE 9 Degree of Numerator Less Than Degree of Denominator

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{11x + 2}{2x^3 - 1} &= \lim_{x \rightarrow -\infty} \frac{(11/x^2) + (2/x^3)}{2 - (1/x^3)} && \text{Divide numerator and denominator by } x^3. \\ &= \frac{0 + 0}{2 - 0} = 0 && \text{See Fig. 2.34.} \end{aligned}$$

We give an example of the case when the degree of the numerator is greater than the degree of the denominator in the next section (Example 8, Section 2.5).

Horizontal Asymptotes

If the distance between the graph of a function and some fixed line approaches zero as a point on the graph moves increasingly far from the origin, we say that the graph approaches the line asymptotically and that the line is an *asymptote* of the graph.

Looking at $f(x) = 1/x$ (See Figure 2.31), we observe that the x -axis is an asymptote of the curve on the right because

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

and on the left because

$$\lim_{x \rightarrow -\infty} \frac{1}{x} = 0.$$

We say that the x -axis is a *horizontal asymptote* of the graph of $f(x) = 1/x$.

DEFINITION Horizontal Asymptote

A line $y = b$ is a **horizontal asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = b.$$

The curve

$$f(x) = \frac{5x^2 + 8x - 3}{3x^2 + 2}$$

sketched in Figure 2.33 (Example 8) has the line $y = 5/3$ as a horizontal asymptote on both the right and the left because

$$\lim_{x \rightarrow \infty} f(x) = \frac{5}{3} \quad \text{and} \quad \lim_{x \rightarrow -\infty} f(x) = \frac{5}{3}.$$

EXAMPLE 10 Substituting a New Variable

Find $\lim_{x \rightarrow \infty} \sin(1/x)$.

Solution We introduce the new variable $t = 1/x$. From Example 6, we know that $t \rightarrow 0^+$ as $x \rightarrow \infty$ (see Figure 2.31). Therefore,

$$\lim_{x \rightarrow \infty} \sin \frac{1}{x} = \lim_{t \rightarrow 0^+} \sin t = 0. \quad \blacksquare$$

The Sandwich Theorem Revisited

The Sandwich Theorem also holds for limits as $x \rightarrow \pm\infty$.

EXAMPLE 11 A Curve May Cross Its Horizontal Asymptote

Using the Sandwich Theorem, find the horizontal asymptote of the curve

$$y = 2 + \frac{\sin x}{x}.$$

Solution We are interested in the behavior as $x \rightarrow \pm\infty$. Since

$$0 \leq \left| \frac{\sin x}{x} \right| \leq \left| \frac{1}{x} \right|$$

and $\lim_{x \rightarrow \pm\infty} |1/x| = 0$, we have $\lim_{x \rightarrow \pm\infty} (\sin x)/x = 0$ by the Sandwich Theorem. Hence,

$$\lim_{x \rightarrow \pm\infty} \left(2 + \frac{\sin x}{x} \right) = 2 + 0 = 2,$$

and the line $y = 2$ is a horizontal asymptote of the curve on both left and right (Figure 2.35).

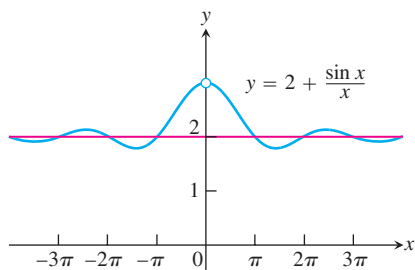


FIGURE 2.35 A curve may cross one of its asymptotes infinitely often (Example 11).

This example illustrates that a curve may cross one of its horizontal asymptotes, perhaps many times. ■

Oblique Asymptotes

If the degree of the numerator of a rational function is one greater than the degree of the denominator, the graph has an **oblique (slanted) asymptote**. We find an equation for the asymptote by dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \rightarrow \pm\infty$. Here's an example.

EXAMPLE 12 Finding an Oblique Asymptote

Find the oblique asymptote for the graph of

$$f(x) = \frac{2x^2 - 3}{7x + 4}$$

in Figure 2.36.

Solution By long division, we find

$$\begin{aligned} f(x) &= \frac{2x^2 - 3}{7x + 4} \\ &= \underbrace{\left(\frac{2}{7}x - \frac{8}{49}\right)}_{\text{linear function } g(x)} + \underbrace{\frac{-115}{49(7x + 4)}}_{\text{remainder}} \end{aligned}$$

As $x \rightarrow \pm\infty$, the remainder, whose magnitude gives the vertical distance between the graphs of f and g , goes to zero, making the (slanted) line

$$g(x) = \frac{2}{7}x - \frac{8}{49}$$

an asymptote of the graph of f (Figure 2.36). The line $y = g(x)$ is an asymptote both to the right and to the left. In the next section you will see that the function $f(x)$ grows arbitrarily large in absolute value as x approaches $-4/7$, where the denominator becomes zero (Figure 2.36). ■

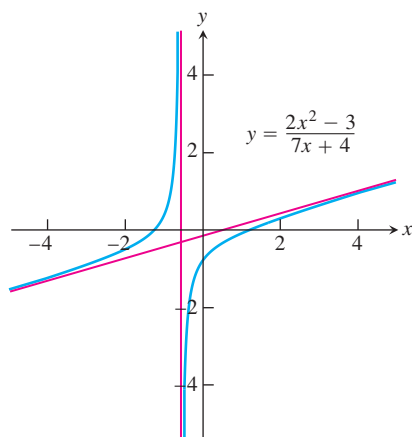


FIGURE 2.36 The function in Example 12 has an oblique asymptote.

2.5

Infinite Limits and Vertical Asymptotes

In this section we extend the concept of limit to *infinite limits*, which are not limits as before, but rather an entirely new use of the term limit. Infinite limits provide useful symbols and language for describing the behavior of functions whose values become arbitrarily large, positive or negative. We continue our analysis of graphs of rational functions from the last section, using vertical asymptotes and dominant terms for numerically large values of x .

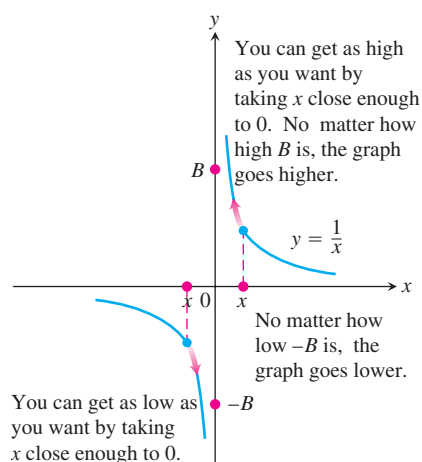


FIGURE 2.37 One-sided infinite limits:

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

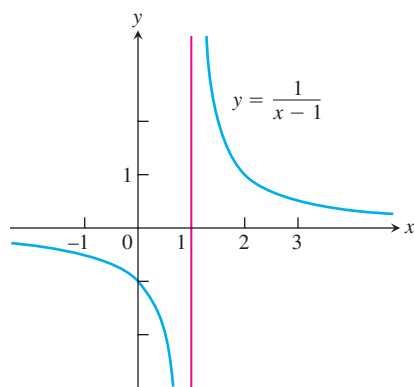


FIGURE 2.38 Near $x = 1$, the function $y = 1/(x - 1)$ behaves the way the function $y = 1/x$ behaves near $x = 0$. Its graph is the graph of $y = 1/x$ shifted 1 unit to the right (Example 1).

Infinite Limits

Let us look again at the function $f(x) = 1/x$. As $x \rightarrow 0^+$, the values of f grow without bound, eventually reaching and surpassing every positive real number. That is, given any positive real number B , however large, the values of f become larger still (Figure 2.37). Thus, f has no limit as $x \rightarrow 0^+$. It is nevertheless convenient to describe the behavior of f by saying that $f(x)$ approaches ∞ as $x \rightarrow 0^+$. We write

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{x} = \infty.$$

In writing this, we are *not* saying that the limit exists. Nor are we saying that there is a real number ∞ , for there is no such number. Rather, we are saying that $\lim_{x \rightarrow 0^+} (1/x)$ *does not exist because $1/x$ becomes arbitrarily large and positive as $x \rightarrow 0^+$* .

As $x \rightarrow 0^-$, the values of $f(x) = 1/x$ become arbitrarily large and negative. Given any negative real number $-B$, the values of f eventually lie below $-B$. (See Figure 2.37.) We write

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

Again, we are not saying that the limit exists and equals the number $-\infty$. There *is* no real number $-\infty$. We are describing the behavior of a function whose limit as $x \rightarrow 0^-$ *does not exist because its values become arbitrarily large and negative*.

EXAMPLE 1 One-Sided Infinite Limits

Find $\lim_{x \rightarrow 1^+} \frac{1}{x - 1}$ and $\lim_{x \rightarrow 1^-} \frac{1}{x - 1}$.

Geometric Solution The graph of $y = 1/(x - 1)$ is the graph of $y = 1/x$ shifted 1 unit to the right (Figure 2.38). Therefore, $y = 1/(x - 1)$ behaves near 1 exactly the way $y = 1/x$ behaves near 0:

$$\lim_{x \rightarrow 1^+} \frac{1}{x - 1} = \infty \quad \text{and} \quad \lim_{x \rightarrow 1^-} \frac{1}{x - 1} = -\infty.$$

Analytic Solution Think about the number $x - 1$ and its reciprocal. As $x \rightarrow 1^+$, we have $(x - 1) \rightarrow 0^+$ and $1/(x - 1) \rightarrow \infty$. As $x \rightarrow 1^-$, we have $(x - 1) \rightarrow 0^-$ and $1/(x - 1) \rightarrow -\infty$. ■

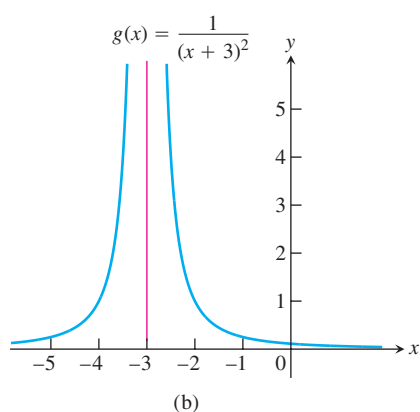
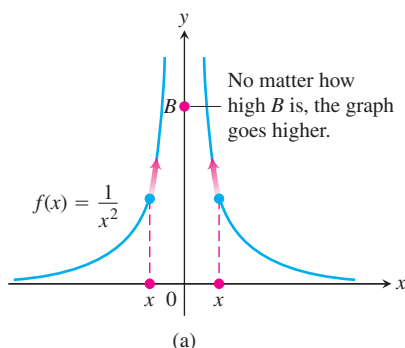


FIGURE 2.39 The graphs of the functions in Example 2. (a) $f(x)$ approaches infinity as $x \rightarrow 0$. (b) $g(x)$ approaches infinity as $x \rightarrow -3$.

EXAMPLE 2 Two-Sided Infinite Limits

Discuss the behavior of

- (a) $f(x) = \frac{1}{x^2}$ near $x = 0$,
 (b) $g(x) = \frac{1}{(x+3)^2}$ near $x = -3$.

Solution

- (a) As x approaches zero from either side, the values of $1/x^2$ are positive and become arbitrarily large (Figure 2.39a):

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$

- (b) The graph of $g(x) = 1/(x+3)^2$ is the graph of $f(x) = 1/x^2$ shifted 3 units to the left (Figure 2.39b). Therefore, g behaves near -3 exactly the way f behaves near 0.

$$\lim_{x \rightarrow -3} g(x) = \lim_{x \rightarrow -3} \frac{1}{(x+3)^2} = \infty.$$

The function $y = 1/x$ shows no consistent behavior as $x \rightarrow 0$. We have $1/x \rightarrow \infty$ if $x \rightarrow 0^+$, but $1/x \rightarrow -\infty$ if $x \rightarrow 0^-$. All we can say about $\lim_{x \rightarrow 0} (1/x)$ is that it does not exist. The function $y = 1/x^2$ is different. Its values approach infinity as x approaches zero from either side, so we can say that $\lim_{x \rightarrow 0} (1/x^2) = \infty$.

EXAMPLE 3 Rational Functions Can Behave in Various Ways Near Zeros of Their Denominators

- (a) $\lim_{x \rightarrow 2} \frac{(x-2)^2}{x^2-4} = \lim_{x \rightarrow 2} \frac{(x-2)^2}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{x-2}{x+2} = 0$
 (b) $\lim_{x \rightarrow 2} \frac{x-2}{x^2-4} = \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(x+2)} = \lim_{x \rightarrow 2} \frac{1}{x+2} = \frac{1}{4}$
 (c) $\lim_{x \rightarrow 2^+} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{x-3}{(x-2)(x+2)} = -\infty$
 (d) $\lim_{x \rightarrow 2^-} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^-} \frac{x-3}{(x-2)(x+2)} = \infty$
 (e) $\lim_{x \rightarrow 2} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2} \frac{x-3}{(x-2)(x+2)}$ does not exist.
 (f) $\lim_{x \rightarrow 2} \frac{2-x}{(x-2)^3} = \lim_{x \rightarrow 2} \frac{-(x-2)}{(x-2)^3} = \lim_{x \rightarrow 2} \frac{-1}{(x-2)^2} = -\infty$

The values are negative for $x > 2$, x near 2.

The values are positive for $x < 2$, x near 2.

See parts (c) and (d).

In parts (a) and (b) the effect of the zero in the denominator at $x = 2$ is canceled because the numerator is zero there also. Thus a finite limit exists. This is not true in part (f), where cancellation still leaves a zero in the denominator.

Precise Definitions of Infinite Limits

Instead of requiring $f(x)$ to lie arbitrarily close to a finite number L for all x sufficiently close to x_0 , the definitions of infinite limits require $f(x)$ to lie arbitrarily far from the ori-

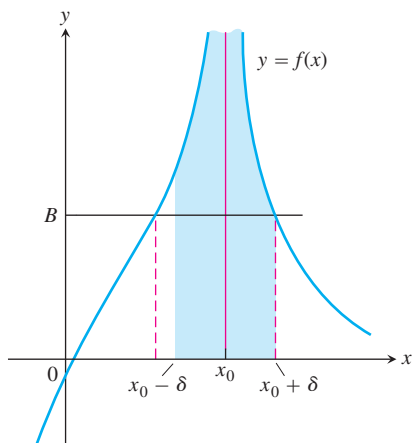


FIGURE 2.40 $f(x)$ approaches infinity as $x \rightarrow x_0$.

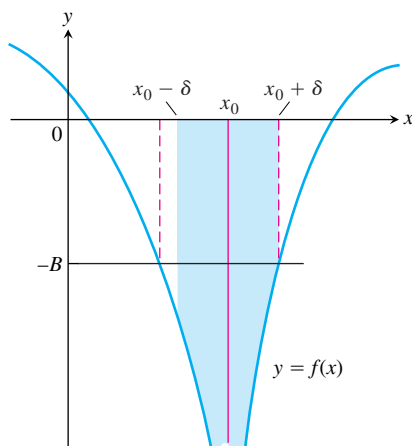


FIGURE 2.41 $f(x)$ approaches negative infinity as $x \rightarrow x_0$.

gin. Except for this change, the language is identical with what we have seen before. Figures 2.40 and 2.41 accompany these definitions.

DEFINITIONS Infinity, Negative Infinity as Limits

1. We say that $f(x)$ **approaches infinity as x approaches x_0** , and write

$$\lim_{x \rightarrow x_0} f(x) = \infty,$$

if for every positive real number B there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \quad \Rightarrow \quad f(x) > B.$$

2. We say that $f(x)$ **approaches negative infinity as x approaches x_0** , and write

$$\lim_{x \rightarrow x_0} f(x) = -\infty,$$

if for every negative real number $-B$ there exists a corresponding $\delta > 0$ such that for all x

$$0 < |x - x_0| < \delta \quad \Rightarrow \quad f(x) < -B.$$

The precise definitions of one-sided infinite limits at x_0 are similar and are stated in the exercises.

EXAMPLE 4 Using the Definition of Infinite Limits

Prove that $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$.

Solution Given $B > 0$, we want to find $\delta > 0$ such that

$$0 < |x - 0| < \delta \quad \text{implies} \quad \frac{1}{x^2} > B.$$

Now,

$$\frac{1}{x^2} > B \quad \text{if and only if} \quad x^2 < \frac{1}{B}$$

or, equivalently,

$$|x| < \frac{1}{\sqrt{B}}.$$

Thus, choosing $\delta = 1/\sqrt{B}$ (or any smaller positive number), we see that

$$|x| < \delta \quad \text{implies} \quad \frac{1}{x^2} > \frac{1}{\delta^2} \geq B.$$

Therefore, by definition,

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$

■

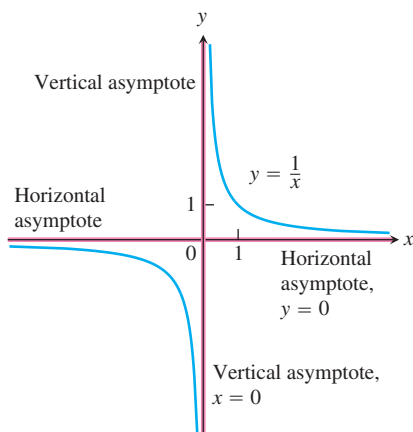


FIGURE 2.42 The coordinate axes are asymptotes of both branches of the hyperbola $y = 1/x$.

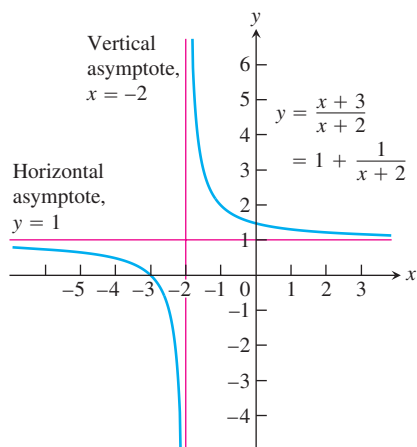


FIGURE 2.43 The lines $y = 1$ and $x = -2$ are asymptotes of the curve $y = (x + 3)/(x + 2)$ (Example 5).

Vertical Asymptotes

Notice that the distance between a point on the graph of $y = 1/x$ and the y -axis approaches zero as the point moves vertically along the graph and away from the origin (Figure 2.42). This behavior occurs because

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

We say that the line $x = 0$ (the y -axis) is a *vertical asymptote* of the graph of $y = 1/x$. Observe that the denominator is zero at $x = 0$ and the function is undefined there.

DEFINITION Vertical Asymptote

A line $x = a$ is a **vertical asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty.$$

EXAMPLE 5 Looking for Asymptotes

Find the horizontal and vertical asymptotes of the curve

$$y = \frac{x + 3}{x + 2}.$$

Solution We are interested in the behavior as $x \rightarrow \pm\infty$ and as $x \rightarrow -2$, where the denominator is zero.

The asymptotes are quickly revealed if we recast the rational function as a polynomial with a remainder, by dividing $(x + 3)$ into $(x + 2)$.

$$\begin{array}{r} 1 \\ x + 2 \overline{) x + 3} \\ \underline{x + 2} \\ 1 \end{array}$$

This result enables us to rewrite y :

$$y = 1 + \frac{1}{x + 2}.$$

We now see that the curve in question is the graph of $y = 1/x$ shifted 1 unit up and 2 units left (Figure 2.43). The asymptotes, instead of being the coordinate axes, are now the lines $y = 1$ and $x = -2$. ■

EXAMPLE 6 Asymptotes Need Not Be Two-Sided

Find the horizontal and vertical asymptotes of the graph of

$$f(x) = -\frac{8}{x^2 - 4}.$$

Solution We are interested in the behavior as $x \rightarrow \pm\infty$ and as $x \rightarrow \pm 2$, where the denominator is zero. Notice that f is an even function of x , so its graph is symmetric with respect to the y -axis.

(a) *The behavior as $x \rightarrow \pm\infty$.* Since $\lim_{x \rightarrow \infty} f(x) = 0$, the line $y = 0$ is a horizontal asymptote of the graph to the right. By symmetry it is an asymptote to the left as well

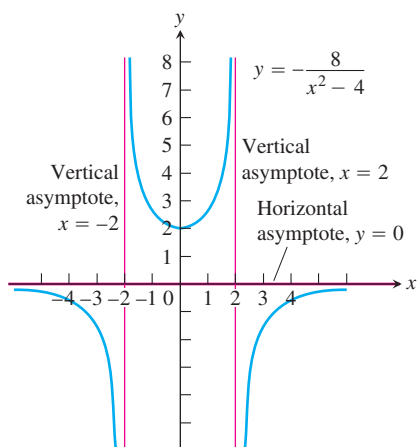


FIGURE 2.44 Graph of $y = -8/(x^2 - 4)$. Notice that the curve approaches the x -axis from only one side. Asymptotes do not have to be two-sided (Example 6).

(Figure 2.44). Notice that the curve approaches the x -axis from only the negative side (or from below).

(b) The behavior as $x \rightarrow \pm 2$. Since

$$\lim_{x \rightarrow 2^+} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 2^-} f(x) = \infty,$$

the line $x = 2$ is a vertical asymptote both from the right and from the left. By symmetry, the same holds for the line $x = -2$.

There are no other asymptotes because f has a finite limit at every other point. ■

EXAMPLE 7 Curves with Infinitely Many Asymptotes

The curves

$$y = \sec x = \frac{1}{\cos x} \quad \text{and} \quad y = \tan x = \frac{\sin x}{\cos x}$$

both have vertical asymptotes at odd-integer multiples of $\pi/2$, where $\cos x = 0$ (Figure 2.45).

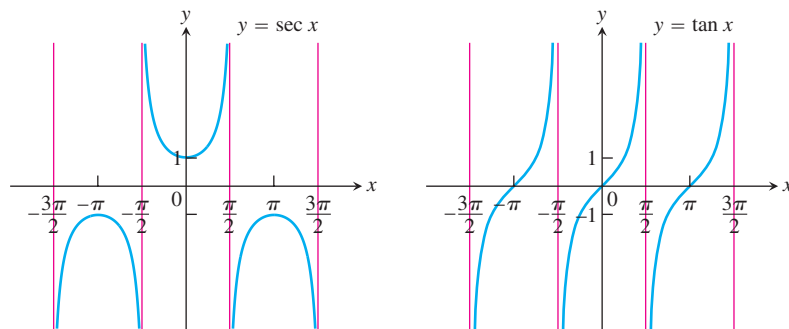


FIGURE 2.45 The graphs of $\sec x$ and $\tan x$ have infinitely many vertical asymptotes (Example 7).

The graphs of

$$y = \csc x = \frac{1}{\sin x} \quad \text{and} \quad y = \cot x = \frac{\cos x}{\sin x}$$

have vertical asymptotes at integer multiples of π , where $\sin x = 0$ (Figure 2.46).

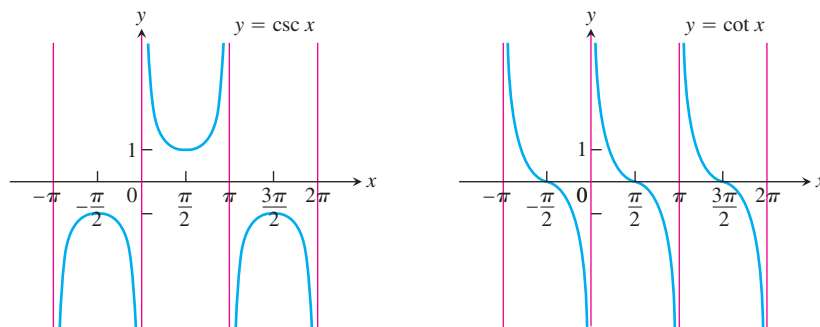


FIGURE 2.46 The graphs of $\csc x$ and $\cot x$ (Example 7). ■